# Quantum Mechanics

## & Quantum Chemistry

*Comprehensive Course Syllabi*

Physics & Theoretical Chemistry Programme

**Series I:** Introductory Quantum Mechanics

Module I.1 — State Space, Operators, and the Bra-Ket Language (6 lectures)

Module I.2 — Bra-Ket Notation and the Language of Quantum Mechanics (12 lectures)

Module I.3 — Quantum Dynamics, Symmetry, and Perturbation Theory (10 lectures)

Module I.4 — Paradigm Physical Systems (12 lectures)

Module I.5 — Introductory Physical Systems: Lecture Menu (12 topics)

**Series II:** Advanced Quantum Mechanics

12 Modules (II.1–II.12), 12 lectures each

**Series III:** Molecular Electronic Structure & Quantum Chemistry

3 Modules (III.1–III.3), approx. 36–40 lectures total

*Each module in Series I & II comprises 12 lectures unless otherwise noted.*

*Hierarchy:* ***Series*** $\supset$ ***Module*** $\supset$ ***Lecture***

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# Preamble and Programme Overview

*This document describes a sequential, research-oriented programme in quantum mechanics and quantum chemistry, organised into a three-level hierarchy:* ***Series*** *(broad thematic strand)* $\supset$ ***Module*** *(self-contained teaching unit, typically 12 lectures)* $\supset$ ***Lecture*** *(single 90-minute session).*

***Series I*** *introduces the algebraic and dynamical foundations of quantum mechanics across three modules: Module I.1 introduces the Hilbert-space language of operators and states; Module I.2 develops the full bra-ket formalism via a dual Dirac–Sakurai approach with five-tier pedagogy; Module I.3 covers quantum dynamics, all pictures of quantum mechanics, perturbation theory, open systems, and example physical systems.*

***Series II*** *deepens the formalism across seven thematic modules.* ***Series III*** *pivots to molecular electronic structure and computational quantum chemistry.*

**Notation conventions used throughout:**

- $\hat{A}, \hat{H}, \hat{U}$ — operators (hat notation)
- $|\psi\rangle, \langle\phi|$ — Dirac ket/bra states
- $\hat{\rho}$ — density operator/matrix
- $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ — commutator
- $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ — anti-commutator

**Hierarchy key:**

- **Series** — top-level strand labelled by Roman numeral (I, II, III)
- **Module** — self-contained teaching block labelled *Series.Number* (e.g. I.1, II.4, III.2)
- **Lecture** — individual 90-min session labelled L$NN$ within its module (e.g. L01, L07, L12)

SERIES I

# Introductory Quantum Mechanics

*Five modules — 40 lectures × 90 minutes*

*Audience: Advanced undergraduates through early graduate students*

*Prerequisites: Linear algebra, classical mechanics, basic complex analysis*

| Module | Title | Lectures |
|---|---|---|
| **I.1** | State Space, Operators, and the Bra-Ket Language<br>Hilbert space; bounded/unbounded operators; adjoint; unitary maps; Hermitian observables; spectral theorem; operator algebra; exponential map; BCH lemma; Wigner's theorem; bra-ket formalism. | 6 |
| **I.2** | Bra-Ket Notation and the Language of Quantum Mechanics<br>States as rays; dual space; inner product; linear operators; eigenkets; projectors; completeness; continuous bases; measurement theory; uncertainty; tensor products; spin; density operators; rigged Hilbert spaces. Dual Dirac–Sakurai approach; five-tier pedagogy. | 12 |
| **I.3** | Quantum Dynamics, Symmetry, and Perturbation Theory<br>Density matrix; pure/mixed states; discrete/continuous bases; uncertainty; symmetries; Noether; time evolution (I & II); Schrödinger, Heisenberg, Dirac pictures; PT; open systems. | 10 |
| **I.4** | Paradigm Physical Systems<br>Free particle; particle in box; harmonic oscillator; Rabi/two-level systems; minimal coupling; Landau levels; Zeeman effect; Stark effect; spin–orbit; fine structure. | 12 |
| **I.5** | Introductory Physical Systems (Lecture Menu)<br>Finite square well; 1D HO; rigid rotor; H-like ions; particle on ring; double well; Kronig–Penney; helium atom; $H_2$ LCAO; resonant tunnelling; linear Stark; Dirac comb. | 12 |
| | **Total** | **40** |

*Each lecture: 90 minutes · Five-tier pedagogy (HS through PhD) · Dual Dirac–Sakurai approach*

# 1 Series I — Introductory Quantum Mechanics

**Series I Overview**

**Audience:** Advanced undergraduates through early graduate students.
**Prerequisites:** Linear algebra, classical mechanics, basic complex analysis.
**Structure:** Five modules totalling 40 lectures × 90 minutes.

| Module | Title | Lectures |
|---|---|---|
| I.1 | State Space, Operators, and the Bra-Ket Language | 6 |
| I.2 | Bra-Ket Notation and the Language of Quantum Mechanics | 12 |
| I.3 | Quantum Dynamics, Symmetry, and Perturbation Theory | 10 |
| I.4 | Paradigm Physical Systems | 12 |
| I.5 | Introductory Physical Systems (Lecture Menu) | 12 |
| | **Total** | **40** |

## 1.1 Module I.1 — State Space, Operators, and the Bra-Ket Language

***Intended audience:*** *Advanced undergraduates with a solid background in classical mechanics and linear algebra.*

***Scope:*** *Foundational module covering the algebraic language of quantum mechanics — from the structure of Hilbert space and operator theory through to the introduction of quantum states, Dirac notation, and the density matrix formalism up to and including the notion of pure and mixed states (§2.3).*

***Duration:*** *6 lectures × 90 minutes (Lectures L01–L06, sections 2.1–2.3). Module I.2 develops the full bra-ket formalism in depth; Module I.3 continues from §2.4 onward.*

### Module Overview

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Hilbert Space of Quantum States | $\mathcal{H}$; norms; completeness; bounded/unbounded operators; adjoint |
| L02 | Unitary Operators & Symmetry | $\hat{U}\hat{U}^\dagger = \hat{1}$; inner-product preservation; rotations; translations |
| L03 | Observables & Spectral Theorem | Hermitian operators; real eigenvalues; orthogonal eigenstates; spectral decomposition |
| L04 | Operator Algebra & Exponential Map | Functions of operators; $e^{i\theta\hat{A}}$; BCH lemma; commutators |
| L05 | Symmetry Operators & Wigner's Theorem | Unitary vs. anti-unitary; Wigner's theorem; projectors $\hat{P}^2 = \hat{P}$ |
| L06 | Bra-Ket Formalism (§2.1–2.2) | Dirac notation; ket/bra duality; inner/outer product; matrix elements |

### Five-Tier Layered Pedagogy

Each lecture is stratified across five audience levels, enabling delivery to a mixed cohort:

| Tier | Cohort | Characteristic content |
|---|---|---|
| Tier 1 | HS | Geometric intuition, analogy with familiar vector spaces, conceptual motivation |
| Tier 2 | Beg. UG | Definitions, worked matrix examples, basic proofs; algebraic facility |
| Tier 3 | Adv. UG | Rigorous derivations, Cauchy–Schwarz, spectral theorem, proof habits |
| Tier 4 | MSc | Operator domains, subtle distinctions (Hermitian vs. self-adjoint), advanced applications |
| Tier 5 | PhD | Functional analysis, deficiency indices, unbounded operators, mathematical physics depth |

### Lecture-by-Lecture Syllabi

#### Module I.1 / L01: The Hilbert Space of Quantum States

**90-min pacing:** (0–20 min) Motivation — why ordinary Euclidean space is insufficient; infinite-dimensional state spaces and physical necessity; (20–45 min) Definition of a Hilbert space $\mathcal{H}$: inner product, norm, completeness (Cauchy sequences); (45–70 min) Bounded vs. unbounded operators; operator norm; adjoint $\hat{A}^\dagger$ defined via $\langle\phi|\hat{A}\psi\rangle = \langle\hat{A}^\dagger\phi|\psi\rangle$; (70–90 min) Anti-Hermitian operators; $[\hat{A}^\dagger]^\dagger = \hat{A}$; worked examples.

**Core learning outcomes:**

- State the definition of a Hilbert space and verify that $L^2(\mathbb{R})$ is one.
- Distinguish bounded from unbounded operators and give physical examples of each.
- Define the adjoint $\hat{A}^\dagger$ and compute it for $2 \times 2$ matrices.
- Classify Hermitian ($\hat{A} = \hat{A}^\dagger$) and anti-Hermitian ($\hat{A} = -\hat{A}^\dagger$) operators.

**Dual-track content:**

- *Geometric track:* Hilbert space as the "right" infinite-dimensional generalisation of $\mathbb{R}^n$; inner product $\langle\cdot|\cdot\rangle$ and angles; completeness as "no missing limits."
- *Algebraic/axiomatic track:* Formal axioms (conjugate symmetry, linearity, positive definiteness); operator theory; norm from inner product $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$.

**Tier differentiation:** HS: inner product as "overlap"; norm as "length"; $\mathbb{R}^2$ and $\mathbb{C}^2$ as tangible examples. Beg. UG: verify axioms in $\mathbb{C}^n$; compute $\hat{A}^\dagger$ for explicit matrices. Adv. UG: prove adjoint is unique; show $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$. MSc: $L^2$ space rigorously; operator norm; boundedness and continuity equivalence. PhD: spectrum of unbounded operators; domain issues; closed operators; Hellinger–Toeplitz theorem.

**Assessment:** Set A — verify Hilbert space axioms for $\mathbb{C}^n$; compute adjoints of given $3 \times 3$ matrices. Set B — prove $(\hat{A}^\dagger)^\dagger = \hat{A}$; show bounded $\Rightarrow$ continuous. Projects: (simple) norm visualiser for $L^2$ functions; (complex) counterexample: unbounded operator with dense domain.

#### Module I.1 / L02: Unitary Operators and Symmetry Transformations

**90-min pacing:** (0–15 min) Recap: inner products and adjoint; (15–40 min) Definition of unitary operator: $\hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \hat{1}$; preservation of inner products and norms $\langle\hat{U}\phi|\hat{U}\psi\rangle = \langle\phi|\psi\rangle$; (40–65 min) Physical incarnations: spatial rotations $\hat{U}(\theta)$, spatial translations $\hat{T}(a)$, time-evolution operator $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ (preview); (65–90 min) Unitary equivalence of operators; spectral invariance under unitary transformations.

**Core learning outcomes:**

- Verify unitarity $\hat{U}^\dagger \hat{U} = \hat{1}$ from definition and examples.
- Prove that unitary maps preserve norms and transition probabilities.
- Identify unitary operators arising from physical symmetries (rotation, translation, phase).
- Understand $\hat{U} = e^{i\hat{G}}$ for Hermitian generator $\hat{G}$ (Stone's theorem preview).

**Dual-track content:**

- *Physical track:* Symmetry $\to$ conservation law (Noether preview); rotations as $SU(2)$ action; phase freedom $e^{i\theta}\hat{1}$ as trivial unitary.
- *Mathematical track:* Unitary group $\mathcal{U}(\mathcal{H})$; isometric isomorphisms; one-parameter unitary groups $\hat{U}(t)$; Stone's theorem (statement).

**Tier differentiation:** HS: rotations as length-preserving maps; "rotating" a quantum state without losing information. Beg. UG: verify $\hat{U}^\dagger \hat{U} = \hat{1}$ for rotation matrices; compute $\hat{U}\,|\psi\rangle$. Adv. UG: show eigenvalues of unitary operators lie on the unit circle; prove spectral invariance. MSc: one-parameter groups; Stone's theorem; generator $\hat{G}$ and Lie algebra connection. PhD: strongly continuous unitary groups on Banach spaces; Hille–Yosida theorem; domain of generator.

**Assessment:** Set A — verify unitarity for $2 \times 2$ rotation and phase matrices; compute $\hat{U}\,|\psi\rangle$. Set B — prove unitary $\Rightarrow$ isometric; show spectrum of $\hat{U}$ lies on $|z| = 1$. Projects: (simple) animate unitary evolution of a two-level state; (complex) matrix exponential $e^{i\theta\hat{G}}$ for various generators.

## Module I.1 / L03: Observables, Hermitian Operators, and the Spectral Theorem

**90-min pacing:** (0–20 min) The measurement postulate: physical observables correspond to Hermitian operators; motivation from real-valued experimental outcomes; (20–45 min) Key properties of Hermitian operators: real eigenvalues, orthogonality of eigenvectors ($\langle a|a'\rangle = \delta_{aa'}$), completeness of eigenbasis; (45–70 min) The spectral theorem (discrete case): $\hat{A} = \sum_a a\,|a\rangle\langle a|$; functions of Hermitian operators $f(\hat{A})$; (70–90 min) Degenerate eigenvalues; continuous spectrum preview; physical examples ($\hat{x}$, $\hat{p}$, $\hat{H}$ for a particle in a box).

**Core learning outcomes:**

- State and apply the measurement postulate linking observables to Hermitian operators.
- Prove that Hermitian operators have real eigenvalues and orthogonal eigenvectors.
- Write the spectral decomposition $\hat{A} = \sum_a a\,|a\rangle\langle a|$ and apply it.
- Compute $f(\hat{A})$ from the spectral decomposition for simple functions.

**Dual-track content:**

- *Physical track (Sakurai):* Preparation and measurement; Stern–Gerlach as paradigm; eigenvalues as the only possible measurement outcomes.
- *Abstract track (Dirac):* Spectral theorem as the operational backbone; resolution of identity $\hat{1} = \sum_a |a\rangle\langle a|$; postulate as *definition* of observable.

**Tier differentiation:** HS: "measuring an observable always gives one of its eigenvalues" — Stern–Gerlach conceptual story. Beg. UG: find eigenvalues/vectors of $2 \times 2$ Hermitian matrices; write spectral form. Adv. UG: prove real eigenvalues and orthogonality; handle degenerate case with Gram–Schmidt. MSc: spectral theorem in full generality; functions of operators; self-adjoint vs. Hermitian distinction preview. PhD: spectral measure and

projection-valued measure (PVM); Borel functional calculus; continuous spectrum treatment.

**Assessment:** Set A — diagonalise Hermitian matrices; compute $e^{i\hat{A}}$ via spectral decomposition. Set B — prove spectral theorem for $n \times n$ Hermitian matrices; degenerate subspace orthogonalisation. Projects: (simple) eigenvalue/eigenvector visualiser for $2 \times 2$ observables; (complex) spectral resolution of the harmonic oscillator Hamiltonian.

## Module I.1 / L04: Operator Algebra and the Exponential Map

**90-min pacing:** (0–15 min) Operator arithmetic: sums $(\hat{A} + \hat{B})\,|\psi\rangle$, products $\hat{A}\hat{B}\,|\psi\rangle$, scalar multiples; non-commutativity; (15–40 min) Functions of operators via power series and spectral decomposition; the matrix exponential $e^{\hat{A}} = \sum_{n=0}^{\infty} \hat{A}^n/n!$ and its convergence; (40–65 min) The exponential map $e^{i\theta\hat{A}}$ for Hermitian $\hat{A}$: unitarity proof; one-parameter families; physical applications (phase, rotation, evolution operators); (65–90 min) Baker–Campbell–Hausdorff (BCH) lemma: $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]+\cdots}$; statement, significance, and low-order applications.

**Core learning outcomes:**

- Perform operator arithmetic and recognise non-commutativity.
- Compute matrix exponentials via diagonalisation and power series.
- Prove $\hat{U} = e^{i\theta\hat{A}}$ is unitary when $\hat{A}$ is Hermitian.
- State the BCH lemma and apply it to first order in the commutator.

**Dual-track content:**

- *Algebraic track:* Operator ring structure; Lie brackets and Lie algebras; BCH as the structure theorem linking Lie algebra to Lie group.
- *Computational track:* Step-by-step matrix exponential evaluation; $e^{i\theta\hat{\sigma}_z}$ and $e^{i\phi\hat{\sigma}_x}$ as worked examples; physical interpretation.

**Tier differentiation:** HS: "exponential of a matrix" — analogy with $e^x$ as infinite sum; rotation by small angles. Beg. UG: compute $e^{i\theta\hat{\sigma}_z}$; verify unitarity numerically. Adv. UG: prove unitarity of $e^{i\theta\hat{A}}$ rigorously; derive BCH to first order. MSc: full BCH series; Zassenhaus formula; group–algebra correspondence; Magnus expansion preview. PhD: convergence of BCH in Banach algebras; analytic extensions; applications in quantum control.

**Assessment:** Set A — compute $e^{i\theta\hat{\sigma}_z}$, $e^{i\phi\hat{\sigma}_y}$; verify unitarity; simple BCH applications. Set B — prove BCH to second order; derive Hadamard identity $e^{\hat{A}}\hat{B}e^{-\hat{A}}$. Projects: (simple) matrix exponential calculator with visualisation; (complex) first-order BCH vs. exact product comparison for non-commuting $2 \times 2$ Hamiltonians.

## Module I.1 / L05: Types of Symmetry Operators and Wigner's Theorem

**90-min pacing:** (0–20 min) Taxonomy of operators: linear vs. anti-linear; unitary vs. anti-unitary; physical motivation from symmetries that reverse arrow of time; (20–50 min) Wigner's theorem: every quantum symmetry is either a unitary or anti-unitary operator on $\mathcal{H}$; statement, intuition, and significance; (50–70 min) Projectors $\hat{P} = |\phi\rangle\langle\phi|$: idempotency $\hat{P}^2 = \hat{P}$, Hermiticity, eigenvalues $\{0, 1\}$; orthogonal projectors and decomposition of identity; (70–90 min) Time-reversal as paradigmatic anti-unitary symmetry; Kramers' theorem preview.

**Core learning outcomes:**

- Distinguish linear/unitary from anti-linear/anti-unitary operators with examples.
- State Wigner's theorem and explain its physical necessity.
- Construct rank-1 projectors $\hat{P} = |\phi\rangle\langle\phi|$ and verify $\hat{P}^2 = \hat{P}$.
- Explain how time-reversal is modelled by an anti-unitary operator.

**Dual-track content:**

- *Physical track:* Symmetry as a map on quantum states preserving transition probabilities; Wigner's theorem as the mathematical conclusion; time-reversal and Kramers degeneracy in spin systems.
- *Mathematical track:* Anti-linear maps $\mathcal{H} \to \mathcal{H}$; anti-unitary condition $\langle\Theta\phi|\Theta\psi\rangle = \langle\psi|\phi\rangle$; projector algebra and projection theorem.

**Tier differentiation:** HS: symmetry as "doing something and the physics looks the same"; time-reversal intuitively. Beg. UG: compute $\hat{P}^2$ for rank-1 projectors; verify $\hat{P}$ Hermitian. Adv. UG: prove eigenvalues of $\hat{P}$ are 0 and 1; orthogonal projector resolution of identity. MSc: Wigner's theorem proof sketch; complex conjugation as anti-unitary; Kramers' theorem statement. PhD: full proof of Wigner's theorem via Uhlhorn's theorem; anti-unitary representations; spin-statistics connection preview.

**Assessment:** Set A — verify $\hat{P}^2 = \hat{P}$ for explicit projectors; identify unitary vs. anti-unitary from action on basis. Set B — prove eigenvalues of a projector are $\{0, 1\}$; show time-reversal squaring gives $\pm\hat{1}$ for integer/half-integer spin. Projects: (simple) symmetry classification quiz for common quantum operators; (complex) Kramers degeneracy verification for spin-$\frac{1}{2}$ Hamiltonian under time-reversal.

## Module I.1 / L06: Bra-Ket Formalism (§2.1–2.2)

**90-min pacing:** (0–20 min) From column vectors to abstract kets: $|\psi\rangle \in \mathcal{H}$; superposition and linearity; (20–45 min) The dual space $\mathcal{H}^*$; bra $\langle\phi|$ as linear functional; inner product $\langle\phi|\psi\rangle$; conjugate symmetry; (45–70 min) Outer product $|\psi\rangle\langle\phi|$; matrix element $\langle\phi|\hat{A}|\psi\rangle$; expectation value $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$; variance $(\Delta A)^2$; (70–90 min) Notation dictionary: abstract ket $\leftrightarrow$ column vector $\leftrightarrow$ wavefunction; connection to previous lectures; bridge to Module I.2.

**Core learning outcomes:**

- Write kets and bras in Dirac notation and perform inner/outer products.
- Evaluate matrix elements $\langle\phi|\hat{A}|\psi\rangle$ for operators from Lectures L01–L05.
- Compute expectation values and variances of observables.
- Translate freely between abstract Dirac notation, column-vector representation, and wavefunction language.

**Dual-track content:**

- *Sakurai (operational) track:* Notation as a concise bookkeeping device; amplitude $\langle a|\psi\rangle$ as measurement probability amplitude; $|\langle a|\psi\rangle|^2$ as Born rule.
- *Dirac (abstract) track:* Bra as element of algebraic dual space $\mathcal{H}^*$; Riesz representation theorem identifies $\langle\psi|$ with $|\psi\rangle$ via inner product.

**Tier differentiation:** HS: "bra" and "ket" as "row" and "column" in a richer language; inner product as overlap. Beg. UG: compute $\langle\phi|\psi\rangle$, outer products, and $\langle\hat{A}\rangle$ for $2 \times 2$ examples. Adv. UG: prove $(|\psi\rangle\langle\phi|)^\dagger = |\phi\rangle\langle\psi|$; work with sums of projectors. MSc: Riesz representation theorem (statement and significance); dual space in infinite dimensions. PhD: bounded vs. unbounded functionals; Dirac notation in the context of rigged Hilbert spaces.

**Assessment:** Set A — inner products, outer products, expectation values; verify Hermiticity of $\hat{x}$ in position representation. Set B — prove variance formula $(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$; show $(\hat{A} \left|\psi\right\rangle \left\langle\phi\right|)^\dagger = \left|\phi\right\rangle \left\langle\psi\right| \hat{A}^\dagger$. Projects: (simple) Dirac notation calculator for spin-$\frac{1}{2}$; (complex) derive Born-rule consequences from inner-product axioms and completeness.

*Continued in Module I.3, Lecture L01: pure and mixed states, density matrix properties, and partial trace (§2.3–2.5).*

## 1.2 Module I.2 — Bra-Ket Notation and the Language of Quantum Mechanics

***Module designation:*** *Series I, Module 2 — develops the full bra-ket formalism in depth, bridging the algebraic foundations of Module I.1 with the dynamical and structural content of Module I.3.*

***Dual-track approach:*** *Dirac (abstract axiomatic) + Sakurai (operational/physical), with a* ***five-tier layered pedagogy*** *spanning HS through PhD.*

***Duration:*** *12 lectures × 90 minutes. Spiral curriculum: seven competency threads revisited progressively across lectures.*

***Assessment:*** *Per-lecture bundles (concept check, Problem Sets A & B, mini-projects, research questions) plus course-level grading. See Section 1.2.*

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | States as Rays & Ket Space | State space; superposition; Hilbert space |
| L02 | Bras, Dual Space & Inner Product | Dual vectors; conjugate linearity; Cauchy–Schwarz |
| L03 | Linear Operators: Adjoint & Hermitian | Adjoint; Hermitian; unitary; normal operators |
| L04 | Eigenkets, Projectors & Spectral Decomp. | Spectral theorem; projectors; completeness |
| L05 | Completeness & Change of Basis | Resolution of identity; unitary basis change |
| L06 | Continuous Bases: $\|x\rangle$, $\|p\rangle$, $\delta$ | Position/momentum eigenstates; delta function |
| L07 | Measurement Theory & Commutation | Postulates; uncertainty; CSCO |
| L08 | Time Evolution & Pictures | Schrödinger, Heisenberg, interaction |
| L09 | Tensor Products & Composite Systems | Entanglement; partial trace; Schmidt |
| L10 | Spin & Finite-Dimensional Spaces | Pauli matrices; SU(2); Bloch sphere |
| L11 | Density Operators & Mixed States | $\hat{\rho}$; purity; von Neumann entropy |
| L12 | Rigged Hilbert Spaces & Foundations | Gel'fand triple; distributions; mathematical rigour |

### Spiral Curriculum — Competency Threads

The module is built around seven interlocking threads, each revisited at increasing depth across the 12 lectures:

| Thread | Content arc |
|---|---|
| **States & Duality** | Kets/bras, inner products, amplitudes (L01–L02) |
| **Representation** | Basis expansion, discrete/continuous, Fourier (L03, L05–L06) |
| **Operators** | Adjoint, Hermitian/unitary, spectral theorem (L03–L04) |
| **Measurement** | Projectors, collapse, POVM preview (L04, L07) |
| **Dynamics** | Unitary evolution, Schrödinger/Heisenberg pictures (L08) |
| **Composition** | Tensor products, entanglement, partial trace (L09, L11) |
| **Foundations** | Distributions, operator domains, rigged Hilbert spaces (L06, L12) |

### Five-Tier Layered Pedagogy

Each lecture is explicitly stratified across five audience levels, enabling delivery to a mixed cohort:

| Tier | Cohort | Characteristic content |
|---|---|---|
| Tier 1 | HS | Complex numbers, amplitudes, vector addition analogy, intuitive motivation |
| Tier 2 | Beg. UG | Normalization, orthogonality, components $c_n = \langle n\|\psi\rangle$, computational facility |
| Tier 3 | Adv. UG | Cauchy–Schwarz, spectral theorem, proof habits, rigorous derivations |
| Tier 4 | MSc | Projective Hilbert space, density operators, measurement subtleties, representation theory |
| Tier 5 | PhD | Rigged Hilbert space, unbounded operators, functional analysis, deficiency indices |

### Lecture-by-Lecture Syllabi

**Module I.2 / L01: Why Dirac Notation? States as Rays & Superposition**

**90-min pacing:** (0–15 min) Motivation — representations vs. abstract states; (15–45 min) Kets as vectors, rays, normalisation, global phase; (45–70 min) Bras (informal), amplitudes, Born rule preview; (70–90 min) Notation dictionary: ket $\leftrightarrow$ column vector $\leftrightarrow$ wavefunction.

**Core learning outcomes:**

- Understand state as equivalence class $|\psi\rangle \sim e^{i\theta}|\psi\rangle$; global phase unobservable.
- Interpret amplitudes and Born-rule probabilities $P(a) = |\langle a|\psi\rangle|^2$.
- Build the notation dictionary in a discrete basis.
- Connect Sakurai (operational) and Dirac (abstract) perspectives.

**Dual-track content:**

- *Sakurai track:* Preparation procedures $\to$ stable frequencies; interference motivates complex amplitudes; $P = |A_1 + A_2|^2$.
- *Dirac track:* $|\psi\rangle$ as ray in $\mathcal{H}$; completeness $\sum_n |n\rangle\langle n| = \hat{1}$; $P(a_i) = |\langle a_i|\psi\rangle|^2$.

**Tier differentiation:** HS: $|z|^2$ as probability, interference analogy. Beg. UG: normalisation $\langle\psi|\psi\rangle = 1$, components. Adv. UG: Cauchy–Schwarz, triangle inequality. MSc: rays and projective Hilbert space $\mathbb{P}(\mathcal{H})$, Wigner preview. PhD: separability, precise Hilbert space statements.

**Assessment:** Set A — normalisation + amplitude problems. Set B — prove Cauchy–Schwarz; phase-invariance exercises. Projects: (simple) basis-expansion calculator; (complex) compare physical meaning across representations.

**Module I.2 / L02: Bras, Dual Space & Inner Product Structure**

**90-min pacing:** (0–20 min) Define bras as linear functionals on $\mathcal{H}$; (20–50 min) Conjugate linearity; inner product axioms; (50–70 min) Probability amplitude $\langle\phi|\psi\rangle$ and Born rule; (70–90 min) Orthonormal basis; Gram–Schmidt sketch.

**Core learning outcomes:**

- Define bra $\langle\psi|$ as element of dual space $\mathcal{H}^*$.

- Master conjugate linearity and inner product rules.
- Derive Born-rule probability from inner product structure.
- Riesz representation theorem (intuition).

**Tier differentiation:** HS: conjugation rules, dot-product analogy. Beg. UG: orthonormal bases, Gram–Schmidt. Adv. UG: Riesz representation theorem intuition, proofs. MSc: anti-linear maps, adjoint rigorously, continuity. PhD: bounded vs. unbounded functionals, Banach space context.

**Assessment:** Set A — compute overlaps, orthogonality checks. Set B — inequalities, norm properties, inner-product geometry. Projects: (simple) visualise state geometry; (complex) derive Born-rule consequences from axioms.

### Module I.2 / L03: Linear Operators: Adjoint, Hermitian & Unitary

**90-min pacing:** (0–20 min) Operators as maps $\mathcal{H} \to \mathcal{H}$; matrix elements; (20–50 min) Adjoint $\hat{A}^\dagger$; Hermitian observables; unitary symmetries; (50–70 min) Expectation values; operator algebra; (70–90 min) Commutators preview; spectral theorem motivation.

**Core learning outcomes:**

- Compute matrix elements $\langle\phi|\,\hat{A}\,|\psi\rangle$.
- Define adjoint: $\langle\phi|\,\hat{A}\,|\psi\rangle = \langle\hat{A}^\dagger\phi|\,|\psi\rangle$.
- Characterise Hermitian ($\hat{A} = \hat{A}^\dagger$) and unitary ($\hat{U}^\dagger\hat{U} = \hat{1}$) operators.
- Prove expectation value of Hermitian operator is real.

**Tier differentiation:** HS: matrices as machines on vectors, $2 \times 2$ examples. Beg. UG: eigenvalues, $\langle\hat{A}\rangle = \langle\psi|\,\hat{A}\,|\psi\rangle$. Adv. UG: proof Hermitian $\Rightarrow$ real expectation; unitary preserves inner product. MSc: spectral theorem preview, commutators, normal operators. PhD: domains of unbounded operators, self-adjoint vs. merely Hermitian.

**Assessment:** Set A — compute adjoint, check Hermiticity, expectation values. Set B — prove spectral properties; operator algebra. Projects: (simple) classify $2 \times 2$ matrices; (complex) commutator algebra and uncertainty preview.

### Module I.2 / L04: Eigenkets, Projectors & Spectral Decomposition

**90-min pacing:** (0–20 min) Eigenvalue equation $\hat{A}\,|a\rangle = a\,|a\rangle$; (20–50 min) Projectors $\hat{P}_a = |a\rangle\langle a|$; properties; (50–75 min) Spectral decomposition $\hat{A} = \sum_a a\,|a\rangle\langle a|$; (75–90 min) Measurement postulate; post-measurement state.

**Core learning outcomes:**

- Use eigenkets for measurement probabilities.
- Construct projectors $\hat{P}_a$ and verify $\hat{P}_a^2 = \hat{P}_a$.
- Write spectral decomposition for discrete observables.
- State measurement postulate (Sakurai formulation).

**Tier differentiation:** HS: "filters" and outcomes intuition — Stern–Gerlach analogy. Beg. UG: compute $P(a) = |\langle a|\psi\rangle|^2$ via projectors. Adv. UG: degeneracy; resolution of identity in eigenbasis. MSc: projection postulate subtleties; Lüders rule. PhD: spectral measures; PVMs and POVMs preview; measurement models.

**Assessment:** Set A — compute probabilities and post-measurement states. Set B — degeneracy, continuous spectrum cases. Projects: (simple) spin-$\frac{1}{2}$ measurement simulation;

(complex) Lüders vs. von Neumann update comparison.

### Module I.2 / L05: Completeness, Change of Basis & Matrix Mechanics

**90-min pacing:** (0–20 min) Resolution of identity; inserting $\hat{1}$; (20–50 min) Basis expansion and change of representation; (50–75 min) Matrix multiplication as inserted identities; (75–90 min) Coordinate transforms; unitary basis changes.

**Core learning outcomes:**

- Use completeness $\sum_n |n\rangle\langle n| = \hat{1}$ to evaluate any bracket.
- Transform between bases using unitary matrices.
- Derive matrix multiplication from operator composition.
- Express operator matrix elements $A_{mn} = \langle m|\hat{A}|n\rangle$.

**Tier differentiation:** HS: coordinate systems analogy, 2D rotation. Beg. UG: compute $A_{mn}$ and transform vectors between bases. Adv. UG: show equivalence of abstract operator and matrix representation. MSc: similarity vs. unitary equivalence; trace and spectrum invariants. PhD: representation theory framing; direct integrals preview.

**Assessment:** Set A — basis expansions, matrix elements. Set B — change-of-basis proofs, unitary equivalence. Projects: (simple) matrix mechanics calculator; (complex) equivalence of Schrödinger and Heisenberg representations.

### Module I.2 / L06: Continuous Bases: $|x\rangle$, $|p\rangle$ & Delta Functions

**90-min pacing:** (0–20 min) Continuous completeness $\int |x\rangle\langle x|\,dx = \hat{1}$; (20–50 min) Wavefunction as $\psi(x) = \langle x|\psi\rangle$; (50–75 min) Momentum basis; $\langle x|p\rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$; (75–90 min) Fourier connection; delta function identities.

**Core learning outcomes:**

- Understand continuous-spectrum orthonormality $\langle x|x'\rangle = \delta(x - x')$.
- Derive wavefunction as projection onto position basis.
- Connect Fourier transform to $\langle x|p\rangle$ inner product.
- Handle delta function distributions carefully.

**Tier differentiation:** HS: what "continuum of outcomes" means physically. Beg. UG: basic integrals; qualitative delta function properties. Adv. UG: careful delta manipulations; normalisation conventions. MSc: distribution theory viewpoint; generalised eigenvectors. PhD: rigged Hilbert spaces motivation; Gel'fand triple preview.

**Assessment:** Set A — wavefunction integrals, Fourier transforms. Set B — distribution theory; momentum-space wavefunctions. Projects: (simple) Gaussian wavepacket in $x/p$ bases; (complex) prove Parseval–Plancherel from completeness.

### Module I.2 / L07: Measurement Theory: Postulates, Compatibility & Uncertainty

**90-min pacing:** (0–20 min) Compatible observables and commutators $[\hat{A}, \hat{B}]$; (20–50 min) Uncertainty relation derivation (Robertson); (50–75 min) Measurement update; expectation values; (75–90 min) POVM preview; state tomography mention.

**Core learning outcomes:**

- Define compatible observables: $[\hat{A}, \hat{B}] = 0 \Leftrightarrow$ common eigenbasis.

- Derive Robertson–Schrödinger uncertainty inequality.
- Compute variances and uncertainty products.
- Distinguish uncertainty from measurement imprecision.

**Tier differentiation:** HS: uncertainty as "spread" not instrument error. Beg. UG: compute variances; simple commutators $[\hat{x}, \hat{p}] = i\hbar$. Adv. UG: derive Robertson–Schrödinger inequality from Cauchy–Schwarz. MSc: POVMs, operational view; state tomography preview. PhD: measurement models (von Neumann coupling), disturbance, contextuality.

**Assessment:** Set A — commutators, variances, uncertainty products. Set B — prove Robertson inequality; compatible observable theorems. Projects: (simple) uncertainty product plotter; (complex) POVM vs. projective measurement comparison.

## Module I.2 / L08: Time Evolution & Pictures of Quantum Mechanics

**90-min pacing:** (0–20 min) Unitary evolution $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$; (20–45 min) Schrödinger picture: states evolve; (45–70 min) Heisenberg picture: operators evolve; (70–90 min) Constants of motion; $[\hat{A}, \hat{H}] = 0 \Leftrightarrow$ conserved.

**Core learning outcomes:**

- Derive $\hat{U}(t)$ from unitarity and time-translation invariance.
- Work fluently in both Schrödinger and Heisenberg pictures.
- Derive Heisenberg equation of motion $\dot{\hat{A}}_H = \frac{i}{\hbar}[\hat{H}, \hat{A}_H]$.
- Identify constants of motion via commutator with $\hat{H}$.

**Tier differentiation:** HS: exponential as "repeated tiny evolution steps." Beg. UG: differentiate $\langle A \rangle$ with time; simple Hamiltonians. Adv. UG: Heisenberg equation of motion; Ehrenfest's theorem. MSc: time-ordering; interaction picture preview; formal solution. PhD: Stone's theorem; one-parameter unitary groups; domain issues.

**Assessment:** Set A — evolve states; compute $\langle A \rangle(t)$. Set B — prove Heisenberg EOM; constants of motion theorems. Projects: (simple) two-level Rabi oscillation; (complex) interaction picture and perturbation theory preview.

## Module I.2 / L09: Tensor Products: Composite Systems & Entanglement

**90-min pacing:** (0–20 min) Tensor product space $\mathcal{H}_A \otimes \mathcal{H}_B$; (20–50 min) Product vs. entangled states; operator action; (50–75 min) Entanglement criterion; Schmidt decomposition; (75–90 min) Partial measurement and reduced descriptions.

**Core learning outcomes:**

- Construct product states $|i\rangle_A \otimes |j\rangle_B$ and general bipartite states.
- Identify entanglement by failure of coefficient factorisation.
- Apply Schmidt decomposition to find Schmidt rank.
- Preview partial trace as reduced description of subsystem.

**Tier differentiation:** HS: "two systems = bigger space" — classical analogy and beyond. Beg. UG: compute product basis components; tensor product of matrices. Adv. UG: Schmidt decomposition; maximally entangled states. MSc: partial trace introduced lightly; entanglement entropy preview. PhD: entanglement measures, operational tasks, reading-based prompts.

**Assessment:** Set A — product basis, operator action on $\mathcal{H}_A \otimes \mathcal{H}_B$. Set B — Schmidt

decomposition; Bell states analysis. Projects: (simple) entanglement checker for 2-qubit states; (complex) Schmidt rank and entanglement entropy.

### Module I.2 / L10: Spin & Finite-Dimensional Hilbert Spaces ($SU(2)$)

**90-min pacing:** (0–20 min) Spin-$\frac{1}{2}$ as 2D Hilbert space; $|+\rangle, |-\rangle$ basis; (20–50 min) Pauli matrices; spin operators $\sigma_x, \sigma_y, \sigma_z$; (50–75 min) Rotation operators $e^{-i\theta\boldsymbol{\sigma}\cdot\hat{n}/2}$; Bloch sphere; (75–90 min) Angular momentum commutation relations preview.

**Core learning outcomes:**

- Work with spin-$\frac{1}{2}$ as prototype finite-dimensional QM.
- Compute eigenkets and probabilities for all spin components.
- Apply rotation operators $U(\theta, \hat{n}) = e^{-i\theta\boldsymbol{\sigma}\cdot\hat{n}/2}$.
- Connect $SU(2)$ group to physical rotations $SO(3)$.

**Tier differentiation:** HS: two-state systems as "qubits" — conceptual motivation. Beg. UG: compute spin measurement probabilities; Stern–Gerlach results. Adv. UG: rotation operators; verify $[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$. MSc: representation theory touch; angular momentum algebra $\hat{J}_\pm$. PhD: $SU(2)$ vs. $SO(3)$, projective representations, Wigner theorem link.

**Assessment:** Set A — spin probabilities, expectation values. Set B — rotation operators; angular momentum algebra. Projects: (simple) Bloch sphere visualiser; (complex) $SU(2) \leftrightarrow SO(3)$ double-cover demonstration.

### Module I.2 / L11: Density Operators, Mixed States & Partial Trace

**90-min pacing:** (0–20 min) Pure vs. mixed states; $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; (20–50 min) $\langle\hat{A}\rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$; properties of $\hat{\rho}$; (50–75 min) Partial trace; reduced density matrices; (75–90 min) Bloch sphere for spin-$\frac{1}{2}$; purity $\mathrm{Tr}(\hat{\rho}^2)$.

**Core learning outcomes:**

- Construct density operator for ensembles and reduced states.
- Compute expectation values via $\mathrm{Tr}(\hat{\rho}\hat{A})$.
- Apply partial trace to obtain subsystem state.
- Distinguish pure ($\mathrm{Tr}(\hat{\rho}^2) = 1$) from mixed ($\mathrm{Tr}(\hat{\rho}^2) < 1$) states.

**Tier differentiation:** HS: "uncertainty because we don't know" vs. genuine quantum spread. Beg. UG: compute trace in matrix form; verify $\hat{\rho}$ properties. Adv. UG: pure vs. mixed; Bloch vector parameterisation of $2 \times 2$ density matrices. MSc: Kraus operators; CPTP maps preview; quantum channels. PhD: open systems, master equations, quantum error correction preview.

**Assessment:** Set A — density matrix construction, $\mathrm{Tr}(\hat{\rho}\hat{A})$. Set B — partial trace, purity, Bloch representation. Projects: (simple) density matrix visualiser for 2-level system; (complex) decoherence model via partial trace.

### Module I.2 / L12: Rigged Hilbert Spaces & Mathematical Foundations

**90-min pacing:** (0–20 min) Why $|x\rangle$ is *not* in the Hilbert space; (20–50 min) Distributions and test functions; Schwartz space; (50–75 min) Self-adjointness vs. Hermiticity revisited; (75–90 min) What Dirac notation hides — safe usage rules.

**Core learning outcomes:**

- Explain why $\langle x|x'\rangle = \delta(x - x')$ requires distribution theory.
- State the Gel'fand triple $\Phi \subset \mathcal{H} \subset \Phi'$.
- Distinguish self-adjoint from Hermitian (deficiency indices).
- Apply Dirac notation safely with awareness of its limits.

**Tier differentiation:** HS: conceptual "idealised states" — measuring infinitely precise position. Beg. UG: cautionary rules for delta functions in calculations. Adv. UG: examples of unbounded operators $(\hat{x}, \hat{p})$; domain issues. MSc: rigged Hilbert space triple $\Phi \subset \mathcal{H} \subset \Phi'$; nuclear spectral theorem. PhD: spectral theorem in full measure-theoretic form; deficiency indices; self-adjoint extensions.

**Assessment:** Set A — delta function identities; distribution manipulations. Set B — self-adjoint extension problems; domain analysis. Projects: (simple) "regularisation" demonstration; (complex) momentum operator on half-line — self-adjoint extensions.

### Assessment Strategy

***Per-Lecture Assessment Bundle*** *(every lecture): concept check (10–15 min, 5–8 short conceptual questions); Problem Set A (simple, 6–10 computational exercises); Problem Set B (complex, 3–6 multi-step derivations and proofs); mini-project simple (1–3 hours); mini-project complex (5–10 hours, open-ended); research questions well-defined (2–4 scoped prompts); research questions fully open (1–2 frontier-style prompts).*

**Course-level grading template:**

- Weekly homework (best 10/12): 40%
- Two mid-course syntheses (after L06 & L10): 20%
- Project (simple or complex track): 25%
- Final oral/whiteboard exam or take-home: 15%

Mixed cohorts: HS/UG → simple track; MSc/PhD → complex track.

### Reading Strategy

**Track A — Core textbooks:** Dirac, *The Principles of Quantum Mechanics* (formalism, notation ethos); Sakurai & Napolitano, *Modern Quantum Mechanics* (measurement, symmetry framing).

**Track B — Support:** Ballentine (postulates/measurement); von Neumann (mathematical structure); functional analysis references (PhD tier).

**Track C — Historical sources:** Born (1926) probability interpretation; Dirac (1927) amplitude formalism; von Neumann (1932) foundations; Stone (1932) unitary groups; Wigner, symmetry principles (MSc/PhD).

## Expansion Plan — Natural Extensions Beyond 12 Lectures

| **Topic** | **Duration** | **Content** |
|---|---|---|
| Angular Momentum & Addition | 2–4 lectures | $\hat{J}_\pm$; Clebsch–Gordan; Wigner–Eckart theorem |
| Harmonic Oscillator & Ladder Operators | 2–3 lectures | $\hat{a}, \hat{a}^\dagger$; coherent states; squeezed states |
| Scattering Theory & $S$-Matrix | 2–4 lectures | Lippmann–Schwinger; T-matrix; partial waves |
| Path Integrals | 2 lectures | Feynman formulation; propagator; connection to Series II Module II.2 |
| Quantum Information Formalism | 2–4 lectures | Quantum channels; CPTP; quantum error correction |
| Advanced Measurement & POVMs | 2 lectures | POVM elements; Naimark's theorem; informationally complete |

## 1.3 Module I.3 — Quantum Dynamics, Perturbation Theory, and Physical Systems

***Scope:*** *Continues directly from Module I.1 (§2.4 onward). Covers the density matrix in full, bases, commutation relations, symmetries and Noether's theorem, the complete theory of time evolution (time-independent, time-dependent, and non-commuting), all three pictures of quantum mechanics, perturbation theory, open systems, and a survey of canonical example systems and physical effects.*

***Duration:*** *10 lectures × 90 minutes (Lecture L01 continued + Lectures L02–L10 of the original Series I programme).*

***Prerequisite:*** *Module I.1; Module I.2 recommended as concurrent or prior study.*

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Pure/Mixed States & Density Matrix | $\hat{\rho}$; purity; partial trace; entanglement entropy |
| L02 | Bases: Discrete and Continuous | Resolution of identity; wavefunctions; Fourier transform |
| L03 | Uncertainty & Commutation Relations | Robertson inequality; CSCO; Zeno effect |
| L04 | Generators, Symmetries & Noether | Stone's theorem; Noether; discrete symmetries |
| L05 | Time Evolution I | $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$; stationary states; probability current |
| L06 | Time Evolution II | Dyson series; Magnus expansion; Floquet preview |
| L07 | Heisenberg Picture | EOM; Ehrenfest; virial theorem; canonical quantisation |
| L08 | Interaction (Dirac) Picture | $S$-matrix; Dyson series; three-picture comparison |
| L09 | Perturbation Theory | RSPT; degenerate PT; variational; Feynman–Hellmann |
| L10 | TDPT, Open Systems & Many-Body | Fermi's golden rule; Lindblad; decoherence |

### Lecture-by-Lecture Syllabi

**Module I.3 — Lecture L01 (cont.): Pure/Mixed States and the Density Matrix (§2.3–2.5)**

**90-min pacing:** (0–20 min) Motivation: why a single ket is insufficient to describe subsystems or statistical ensembles; pure state $\hat{\rho} = |\psi\rangle\langle\psi|$ as a special case; (20–45 min) Mixed states: statistical mixture $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; physical distinction between classical ignorance and genuine quantum superposition; (45–70 min) Properties of the density operator: $\hat{\rho}^\dagger = \hat{\rho}$, $\mathrm{Tr}\,\hat{\rho} = 1$, $\hat{\rho} \geq 0$; purity $\gamma = \mathrm{Tr}\,\hat{\rho}^2 \leq 1$ (equality iff pure); expectation values $\langle\hat{A}\rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$; (70–90 min) Bipartite systems $\mathcal{H}_A \otimes \mathcal{H}_B$; partial trace $\hat{\rho}_A = \mathrm{Tr}_B\,\hat{\rho}_{AB}$; entanglement entropy $S = -\mathrm{Tr}(\hat{\rho}_A \ln \hat{\rho}_A)$ (preview).

**Core learning outcomes:**

- Construct $\hat{\rho}$ for pure states, mixtures, and subsystems.
- Verify the three defining properties ($\hat{\rho}^\dagger = \hat{\rho}$, $\mathrm{Tr}\,\hat{\rho} = 1$, $\hat{\rho} \geq 0$) for given density matrices.
- Compute expectation values and purity; distinguish pure from mixed states.
- Apply partial trace to obtain reduced density matrices and compute entanglement entropy.

**Dual-track content:**

- *Statistical/physical track:* Ensemble interpretation of $\hat{\rho}$; classical vs. quantum uncertainty; Bloch sphere picture for spin-$\frac{1}{2}$; why a mixed state cannot be "purified" within the same Hilbert space.
- *Mathematical/algebraic track:* $\hat{\rho}$ as a positive semi-definite trace-class operator; spectral decomposition; partial trace as unique linear map satisfying $\mathrm{Tr}_B(\hat{\rho}_{AB}(\hat{1} \otimes \hat{O}_B)) = \mathrm{Tr}(\hat{\rho}_B \hat{O}_B)$.

**Tier differentiation:** HS: "classical probability of being in a state" vs. genuine quantum superposition — the key conceptual divide; coin-flip analogy and its limits. Beg. UG: compute $\hat{\rho}$, $\mathrm{Tr}\,\hat{\rho}^2$, and $\langle \hat{A} \rangle$ for $2 \times 2$ examples; verify properties numerically. Adv. UG: prove $\mathrm{Tr}\,\hat{\rho}^2 \leq 1$ from spectral decomposition; Bloch vector parameterisation for spin-$\frac{1}{2}$; partial trace calculation. MSc: Purification of mixed states; Schmidt decomposition connection; entanglement entropy and its properties (sub-additivity preview). PhD: Trace-class operators in infinite dimensions; von Neumann entropy; quantum relative entropy; complete positivity preview.

**Assessment:** Set A — construct $\hat{\rho}$ for explicit mixtures; compute purity and $\langle \hat{\sigma}_z \rangle$; partial trace for 2-qubit Bell states. Set B — prove $\mathrm{Tr}\,\hat{\rho}^2 \leq (\mathrm{Tr}\,\hat{\rho})^2$ with equality iff rank-1; show entanglement entropy vanishes for product states. Projects: (simple) Bloch sphere trajectory under dephasing noise; (complex) purification of a given mixed state and verification of entanglement entropy.

## Module I.3 / L02: Bases: Discrete and Continuous

**90-min pacing:** (0–20 min) Discrete orthonormal bases: $\langle e_i | e_j \rangle = \delta_{ij}$; resolution of identity $\hat{1} = \sum_i |e_i\rangle \langle e_i|$; change of basis via unitary matrices; matrix representation in a given basis; (20–45 min) Continuous bases: position eigenstates $\hat{x}\,|x\rangle = x\,|x\rangle$; continuous orthonormality $\langle x|x'\rangle = \delta(x - x')$; completeness $\hat{1} = \int dx\, |x\rangle \langle x|$; wavefunction as $\psi(x) = \langle x|\psi\rangle$; (45–70 min) Momentum representation: $\tilde{\psi}(p) = \langle p|\psi\rangle$; $\langle x|p\rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$; Fourier transform as change of basis; (70–90 min) The Dirac delta distribution: properties, Fourier representation, regularisation; over-completeness and coherent states (preview).

**Core learning outcomes:**

- Use resolution of identity in discrete and continuous forms to evaluate any bracket.
- Derive the wavefunction $\psi(x) = \langle x|\psi\rangle$ as a projection.
- Perform basis changes between position and momentum representations via Fourier transforms.
- Manipulate delta functions and understand their distributional nature.

**Dual-track content:**

- *Computational track:* Concrete calculation of Fourier transforms; Gaussian wavepackets in $x$ and $p$; Parseval–Plancherel theorem; practical use of $\hat{1}$ insertions to convert between representations.
- *Mathematical track:* Generalised eigenvectors; distribution theory and Schwartz space; over-complete frames and the resolution of identity for coherent states; Gel'fand triple motivation.

**Tier differentiation:** HS: "snapshots" of a state in different languages (position vs. momentum); why the wavefunction $\psi(x)$ is just one way to represent the same ket. Beg. UG: compute $\psi(x)$ from a given ket in a discrete basis; evaluate simple Fourier transforms; use delta function properties. Adv. UG: prove Parseval–Plancherel from completeness; handle delta function carefully in operator matrix elements. MSc: distribution theory viewpoint;

generalised eigenvectors and their domain; Fourier transform as unitary map on $L^2(\mathbb{R})$. PhD: Gel'fand triple $\Phi \subset L^2 \subset \Phi'$; nuclear spectral theorem; coherent states and frame theory; wavelet transforms.

**Assessment:** Set A — compute $\psi(x)$ and $\tilde{\psi}(p)$ for Gaussian and square states; evaluate $\langle x|\hat{p}|\psi\rangle$ via completeness insertion. Set B — prove Fourier transform is unitary on $L^2$; delta function identities and distributional derivatives. Projects: (simple) Gaussian wavepacket visualiser in $x$ and $p$; (complex) uncertainty product $\Delta x\,\Delta p$ for a family of wavepackets and saturation condition.

## Module I.3 / L03: Uncertainty and Commutation Relations

**90-min pacing:** (0–20 min) Commutators as the algebraic fingerprint of incompatibility; $[\hat{x}, \hat{p}] = i\hbar$; general commutator identities and Jacobi identity; (20–45 min) The generalised uncertainty principle: variance definition $(\Delta A)^2 = \langle \hat{A}^2\rangle - \langle \hat{A}\rangle^2$; Robertson inequality $\Delta A\,\Delta B \geq \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle|$ derived via Cauchy–Schwarz; Schrödinger's improvement with anti-commutator term; (45–65 min) Canonical commutation relations (CCRs): $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$; Stone–von Neumann theorem (statement); complete set of commuting observables (CSCO) and quantum numbers; (65–90 min) Compatible and incompatible measurements; projection postulate; sequential measurements; quantum Zeno effect; energy–time uncertainty (careful treatment of $\Delta E\,\Delta t$).

**Core learning outcomes:**

- Derive Robertson's uncertainty inequality from the Cauchy–Schwarz inequality.
- Evaluate commutators for polynomial functions of $\hat{x}$ and $\hat{p}$.
- Define a CSCO and explain how it labels quantum states uniquely.
- Distinguish compatible from incompatible observables and their measurement implications.

**Dual-track content:**

- *Physical track (Sakurai):* Uncertainty as an intrinsic property of quantum states, not of the measuring apparatus; Stern–Gerlach sequential measurement as the paradigm for incompatibility; energy–time uncertainty as a statement about state lifetime.
- *Mathematical track:* Derivation of Robertson–Schrödinger inequality; Cauchy–Schwarz in Hilbert space; conditions for saturation (Gaussian states); Stone–von Neumann uniqueness of CCR representation.

**Tier differentiation:** HS: "you cannot know both position and momentum exactly" — why it is not about clumsy measurements; wave analogy with bandwidth. Beg. UG: compute $[\hat{x}, \hat{p}^2]$, $[\hat{x}^2, \hat{p}]$ from CCR; evaluate $\Delta x\,\Delta p$ for Gaussian. Adv. UG: full derivation of Robertson inequality; conditions for saturation; prove that $[\hat{A}, \hat{B}] = 0 \Leftrightarrow$ common eigenbasis (non-degenerate case). MSc: Schrödinger's stronger uncertainty relation; CSCO for hydrogen; Stone–von Neumann theorem and its physical content. PhD: Weyl form of CCR; non-equivalent representations; quantum Zeno effect rigorously; information-theoretic uncertainty relations (entropic).

**Assessment:** Set A — evaluate commutators; compute variances for harmonic oscillator eigenstates; verify Robertson inequality numerically. Set B — derive Robertson inequality from scratch; prove saturation condition; CSCO construction for a particle in 3D central potential. Projects: (simple) uncertainty product plotter for parameterised family of states; (complex) entropic uncertainty relation vs. Robertson for conjugate bases.

## Module I.3 / L04: Infinitesimal Generators, Symmetries, and Noether's Theorem

**90-min pacing:** (0–20 min) One-parameter unitary groups: $\hat{U}(\theta) = e^{i\theta\hat{G}}$; Stone's theorem identifying the generator $\hat{G}$; momentum as generator of spatial translations, angular momentum of rotations, Hamiltonian of time translations; (20–45 min) Symmetry of the Hamiltonian: $[\hat{H}, \hat{U}] = 0 \Leftrightarrow [\hat{H}, \hat{G}] = 0$; conservation of $\hat{G}$ in the Heisenberg picture as Noether's theorem; examples: translation invariance $\to$ momentum conservation, rotational invariance $\to$ angular momentum conservation; (45–65 min) Discrete symmetries: parity $\hat{\Pi}$, charge conjugation $\hat{C}$, time reversal $\hat{T}$ (Wigner anti-unitary); $CPT$ theorem (statement and significance); (65–90 min) Symmetry breaking: explicit vs. spontaneous; degenerate ground states; Goldstone modes (qualitative).

**Core learning outcomes:**

- Identify the generator of a given one-parameter unitary group.
- Prove Noether's theorem in quantum mechanics via the Heisenberg equation.
- Classify discrete symmetries and their representations (unitary vs. anti-unitary).
- Distinguish explicit from spontaneous symmetry breaking and describe physical consequences.

**Dual-track content:**

- *Physical track:* Conservation laws as a direct consequence of symmetry; physical examples (translation $\to$ momentum, rotation $\to$ angular momentum, time $\to$ energy); spontaneous symmetry breaking in ferromagnets and superconductors (qualitative).
- *Mathematical track:* Lie group – Lie algebra correspondence via exponential map; Stone's theorem; adjoint action and structure constants; Wigner's theorem applied to discrete symmetries.

**Tier differentiation:** HS: "if physics looks the same after a transformation, something is conserved" — Noether's theorem as the deepest connection between symmetry and conservation. Beg. UG: verify $[\hat{H}, \hat{p}] = 0$ for translationally invariant $\hat{H}$; check parity of simple wavefunctions. Adv. UG: prove Noether's theorem from Heisenberg equation; derive parity selection rules; show $\hat{\Pi}\hat{x}\hat{\Pi}^{-1} = -\hat{x}$. MSc: Lie algebra generators; structure constants; $\mathfrak{su}(2)$ algebra; full Wigner classification of discrete symmetries. PhD: Projective representations; central extensions; spontaneous symmetry breaking and Goldstone theorem; Wigner–Eckart theorem preview.

**Assessment:** Set A — identify generators for rotations, translations, phase shifts; verify conservation laws for given Hamiltonians; parity of wavefunctions. Set B — prove Noether's theorem in the Heisenberg picture; classify symmetries of the hydrogen Hamiltonian; analyse $CPT$ invariance. Projects: (simple) symmetry classification tool for toy Hamiltonians; (complex) spontaneous symmetry breaking in double-well potential — ground state degeneracy and perturbative lifting.

## Module I.3 / L05: Time Evolution I — Time-Independent Hamiltonians

**90-min pacing:** (0–15 min) The time-evolution postulate: $i\hbar\partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$; linearity and unitarity as fundamental requirements; (15–40 min) The time evolution operator $\hat{U}(t, t_0)$: definition, group property $\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1)\hat{U}(t_1, t_0)$, unitarity $\hat{U}^\dagger\hat{U} = \hat{1}$; time-independent case $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ via spectral decomposition; (40–65 min) Stationary states: $\hat{H} |E_n\rangle = E_n |E_n\rangle$; general solution as superposition with phase factors $e^{-iE_nt/\hbar}$; expectation values of time-independent operators in energy eigenstates; (65–90 min) Time-reversal symmetry: anti-unitary $\hat{T}$; Kramers degeneracy for half-integer spin; $T$-invariant Hamil-

tonians; Schrödinger equation in position representation: $\hat{p} = -i\hbar\nabla$; probability current $\mathbf{j} = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*)$; continuity equation $\partial_t|\psi|^2 + \nabla\cdot\mathbf{j} = 0$.

**Core learning outcomes:**

- Derive $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ from the time-evolution postulate.
- Evolve an arbitrary initial state by expanding in the energy eigenbasis.
- Identify and exploit stationary state structure for computing expectation values.
- Derive the probability current and continuity equation in position representation.

**Dual-track content:**

- *Wave mechanics track:* Schrödinger equation as a PDE; boundary conditions; probability current; stationary states as separable solutions $\psi(x,t) = \phi(x)e^{-iEt/\hbar}$.
- *Abstract operator track:* $\hat{U}$ from Stone's theorem; spectral decomposition of $e^{-i\hat{H}t/\hbar}$; time translation group; anti-unitary time reversal and Kramers theorem.

**Tier differentiation:** HS: "the Hamiltonian is the quantum engine of time evolution" — energy eigenstates oscillate simply; superpositions interfere as time goes on. Beg. UG: evolve a two-level system; compute $\langle\hat{\sigma}_z\rangle(t)$; derive Rabi oscillation frequency. Adv. UG: full derivation of $\hat{U}$; probability current derivation; prove unitarity of $e^{-i\hat{H}t/\hbar}$ for Hermitian $\hat{H}$. MSc: time-reversal symmetry; Kramers degeneracy proof for half-integer spin; compare time-independent and time-dependent treatments. PhD: Stone's theorem; generator domain; self-adjoint extensions and their role in determining the evolution; Zeno's paradox and its resolution.

**Assessment:** Set A — evolve superposition states; compute $\langle\hat{E}\rangle(t)$, $\langle\hat{x}\rangle(t)$; derive probability current for plane waves. Set B — prove group property of $\hat{U}$; unitarity; derive continuity equation; analyse Kramers degeneracy for spin-$\frac{1}{2}$ particle. Projects: (simple) time evolution visualiser for a particle in a box; (complex) Kramers degeneracy breaking by explicit $T$-violation and energy level splitting analysis.

## Module I.3 / L06: Time Evolution II — Time-Dependent and Non-Commuting Hamiltonian

**90-min pacing:** (0–20 min) Failure of $\hat{U} = e^{-i\hat{H}t/\hbar}$ when $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$; integral equation for $\hat{U}(t,t_0)$; iterative solution and the Dyson series; (20–45 min) Time-ordered exponential $\mathcal{T}\exp\left(-\frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t')dt'\right)$; Dyson's time-ordering operator $\mathcal{T}$; first- and second-order terms explicitly; (45–65 min) Piecewise-constant approximation: product formula $\hat{U} \approx \prod_k e^{-i\hat{H}(t_k)\Delta t/\hbar}$; connection to Trotter–Suzuki decomposition and quantum simulation; (65–90 min) Magnus expansion: exponentiation of the Dyson series; Magnus cumulants $\Omega_1, \Omega_2, \ldots$; unitarity preservation; convergence radius; physical example: driven spin-$\frac{1}{2}$ in an oscillating field; preview of Berry phase.

**Core learning outcomes:**

- Derive the Dyson series for a time-dependent Hamiltonian.
- Write the time-ordered exponential and evaluate its first two terms.
- Explain why the Magnus expansion preserves unitarity while the Dyson series does not term-by-term.
- Apply the piecewise-constant approximation to a simple driven system.

**Dual-track content:**

- *Physical track:* Driven two-level system (Rabi problem exact vs. perturbative); rotating wave approximation origin; geometric phase accumulation as a physical effect of non-

commutativity; quantum control applications.

- *Mathematical track:* Time-ordering as a resummation of a Neumann series; Magnus expansion as the unique Hermitian exponent; Zassenhaus-type decompositions; convergence of Magnus expansion (Baker norm criterion).

**Tier differentiation:** HS: "when the Hamiltonian changes in time, the future depends on the exact history" — analogy with a spinning top being poked at different times. Beg. UG: first-order Dyson term; transition probability formula; numerical piecewise evolution for a two-level system. Adv. UG: full Dyson series derivation; prove time-ordering is necessary; compute $\Omega_1$ and $\Omega_2$ of Magnus expansion. MSc: Magnus expansion convergence; Trotter–Suzuki product formula and its error; connection to quantum simulation circuits. PhD: Magnus expansion in Lie algebraic language; Floquet theory for periodic Hamiltonians; geometric phases and holonomy in the Magnus framework.

**Assessment:** Set A — compute first-order Dyson term for oscillating Hamiltonian; piecewise evolution of a qubit; transition probability to first order. Set B — derive Dyson series from integral equation; prove $\Omega_1$ of Magnus expansion; convergence criterion application. Projects: (simple) numerical Dyson vs. Magnus evolution comparison for driven qubit; (complex) Floquet Hamiltonian for periodically driven two-level system and quasi-energy spectrum.

## Module I.3 / L07: Pictures of Quantum Mechanics and the Heisenberg Picture

**90-min pacing:** (0–15 min) The equivalence-class of pictures: physical predictions (transition probabilities) are picture-independent; the Schrödinger picture reviewed — states evolve, operators fixed; (15–40 min) The Heisenberg picture: operators carry the time dependence, $\hat{A}_H(t) = \hat{U}^\dagger(t)\hat{A}\hat{U}(t)$; states static $|\psi_H\rangle = |\psi(t_0)\rangle$; Heisenberg equation of motion $\dot{\hat{A}}_H = \frac{i}{\hbar}[\hat{H}, \hat{A}_H] + (\partial_t \hat{A})_H$; (40–65 min) Consequences: Ehrenfest's theorem $\frac{d}{dt}\langle\hat{x}\rangle = \langle\hat{p}\rangle/m$, $\frac{d}{dt}\langle\hat{p}\rangle = -\langle\nabla V\rangle$; virial theorem; constants of motion $[\hat{H}, \hat{A}] = 0 \Rightarrow \dot{\hat{A}}_H = 0$; connection to Noether (Lecture L04); (65–90 min) Canonical quantisation in the Heisenberg picture: classical Poisson brackets $\to$ commutators, $\{f, g\}_{\mathrm{PB}} \to \frac{1}{i\hbar}[\hat{f}, \hat{g}]$; Dirac's quantisation rule; ordering ambiguities.

**Core learning outcomes:**

- Transform operators and states between Schrödinger and Heisenberg pictures.
- Derive and apply the Heisenberg equation of motion.
- Prove and use Ehrenfest's theorem and the virial theorem.
- Apply Dirac's canonical quantisation rule and identify operator ordering issues.

**Dual-track content:**

- *Physical track (Sakurai):* Heisenberg picture as the quantum analogue of classical mechanics — operators play the role of dynamical variables; Ehrenfest's theorem as the quantum–classical correspondence; constants of motion as quantum-mechanical conservation laws.
- *Mathematical/structural track:* Heisenberg equation as a derivation on the operator algebra; relationship to Lie bracket structure; canonical quantisation as a map from Poisson manifold to non-commutative algebra.

**Tier differentiation:** HS: "in one picture the states move, in the other the measuring devices move" — the physics is the same; analogy with active vs. passive coordinate transformations. Beg. UG: transform $\hat{x}(t)$ and $\hat{p}(t)$ for a harmonic oscillator; verify Ehrenfest's theorem

numerically. Adv. UG: full derivation of Heisenberg EOM; virial theorem proof; solve $\hat{x}_H(t)$ and $\hat{p}_H(t)$ for particle in a quadratic potential. MSc: Dirac quantisation rule; operator ordering ambiguities (Weyl ordering); connection to classical integrability and action–angle variables. PhD: Algebraic quantum mechanics; Heisenberg picture in $C^*$-algebra language; Moyal bracket and Wigner function as semi-classical bridge.

**Assessment:** Set A — solve Heisenberg EOM for harmonic oscillator; compute $\langle\hat{x}\rangle(t)$ and $\langle\hat{p}\rangle(t)$; verify Ehrenfest for a cubic potential. Set B — prove virial theorem; apply canonical quantisation to a classical system; analyse ordering ambiguities for $\hat{x}^2\hat{p}$. Projects: (simple) comparison of Schrödinger and Heisenberg pictures for a driven harmonic oscillator; (complex) Moyal bracket vs. quantum commutator for polynomial observables.

## Module I.3 / L08: The Dirac (Interaction) Picture

**90-min pacing:** (0–15 min) Motivation: perturbative treatment of $\hat{V}(t)$ while exploiting exact knowledge of $\hat{H}_0$ dynamics; splitting $\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$; (15–40 min) Definition of the interaction picture: states $|\psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar}|\psi_S(t)\rangle$; operators $\hat{V}_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{V}\, e^{-i\hat{H}_0 t/\hbar}$; interaction picture Schrödinger equation $i\hbar\partial_t|\psi_I\rangle = \hat{V}_I(t)|\psi_I\rangle$; (40–65 min) Interaction picture propagator $\hat{U}_I(t, t_0)$; Dyson series in the interaction picture; perturbative expansion of transition amplitudes; $S$-matrix $= \hat{U}_I(\infty, -\infty)$; physical interpretation as scattering operator; (65–90 min) Comparison table of all three pictures: state and operator evolution, equations of motion, preferred use-cases; bridge to time-dependent perturbation theory (L09–L10).

**Core learning outcomes:**

- Transform states and operators into and out of the interaction picture.
- Derive the interaction picture Schrödinger equation from the full equation.
- Write the Dyson series in the interaction picture to any desired order.
- Connect the $S$-matrix to the interaction picture propagator.

**Dual-track content:**

- *Physical track:* Interaction picture as the natural language of perturbation theory and scattering; $S$-matrix elements as transition amplitudes; relationship to measurable cross-sections and decay rates.
- *Mathematical track:* Interaction picture as a unitary equivalence transformation; two-sided action of free evolution; composition with Dyson series to produce $S$-matrix; time-ordering and causality.

**Tier differentiation:** HS: "park the easy part of the Hamiltonian in the operators, and let the hard part drive the states" — interaction picture as a practical bookkeeping device. Beg. UG: transform a simple interaction $\hat{V} = \lambda\hat{x}$ into the interaction picture for a harmonic oscillator $\hat{H}_0$; first-order Dyson term. Adv. UG: full transformation procedure; Dyson series to second order; prove equivalence of all three pictures for physical predictions. MSc: $S$-matrix and cross-sections; optical theorem; Lippmann–Schwinger equation preview; Feynman diagram structure. PhD: $S$-matrix in QFT context; adiabatic switching; Gell-Mann–Low theorem; LSZ reduction formula (conceptual).

**Assessment:** Set A — transform given operators and states to interaction picture; compute first-order transition amplitude for a harmonic perturbation. Set B — derive interaction picture EOM; prove unitarity of $S$-matrix; second-order Dyson series for a constant perturbation. Projects: (simple) three-picture comparison for a driven two-level system; (complex) $S$-matrix for two-level system under short pulse: exact vs. first/second-order.

## Module I.3 / L09: Perturbation Theory

**90-min pacing:** (0–20 min) Setup: $\hat{H} = \hat{H}_0 + \lambda\hat{V}$; known eigenstates $\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle$; perturbative ansatz $E_n = \sum_k \lambda^k E_n^{(k)}$, $|n\rangle = \sum_k \lambda^k |n^{(k)}\rangle$; non-degenerate case; (20–45 min) Rayleigh–Schrödinger PT: first-order energy $E_n^{(1)} = \langle n^{(0)}| \hat{V} |n^{(0)}\rangle$; first-order state correction $|n^{(1)}\rangle = \sum_{m\neq n} \frac{\langle m^{(0)}|\hat{V}|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle$; second-order energy; convergence and breakdown conditions; (45–65 min) Degenerate perturbation theory: identify degenerate subspace; block diagonalise $\hat{V}$ within it; secular equation; correct zeroth-order states; worked example (hydrogen Stark effect); (65–90 min) Born–Oppenheimer separation; Feynman–Hellmann theorem $\partial_\lambda E_n = \langle n|\partial_\lambda \hat{H}|n\rangle$; adiabatic theorem (statement); Ritz variational principle as complementary tool.

**Core learning outcomes:**

- Derive first- and second-order energy corrections in non-degenerate RSPT.
- Compute first-order state corrections and apply them to matrix elements.
- Handle degenerate perturbation theory by diagonalising $\hat{V}$ in the degenerate subspace.
- Apply the Feynman–Hellmann theorem and the Ritz variational principle.

**Dual-track content:**

- *Physical/applied track:* Hydrogen atom in electric field (Stark effect); fine structure as perturbation; Born–Oppenheimer in molecular physics; variational quantum eigensolvers (preview).
- *Mathematical/formal track:* Kato–Rellich theorem on analytic perturbations; convergence of Rayleigh–Schrödinger series; resolvent operator $\hat{R}_0(z) = (z - \hat{H}_0)^{-1}$ and Brillouin–Wigner PT.

**Tier differentiation:** HS: "a small push changes energy levels slightly" — first-order PT as the quantum analogue of Taylor expansion for energy levels. Beg. UG: first-order energy corrections for simple perturbations; linear Stark effect for hydrogen $n = 2$ by diagonalisation. Adv. UG: second-order energy; convergence conditions; full degenerate PT procedure; Feynman–Hellmann theorem proof. MSc: Kato–Rellich theorem; Brillouin–Wigner PT; Born–Oppenheimer rigorously; adiabatic theorem statement and proof sketch. PhD: Resolvent formalism; Dyson equation for the resolvent; divergence of perturbation series and resummation; PT for resonances.

**Assessment:** Set A — first-order corrections for anharmonic oscillator $\lambda\hat{x}^4$; linear and quadratic Stark effect for hydrogen; variational bound for ground state. Set B — second-order energy for harmonic oscillator shifted by linear term; degenerate PT for $n = 2$ hydrogen in electric field; Feynman–Hellmann for HO. Projects: (simple) PT vs. exact energy levels for anharmonic oscillator as a function of $\lambda$; (complex) Brillouin–Wigner vs. Rayleigh–Schrödinger comparison and convergence.

## Module I.3 / L10: Time-Dependent PT, Open Systems, and Many-Body Preview

**90-min pacing:** (0–20 min) Time-dependent perturbation theory: first-order transition amplitude $c_f^{(1)}(t) = -\frac{i}{\hbar}\int_{t_0}^{t} dt'\, e^{i\omega_{fi}t'} \langle f| \hat{V}(t') |i\rangle$; transition probability $|c_f^{(1)}|^2$; sudden and adiabatic limits; (20–40 min) Fermi's golden rule: continuous spectrum; density of states $\rho(E_f)$; $\Gamma_{i\to f} = \frac{2\pi}{\hbar} |\langle f| \hat{V} |i\rangle|^2 \rho(E_f)$; derivation from first-order TDPT via sinc function limit; selection rules; (40–65 min) Open quantum systems: closed (unitary) vs. open (environment coupling); Lindblad master equation $\dot{\hat{\rho}} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] + \sum_k (\hat{L}_k \hat{\rho} \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \hat{\rho}\})$; jump operators; decay channels; dephasing; (65–90 min) Decoherence and the measurement problem:

environment-induced superselection; pointer states; many-body preview: two-electron system, exchange interaction, singlet and triplet states, para- and ortho-helium.

**Core learning outcomes:**

- Derive Fermi's golden rule from first-order TDPT.
- Write the Lindblad master equation and identify physical contributions of each term.
- Explain decoherence as environment-induced suppression of off-diagonal density matrix elements.
- Construct two-electron spin states (singlet and triplet) and explain their exchange origin.

**Dual-track content:**

- *Physical/applied track:* Fermi's golden rule in atomic spectroscopy, nuclear decay, and solid-state (electron–phonon scattering); Lindblad equation in quantum optics (spontaneous emission) and quantum computing (gate errors); decoherence as the bridge between quantum and classical.
- *Mathematical/structural track:* Derivation of Fermi's golden rule rigorously; Lindblad equation from complete positivity and the GKSL theorem; pointer states as preferred bases selected by the environment.

**Tier differentiation:** HS: "open systems" as a leaky box — energy and information escaping to the environment; radioactive decay as Fermi's golden rule in action. Beg. UG: compute $\Gamma_{i\to f}$ for a constant perturbation; simple Lindblad evolution (amplitude damping) for a qubit. Adv. UG: derive Fermi's golden rule from TDPT; verify Lindblad equation preserves $\mathrm{Tr}\,\hat{\rho} = 1$ and positivity; singlet/triplet construction. MSc: GKSL theorem and complete positivity; Kraus representation; decoherence in the spin-boson model; Born–Markov approximation. PhD: Nakajima–Zwanzig projection technique; non-Markovian dynamics; Floquet–Lindblad equation; entanglement in many-body systems.

**Assessment:** Set A — compute $\Gamma_{i\to f}$ for an atom in an oscillating field; simple Lindblad evolution; singlet/triplet state construction and symmetry. Set B — derive Fermi's golden rule; prove Lindblad equation trace-preserving; exchange interaction from antisymmetry of two-electron wavefunction. Projects: (simple) decoherence visualiser: off-diagonal decay under Lindblad dephasing; (complex) Born–Markov derivation of Lindblad equation for a qubit coupled to a harmonic bath.

### Assessment Strategy

***Per-Lecture Assessment Bundle*** *(every lecture): concept check (10–15 min, 5–8 short conceptual questions); Problem Set A (simple, 6–10 computational exercises); Problem Set B (complex, 3–6 multi-step derivations and proofs); mini-project simple (1–3 hours); mini-project complex (5–10 hours, open-ended); research questions well-defined (2–4 scoped prompts); research questions fully open (1–2 frontier-style prompts).*

**Course-level grading template:**

- Weekly homework (best 8/10): 40%
- Mid-course synthesis (after L06): 20%
- Project (simple or complex track): 25%
- Final oral/whiteboard exam or take-home: 15%

Mixed cohorts: HS/UG → simple track; MSc/PhD → complex track.

### Reading Strategy

**Track A — Core textbooks:** Sakurai & Napolitano, *Modern Quantum Mechanics* Ch. 2–5 (dynamics, symmetry, perturbation theory); Shankar, *Principles of Quantum Mechanics* (perturbation theory and example systems).

**Track B — Support:** Messiah, *Quantum Mechanics* (density matrix, open systems); Cohen-Tannoudji, Diu & Laloë (time-dependent PT, Fermi's golden rule).

**Track C — Specialised:** Lindblad (1976) original paper on master equations (MSc/PhD); Born & Oppenheimer (1927) original paper.

## 1.4 Module I.4 — Paradigm Physical Systems

**Series I / Module I.4** *Paradigm Physical Systems*

**Duration:** 12 lectures × 90 minutes (Lectures L01–L12). Each lecture develops one canonical quantum system in full depth, applying the formalism of Modules I.1–I.3. Lectures are self-contained and may be taught in any order after Module I.3 L10.

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Free Particle & Particle in a Box | Plane waves; wave packets; quantisation; DOS in 1/2/3D |
| L02 | Simple Bloch Electron | Bloch's theorem; BZ; nearly-free electron gap; tight-binding |
| L03 | Harmonic Oscillator | Ladder operators; coherent states; anharmonic PT |
| L04 | Rabi Model & Two-Level Systems | Jaynes–Cummings; Rabi oscillations; RWA; $\pi$-pulses |
| L05 | Minimal Coupling & Landau Levels | $(\hat{\mathbf{p}} - e\mathbf{A})^2/2m$; gauge invariance; Landau levels |
| L06 | Zeeman Effect | Normal and anomalous; Paschen–Back; Breit–Rabi |
| L07 | Stark Effect | Linear and quadratic; parabolic coordinates; polarisability |
| L08 | Spin–Orbit & Fine Structure | Thomas precession; $\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$; $E_{nj}^{\mathrm{FS}}$ |
| L09 | Hydrogen Atom | Coulomb potential; radial equation; $\psi_{nlm}$; $SO(4)$ symmetry |
| L10 | Scattering from a Potential Barrier | Step potential; transmission; tunnelling; $T$ and $R$ coefficients |
| L11 | Scattering from a Potential Well | Finite square well; bound states; resonance transmission |
| L12 | Delta-Function Potential & WKB | $\delta(x)$ bound state; WKB quantisation; tunnelling rate |

### Lecture-by-Lecture Syllabi

**Module I.4 / L01: Free Particle and Particle in a Box**

**90-min pacing:** (0–20 min) Free particle: $\hat{H} = \hat{p}^2/2m$; plane-wave solutions $\psi_k(x) = e^{ikx}/\sqrt{L}$ (periodic BC) or $e^{ikx}/\sqrt{2\pi}$ (continuum); energy $E_k = \hbar^2 k^2/2m$; continuum of states; momentum eigenstates; (20–45 min) Wave packets: $\psi(x,t) = \int \tilde{\phi}(k) e^{ikx - i\omega_k t} dk/\sqrt{2\pi}$; group velocity $v_g = d\omega/dk|_{k_0} = \hbar k_0/m$; phase velocity $v_p = \omega/k = \hbar k/2m$; dispersion broadening; Gaussian wave packet: exact solution; $\Delta x\, \Delta p = \hbar/2$ at $t = 0$, grows as $t \to \infty$; (45–70 min) Infinite square well: $V = 0$ for $0 < x < L$, $\infty$ otherwise; solutions $\psi_n = \sqrt{2/L}\sin(n\pi x/L)$; energies $E_n = n^2\pi^2\hbar^2/2mL^2$; zero-point energy $E_1 > 0$; parity; orthonormality; completeness; expectation values; (70–90 min) 3D box; degeneracy; density of states in 1D, 2D, 3D $g(E) = (L/\pi\hbar)\sqrt{m/2E}$ in 1D, $g(E) = mL^2/\pi\hbar^2$ in 2D (constant!), $g(E) = V(2m)^{3/2}E^{1/2}/2\pi^2\hbar^3$ in 3D; van Hove singularities.

**Core learning outcomes:**

- Solve the free-particle and infinite-square-well Schrödinger equations.
- Construct and propagate a Gaussian wave packet; compute group and phase velocities.
- Derive the density of states in 1D, 2D, and 3D.
- Interpret zero-point energy physically and compute $\langle x \rangle, \langle p \rangle, \langle E \rangle$.

**Dual-track content:**

- *Physical track:* Particle in a box as the prototype for all confined quantum systems; zero-point energy as the origin of atomic stability; wave packet spreading as a measurable quantum effect.
- *Mathematical track:* Fourier analysis as change of basis; density of states from counting states in $k$-space; van Hove singularities from $\nabla_k E = 0$.

**Tier differentiation:** HS: "the particle cannot be at rest inside the box" — zero-point energy and quantum confinement. Beg. UG: $\psi_n$, $E_n$, $\langle x \rangle$, $\langle p \rangle$; Gaussian wave packet; group velocity. Adv. UG: 3D box; density of states; van Hove singularities; wave-packet dispersion derivation. MSc: Airy functions for a linear potential; finite square well; transfer matrix method. PhD: self-adjoint extensions of $\hat{p}$ on $[0, L]$; particle in a box in curved space; resonances and quasi-bound states (complex scaling).

**Assessment:** Set A — $E_n$, $\psi_n$, $\langle x \rangle$, $\Delta x\, \Delta p$ for 1D box; group velocity of Gaussian packet. Set B — density of states in 2D and 3D; wave-packet spreading: $\Delta x(t)$ for Gaussian. Projects: (simple) Gaussian wave packet animation: $|\psi(x,t)|^2$ spreading and oscillation; (complex) van Hove singularities in 1D and 3D DOS: logarithmic vs. power-law divergence.

## Module I.4 / L02: The Simple Bloch Electron

**90-min pacing:** (0–20 min) Periodic potential $V(x+a) = V(x)$; translation operator $\hat{T}_a |x\rangle = |x+a\rangle$; $[\hat{H}, \hat{T}_a] = 0$ when $V$ is periodic; simultaneous eigenstates; (20–45 min) Bloch's theorem: $\psi_{nk}(x) = e^{ikx} u_{nk}(x)$ with $u_{nk}(x+a) = u_{nk}(x)$; crystal momentum $\hbar k$; first Brillouin zone $k \in (-\pi/a, \pi/a]$; band index $n$; reduced, extended, and repeated zone schemes; (45–65 min) Nearly-free electron model: treat $V$ as a small perturbation; energy near zone boundary $k = \pm\pi/a$ splits into two bands; band gap $2|V_G|$ at the zone boundary ($V_G$ = Fourier component of $V$); effective mass $m^* = (d^2E/dk^2)^{-1}\hbar^2$; heavy and light bands; (65–90 min) Tight-binding model (1D, one orbital per site): $E(k) = \varepsilon_0 - 2t\cos(ka)$; bandwidth $4t$; effective mass at $k = 0$; connection to hopping integrals; Wannier functions; metals, insulators, and semiconductors from band filling.

**Core learning outcomes:**

- State and derive Bloch's theorem from translation symmetry.
- Identify crystal momentum, the Brillouin zone, and band gaps.
- Compute band gaps from the nearly-free electron model.
- Solve the 1D tight-binding model and extract the effective mass.

**Dual-track content:**

- *Physical/materials track:* Connection to real materials: Si, GaAs, graphene; band filling and electrical conductivity; effective mass in semiconductor devices; Wannier functions as chemically intuitive localised orbitals.
- *Mathematical track:* Bloch's theorem from the representation theory of $\mathbb{Z}$; Floquet–Bloch analogy (time vs. space periodicity); Wannier–Kohn theorem; Berry phase of Bloch bands (preview of topology).

**Tier differentiation:** HS: "a periodic lattice turns continuous energy into discrete bands" — the origin of conductors and insulators. Beg. UG: Bloch's theorem; BZ; nearly-free electron gap; tight-binding $E(k)$. Adv. UG: derivation of Bloch from $[\hat{H}, \hat{T}_a] = 0$; effective mass derivation; Wannier functions. MSc: Peierls instability; Su–Schrieffer–Heeger model; Berry phase of a band; Zak phase. PhD: Wannier functions and maximally localised orbitals; topological invariants (Chern number); non-Abelian Berry connection.

**Assessment:** Set A — Bloch wave for a cosine potential; BZ boundaries; tight-binding $E(k)$ and bandwidth. Set B — derive Bloch's theorem; nearly-free electron gap $2|V_G|$; effective mass at $k = 0$. Projects: (simple) tight-binding band structure for 1D, 2D square, and graphene lattices; (complex) Berry phase (Zak phase) for a 1D SSH model: topological vs. trivial phase.

## Module I.4 / L03: The Harmonic Oscillator

**90-min pacing:** (0–20 min) Hamiltonian $\hat{H} = \hat{p}^2/2m + m\omega^2\hat{x}^2/2$; algebraic approach: ladder operators $\hat{a} = \sqrt{m\omega/2\hbar}(\hat{x} + i\hat{p}/m\omega)$, $\hat{a}^\dagger = \sqrt{m\omega/2\hbar}(\hat{x} - i\hat{p}/m\omega)$; $[\hat{a}, \hat{a}^\dagger] = 1$; $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$; (20–45 min) Spectrum: $E_n = (n + \frac{1}{2})\hbar\omega$; number states $|n\rangle$; $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$, $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$; matrix elements $\langle m|\,\hat{x}\,|n\rangle = \sqrt{\hbar/2m\omega}(\sqrt{n}\delta_{m,n-1} + \sqrt{n+1}\delta_{m,n+1})$; position wavefunctions via $\hat{a}\,|0\rangle = 0$: Hermite polynomials; (45–65 min) Coherent states: $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$; $|\alpha\rangle = e^{-|\alpha|^2/2}\sum_n \alpha^n\,|n\rangle\,/\sqrt{n!}$; over-completeness; Poisson photon number distribution; minimum uncertainty; classical-like oscillation $\langle\hat{x}\rangle = x_{\mathrm{cl}}(t)$; (65–90 min) Anharmonic oscillator: $\hat{V}' = \lambda\hat{x}^3 + \mu\hat{x}^4$; first- and second-order energy corrections via ladder operators; selection rules $\Delta n = \pm1, \pm2, \pm3$; Morse potential as a realistic anharmonic oscillator; vibrational spectroscopy.

**Core learning outcomes:**

- Derive the HO spectrum algebraically using ladder operators.
- Compute matrix elements of $\hat{x}$ and $\hat{p}$ in the number basis.
- Construct and interpret coherent states as minimum-uncertainty classical-like states.
- Apply first-order PT to compute anharmonic energy corrections.

**Dual-track content:**

- *Physical track:* HO as the universal model near a potential minimum; zero-point energy in He (stays liquid at atmospheric pressure); photons as HO quanta; phonons in solids; vibrational spectra of diatomic molecules.
- *Mathematical track:* Ladder operator algebra as $\mathfrak{su}(1,1)$ or Heisenberg–Weyl; coherent states as displaced vacuum; Glauber–Sudarshan $P$-representation; generating function for Hermite polynomials.

**Tier differentiation:** HS: HO as the "building block of physics" — springs, pendula, photons, phonons. Beg. UG: spectrum; ladder operators; zero-point energy; Hermite polynomials. Adv. UG: matrix elements; coherent states; Poisson distribution; anharmonic PT. MSc: squeezed states; Wigner function; $P$-function; Husimi $Q$-function. PhD: quantisation of EM field from HO; non-linear optics from anharmonicity; exactly solvable anharmonic potentials (Morse, Pöschl–Teller).

**Assessment:** Set A — $E_n$, $\langle m|\,\hat{x}^2\,|n\rangle$, $\langle\hat{x}\rangle(t)$ for coherent state; first-order anharmonic correction to $E_n$ for $\lambda\hat{x}^4$. Set B — derive $|n\rangle = (\hat{a}^\dagger)^n\,|0\rangle\,/\sqrt{n!}$; prove coherent state over-completeness; second-order anharmonic for $\lambda\hat{x}^3$. Projects: (simple) Wigner function of $|n\rangle$, $|\alpha\rangle$, and superposition states; (complex) Morse potential: exact vs. perturbative vibrational levels for HCl.

## Module I.4 / L04: Rabi Model and Two-Level Systems

**90-min pacing:** (0–20 min) Two-level system (TLS) Hamiltonian: $\hat{H}_0 = \frac{\hbar\omega_0}{2}\hat{\sigma}_z$; energy splitting $\hbar\omega_0$; Bloch sphere representation; general state $|\psi\rangle = \cos(\theta/2)\,|\uparrow\rangle + e^{i\phi}\sin(\theta/2)\,|\downarrow\rangle$; (20–45 min) Driven TLS: oscillating field $\hat{H}' = -dE_0\cos(\omega t)\hat{\sigma}_x$; transformation to rotating

frame; rotating wave approximation (RWA): drop counter-rotating term; resonance condition $\omega = \omega_0$; Rabi Hamiltonian in rotating frame: $\hat{H}_{\text{rot}} = \frac{\hbar\Delta}{2}\hat{\sigma}_z + \frac{\hbar\Omega_R}{2}\hat{\sigma}_x$ with detuning $\Delta = \omega - \omega_0$ and Rabi frequency $\Omega_R = dE_0/\hbar$; (45–65 min) Exact Rabi solution: generalised Rabi frequency $\tilde{\Omega} = \sqrt{\Delta^2 + \Omega_R^2}$; population inversion $P_\uparrow(t) = (\Omega_R/\tilde{\Omega})^2 \sin^2(\tilde{\Omega}t/2)$; $\pi$-pulse ($\Omega_R t = \pi$) and $\pi/2$-pulse for complete inversion/superposition; (65–90 min) Jaynes–Cummings model: full quantum treatment of TLS + single EM mode; $\hat{H}_{\text{JC}} = \hbar\omega_c \hat{a}^\dagger\hat{a} + \frac{\hbar\omega_0}{2}\hat{\sigma}_z + \hbar g(\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger\hat{\sigma}_-)$; dressed states; collapse and revival of Rabi oscillations; photon number states.

**Core learning outcomes:**

- Derive the rotating frame Hamiltonian and apply the RWA.
- Solve the Rabi problem exactly for arbitrary detuning.
- Compute the population inversion and identify $\pi$- and $\pi/2$-pulses.
- Write the Jaynes–Cummings Hamiltonian and identify dressed states.

**Dual-track content:**

- *Physical/applications track:* NMR and MRI as Rabi oscillations of nuclear spins; qubit gates in quantum computing as $\pi/\pi/2$ pulses; atom trapping and cooling; dressed-state picture for AC Stark shift.
- *Mathematical track:* $SU(2)$ structure of TLS; Bloch equation of motion; Jaynes–Cummings exact diagonalisation in $\{|n,\uparrow\rangle, |n+1,\downarrow\rangle\}$ subspace.

**Tier differentiation:** HS: "a laser pulse can flip a quantum spin — that's the basis of NMR and quantum computers." Beg. UG: driven TLS; resonance; Rabi oscillations; $\pi$-pulse. Adv. UG: RWA derivation; off-resonance Rabi; Bloch–Siegert shift. MSc: Jaynes–Cummings dressed states; collapse and revival; vacuum Rabi splitting. PhD: beyond RWA (Rabi model vs. JC); ultra-strong coupling; quantum phase transition in Rabi model.

**Assessment:** Set A — Rabi frequency for given field strength; $\pi$-pulse time; on-resonance inversion. Set B — derive rotating frame Hamiltonian; off-resonance Rabi: $P_\uparrow(t)$ for $\Delta = \Omega_R$. Projects: (simple) Bloch sphere trajectory under $\pi/2$–wait–$\pi/2$ (Ramsey spectroscopy); (complex) Jaynes–Cummings: collapse and revival of Rabi oscillations for Fock and coherent states.

## Module I.4 / L05: Minimal Coupling Hamiltonian and Landau Levels

**90-min pacing:** (0–20 min) Classical mechanics in EM fields: Lorentz force; canonical momentum $\mathbf{p}_{\text{can}} = m\mathbf{v} + e\mathbf{A}$; Hamiltonian $H = (\mathbf{p}_{\text{can}} - e\mathbf{A})^2/2m + e\phi$; quantisation: $\hat{H} = (\hat{\mathbf{p}} - e\mathbf{A})^2/2m + e\phi$; gauge invariance: $\mathbf{A} \to \mathbf{A} + \nabla\chi$, $\psi \to e^{ie\chi/\hbar}\psi$; (20–45 min) Uniform magnetic field $\mathbf{B} = B\hat{z}$; symmetric gauge $\mathbf{A} = \frac{B}{2}(-y, x, 0)$ or Landau gauge $\mathbf{A} = (0, Bx, 0)$; Landau gauge: $\hat{H} = \frac{1}{2m}[\hat{p}_x^2 + (\hat{p}_y - eBx)^2]$; reduce to HO: $\hat{H} = \hat{p}_x^2/2m + \frac{m\omega_c^2}{2}(x - x_0)^2$ with cyclotron frequency $\omega_c = eB/m$ and guiding centre $x_0 = \hbar k_y/m\omega_c$; (45–65 min) Landau levels: $E_n = (n+\frac{1}{2})\hbar\omega_c$; massive degeneracy; density of states: one state per area $2\pi l_B^2$ where $l_B = \sqrt{\hbar/eB}$ (magnetic length); filling factor $\nu = N_e/(eBA/h)$; quantum Hall effect preview; (65–90 min) Aharonov–Bohm effect: $\oint \mathbf{A} \cdot d\mathbf{l} = \Phi$ (enclosed flux); phase $e\Phi/\hbar$; physical despite $\mathbf{B} = 0$ in accessible region; experimental observation via electron interferometry; connection to Berry phase.

**Core learning outcomes:**

- Write the minimal coupling Hamiltonian and explain gauge invariance.
- Solve for Landau levels and derive their degeneracy.

- Define the magnetic length and filling factor.
- Explain the Aharonov–Bohm effect and its topological character.

**Dual-track content:**

- *Physical track:* Landau levels as the quantum basis for the quantum Hall effect; cyclotron resonance in semiconductors; Shubnikov–de Haas oscillations; Aharonov–Bohm interference in mesoscopic rings.
- *Mathematical track:* Gauge theory as $U(1)$ fibre bundle; minimal coupling as covariant derivative; Aharonov–Bohm as holonomy; magnetic translation group.

**Tier differentiation:** HS: "a magnetic field forces electrons to move in circles — quantised, those circles become Landau levels." Beg. UG: minimal coupling Hamiltonian; gauge transformation; Landau level spectrum. Adv. UG: gauge derivation; degeneracy; filling factor; Aharonov–Bohm phase. MSc: magnetic length; QHE overview; Laughlin's gauge argument; de Haas–van Alphen. PhD: magnetic translation group; Hofstadter butterfly; fractional QHE; composite fermions.

**Assessment:** Set A — gauge transform $\mathbf{A} \to \mathbf{A} + \nabla\chi$; Landau level $E_n$; cyclotron frequency for electron. Set B — Landau level degeneracy derivation; Aharonov–Bohm phase for a given flux. Projects: (simple) Landau level DOS vs. magnetic field: SdH oscillation preview; (complex) Aharonov–Bohm interference in a ring: conductance oscillations vs. flux.

## Module I.4 / L06: Zeeman Effect

**90-min pacing:** (0–20 min) Magnetic moment from orbital motion: $\hat{\boldsymbol{\mu}}_L = -\frac{e}{2m}\hat{\mathbf{L}} = -\mu_B\hat{\mathbf{L}}/\hbar$; Bohr magneton $\mu_B = e\hbar/2m$; interaction Hamiltonian $\hat{H}_Z = -\hat{\boldsymbol{\mu}}\cdot\mathbf{B}$; (20–45 min) Normal Zeeman effect (spin ignored): $\hat{H}' = \mu_B B\hat{L}_z/\hbar$; first-order PT: $E^{(1)}_{nlm} = \mu_B m B$; triplet splitting $\Delta E = \mu_B B$; Lorentz classical model gives the same result; experimental observation in Cd, Mg; (45–65 min) Anomalous Zeeman effect (spin included): total magnetic moment $\hat{\boldsymbol{\mu}} = -\mu_B(\hat{\mathbf{L}} + g_s\hat{\mathbf{S}})/\hbar$ with $g_s \approx 2$; Landé $g$-factor from projection: $g_J = 1 + \frac{j(j+1)+s(s+1)-l(l+1)}{2j(j+1)}$; energy $E^{(1)}_{njm_j} = g_J\mu_B m_j B$; pattern of anomalous Zeeman in Na D-lines; (65–90 min) Paschen–Back effect: strong field limit where $\mu_B B \gg$ spin–orbit splitting; good quantum numbers switch from $j, m_j$ to $m_l, m_s$; linear splitting in $m_l + 2m_s$; Breit–Rabi diagram: interpolation between weak and strong field; applications: laser cooling and trapping (magneto-optical trap); EPR/ESR spectroscopy.

**Core learning outcomes:**

- Derive the normal Zeeman splitting from first-order PT.
- Compute the Landé $g$-factor and explain anomalous Zeeman splitting.
- Identify the good quantum numbers in weak and strong field limits.
- Describe the Breit–Rabi interpolation and its applications.

**Dual-track content:**

- *Spectroscopy track:* Normal and anomalous Zeeman patterns; Na D-line splitting; astrophysical Zeeman effect (magnetic fields in sunspots and neutron stars); magneto-optical trap level structure.
- *Theoretical track:* Landé $g$-factor from Wigner–Eckart theorem; Breit–Rabi exact solution for two-level mixing; $g$-factor anomaly $g_s - 2$ from QED (preview).

**Tier differentiation:** HS: "a magnetic field splits spectral lines — Zeeman used it to discover spin (indirectly)." Beg. UG: normal Zeeman; Landé $g_J$; triplet pattern; anomalous

Zeeman for Na. Adv. UG: Landé $g$-factor derivation from projection; Paschen–Back limit. MSc: Breit–Rabi exact solution; astrophysical applications; magneto-optical trap. PhD: $g_s - 2$ from QED one-loop; nuclear Zeeman; molecular Zeeman in diamagnetic molecules.

**Assessment:** Set A — normal Zeeman triplet for Cd; Landé $g_J$ for $^2P_{3/2}$; anomalous Zeeman pattern for Na D. Set B — Landé $g_J$ derivation from Wigner–Eckart; Paschen–Back level energies for H $n = 2$. Projects: (simple) Zeeman energy level diagram for H $n = 2$ vs. field strength (weak→Paschen–Back); (complex) Breit–Rabi diagram for $^{87}$Rb $F = 1, 2$ hyperfine states: magneto-optical trap.

## Module I.4 / L07: Stark Effect

**90-min pacing:** (0–20 min) Electric field perturbation: $\hat{H}' = -e\mathbf{E} \cdot \hat{\mathbf{r}} = -eE\hat{z}$ (field along $z$); why first-order vanishes for non-degenerate eigenstates of definite parity: $\langle nlm|z|nlm\rangle = 0$ by parity; exception: degenerate $n = 2$ states of hydrogen; (20–45 min) Linear Stark effect in hydrogen $n = 2$: four degenerate states $\{2s, 2p_{-1}, 2p_0, 2p_1\}$; only $\langle 2s|\hat{H}'|2p_0\rangle = -3eEa_0$ is non-zero (by selection rules and parity); diagonalise $4 \times 4$ matrix; four eigenstates with energies $\{-3eEa_0, 0, 0, +3eEa_0\}$; physical picture: mixed $s$-$p$ states with permanent electric dipole moment; (45–65 min) Quadratic Stark effect for ground state of H: $E_0^{(2)} = \sum_{n\neq 0} |\langle n|\hat{H}'|0\rangle|^2/(E_0 - E_n)$; evaluate using sum rules: $E_0^{(2)} = -\frac{9}{4}a_0^3E^2$ (exact); polarisability $\alpha = -\partial^2E/\partial E^2 = \frac{9}{2}a_0^3$ (in Gaussian units); (65–90 min) Parabolic coordinates for the Stark effect in hydrogen: exact separation in $(\xi, \eta, \phi)$; quantum numbers $n_1, n_2, m$; $E^{(1)}_{n_1n_2m} = -3neEa_0(n_1 - n_2)/2$; connection to classical ionisation (barrier suppression); strong-field ionisation and above-threshold ionisation preview.

**Core learning outcomes:**

- Explain why the first-order Stark shift vanishes for non-degenerate parity eigenstates.
- Compute the linear Stark effect for hydrogen $n = 2$ by degenerate PT.
- Derive the quadratic Stark shift and polarisability for the hydrogen ground state.
- Describe parabolic coordinates and the exact separation of the Stark Hamiltonian.

**Dual-track content:**

- *Physical track:* Stark effect in molecular spectroscopy (polarity measurement); DC Stark shift for qubit control; electric-field ionisation; Rydberg atoms in electric fields; barrier-suppression ionisation in intense laser fields.
- *Mathematical track:* Sum-rule approach to quadratic Stark; parabolic coordinates as a natural frame for the Coulomb + linear potential problem; connection to classical action variables.

**Tier differentiation:** HS: "an electric field shifts energy levels — the Stark effect measures molecular polarity." Beg. UG: parity argument for vanishing first order; $n = 2$ linear Stark; polarisability. Adv. UG: quadratic Stark derivation; sum rules; parabolic quantum numbers. MSc: exact parabolic solution; tunnelling ionisation (Keldysh parameter); above-threshold ionisation. PhD: strong-field ionisation rates (ADK formula); high-harmonic generation; attosecond physics.

**Assessment:** Set A — parity argument; $n = 2$ Stark matrix elements; polarisability of H ground state. Set B — derive quadratic Stark via sum rules; parabolic quantum numbers for $n = 3$ H. Projects: (simple) Stark energy levels for H $n = 2, 3$: linear splitting vs. field; (complex) polarisability of alkali atoms from sum rules: compare Li, Na, K.

## Module I.4 / L08: Spin–Orbit Coupling and Fine Structure of Hydrogen

**90-min pacing:** (0–20 min) Physical origin of spin–orbit coupling: in the rest frame of the electron, the nucleus orbits and creates a magnetic field $\mathbf{B}_{\text{eff}} = \mathbf{v} \times \mathbf{E}/c^2$; magnetic moment of the electron $\hat{\boldsymbol{\mu}}_s = -g_s \mu_B \hat{\mathbf{S}}/\hbar$; naive SO Hamiltonian $\hat{H}_{\text{SO}}^{\text{naive}} = \mu_B \mathbf{B}_{\text{eff}} \cdot \hat{\mathbf{S}}/\hbar$; Thomas precession: relativistic correction reduces by factor 2; correct result $\hat{H}_{\text{SO}} = \frac{1}{2m^2c^2}\frac{1}{r}\frac{dV}{dr}\hat{\mathbf{L}}\cdot\hat{\mathbf{S}} = \xi(r)\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}$; (20–45 min) Evaluation in hydrogen: $\hat{\mathbf{L}}\cdot\hat{\mathbf{S}} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$; eigenvalue in $|ljm_j\rangle$ basis: $\frac{\hbar^2}{2}[j(j+1) - l(l+1) - s(s+1)]$; $\langle r^{-3}\rangle_{nl} = \frac{2}{n^3 a_0^3 l(l+\frac{1}{2})(l+1)}$; first-order SO energy for $l > 0$: $E_{\text{SO}}^{(1)} = \frac{\alpha^2 E_n}{2n}\frac{j(j+1)-l(l+1)-3/4}{l(l+\frac{1}{2})(l+1)}$; (45–65 min) Other fine-structure corrections: relativistic kinetic energy $\hat{H}'_{\text{KE}} = -\hat{p}^4/8m^3c^2$; Darwin term $\hat{H}_D = \frac{\pi\hbar^2}{2m^2c^2}|e|\delta^3(\mathbf{r})$ (only for $l = 0$); total fine structure: $E_{nj}^{\text{FS}} = -\frac{\alpha^2 E_n}{2n^2}\left[\frac{4n}{j+1/2} - 3\right]$; (65–90 min) Fine structure of hydrogen: energy level diagram including all three corrections; $2s_{1/2}$ and $2p_{1/2}$ are degenerate in Dirac theory; Lamb shift lifts this (QED); comparison with Dirac equation prediction; $LS$ vs. $jj$ coupling in heavy atoms; hyperfine structure from nuclear spin (preview); precision measurements and the fine structure constant.

**Core learning outcomes:**

- Derive the spin–orbit Hamiltonian including the Thomas precession factor.
- Evaluate SO energy corrections using $\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}$ in the $|ljm_j\rangle$ basis.
- Identify all three fine-structure corrections and compute the total $E_{nj}^{\text{FS}}$.
- Explain the Lamb shift qualitatively and its significance for QED.

**Dual-track content:**

- *Physical track:* Fine structure as the gateway to relativistic QM; Thomas precession as a measurable relativistic effect; $LS$ vs. $jj$ coupling in the periodic table; Lamb shift as the first clear sign of vacuum fluctuations (QED); precision spectroscopy tests.
- *Mathematical track:* Derivation from Dirac equation in the non-relativistic limit; $1/r\, dV/dr$ expectation value via Kramers–Pasternack relation; Foldy–Wouthuysen transformation.

**Tier differentiation:** HS: "relativity and quantum mechanics together split hydrogen's spectral lines into fine structure." Beg. UG: SO Hamiltonian; $\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}$ in $j$-basis; energy diagram. Adv. UG: Thomas precession; Darwin term; all three corrections; $E_{nj}^{\text{FS}}$. MSc: Lamb shift (qualitative QED); $jj$ coupling in heavy atoms; Kramers–Pasternack. PhD: Foldy–Wouthuysen; Dirac equation non-relativistic limit; QED radiative corrections; Breit equation for two-electron systems.

**Assessment:** Set A — SO energy for H $2p$ with $j = 3/2$ and $j = 1/2$; fine-structure splitting; Darwin correction for $ns$ states. Set B — derive $\hat{H}_{\text{SO}}$ including Thomas factor; Kramers–Pasternack for $\langle r^{-3}\rangle$; fine-structure formula $E_{nj}^{\text{FS}}$. Projects: (simple) hydrogen fine-structure energy level diagram: $n = 1, 2, 3$ including all three corrections; (complex) Foldy–Wouthuysen transformation: derive non-relativistic limit of Dirac Hamiltonian.

## Module I.4 / L09: The Hydrogen Atom

**90-min pacing:** (0–20 min) Hydrogen Hamiltonian in atomic units: $\hat{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r}$; separation in spherical coordinates: $\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)Y_l^m(\theta,\phi)$; reduction of radial equation via $u_{nl}(r) = rR_{nl}(r)$; effective potential $V_{\text{eff}}(r) = -1/r + l(l+1)/2r^2$; (20–45 min) Power-series solution: associated Laguerre polynomials; termination condition gives quantisation $E_n = -1/(2n^2)$ a.u. $= -13.6/n^2$ eV; principal quantum number $n = 1, 2, 3, \ldots$; orbital

$l = 0, \ldots, n-1$; magnetic $m = -l, \ldots, l$; radial wavefunctions $R_{nl}(r)$ for $n = 1, 2, 3$; radial nodes $n-l-1$; angular nodes $l$; (45–65 min) Degeneracy: $n^2$-fold degeneracy (ignoring spin); explanation via hidden $SO(4)$ symmetry; Runge–Lenz vector $\hat{\mathbf{A}}$; $[\hat{H}, \hat{\mathbf{A}}] = 0$; $\mathfrak{so}(4)$ algebra; algebraic derivation of $1/n^2$ spectrum without solving the radial equation; (65–90 min) Expectation values and radial properties: $\langle r \rangle_{nl} = a_0[3n^2 - l(l+1)]/2Z$; $\langle r^{-1} \rangle = Z/n^2 a_0$; $\langle r^{-3} \rangle$ (Kramers–Pasternack); most probable radius; radial probability density $4\pi r^2 |R_{nl}|^2$; connection to fine structure (L08) and to multi-electron atoms.

**Core learning outcomes:**

- Separate the hydrogen Schrödinger equation and solve the radial part.
- State the energy spectrum, quantum numbers, and degeneracy.
- Compute radial expectation values and interpret radial probability densities.
- Explain the hidden $SO(4)$ symmetry and its algebraic consequences.

**Dual-track content:**

- *Physical track:* Hydrogen as the prototype for all atomic structure; comparison with experimental spectroscopy (Lyman, Balmer, Paschen series); connection to fine structure (L08), Zeeman (L06), and Stark (L07) effects.
- *Mathematical track:* Runge–Lenz vector and $\mathfrak{so}(4)$ algebra; algebraic derivation of the spectrum; parabolic coordinates as an alternative separation; $N$-dimensional hydrogen and supersymmetric QM.

**Tier differentiation:** HS: hydrogen orbitals as the shapes of atomic physics; spectral series and the Bohr model vs. exact QM. Beg. UG: radial wavefunctions; probability densities; $E_n = -13.6/n^2$ eV; quantum numbers. Adv. UG: series solution and termination; $\langle r^k \rangle_{nl}$; $SO(4)$ symmetry sketch. MSc: Runge–Lenz vector algebra; parabolic coordinates; algebraic derivation of spectrum. PhD: $N$-dimensional hydrogen; supersymmetric QM for hydrogen; muonic hydrogen and reduced mass.

**Assessment:** Set A — $R_{10}, R_{21}$; probability density; most probable radius; $\langle r \rangle_{2p}$; $E_n$. Set B — termination condition derivation; prove $[\hat{H}, \hat{\mathbf{A}}] = 0$; algebraic spectrum from $\mathfrak{so}(4)$. Projects: (simple) radial probability density plotter for all orbitals with $n \leq 3$; (complex) Runge–Lenz algebra: derive energy spectrum of hydrogen without the Schrödinger equation.

## Module I.4 / L10: Scattering from a Potential Barrier

**90-min pacing:** (0–20 min) Setup: particle with energy $E$ incident from the left on a rectangular potential barrier $V_0 > 0$ of width $a$; three regions with different wavevectors: $k_1 = \sqrt{2mE}/\hbar$ (regions I, III), $k_2 = \sqrt{2m|E - V_0|}/\hbar$ (region II); two sub-cases: $E > V_0$ (above barrier) and $E < V_0$ (tunnelling); (20–50 min) Matching boundary conditions: continuity of $\psi$ and $\psi'$ at $x = 0$ and $x = a$; matrix transfer method; transmission amplitude $t$ and reflection amplitude $r$; transmission coefficient $T = |t|^2$ and reflection coefficient $R = |r|^2$; verify $T + R = 1$; above-barrier case $E > V_0$: oscillatory $T(E)$ with resonances at $k_2 a = n\pi$; (50–70 min) Tunnelling case $E < V_0$: $k_2 \to i\kappa_2 = i\sqrt{2m(V_0 - E)}/\hbar$; $T = \left[1 + \frac{(k_1^2+\kappa_2^2)^2}{4k_1^2\kappa_2^2} \sinh^2(\kappa_2 a)\right]^{-1}$; thin-barrier limit: $T \approx e^{-2\kappa_2 a}$; Gamow factor; applications: alpha decay, STM, nuclear fusion; (70–90 min) Step potential (limiting case $a \to 0$, $V_0 \to \infty$): $T = 4k_1k_2/(k_1+k_2)^2$, $R = (k_1-k_2)^2/(k_1+k_2)^2$; above-barrier reflection (quantum effect with no classical analogue); energy dependence of $T$; time delay; phase of reflection coefficient; Hartman effect.

**Core learning outcomes:**

- Apply boundary conditions to derive $T$ and $R$ for a rectangular barrier.
- Derive the tunnelling transmission coefficient and the Gamow factor.
- Explain above-barrier reflection as a quantum effect with no classical analogue.
- Identify resonance transmission conditions and their physical origin.

**Dual-track content:**

- *Physical/applications track:* STM tunnelling current; alpha-decay Gamow factor; nuclear quantum tunnelling in fusion; Josephson junction; resonant tunnelling diode.
- *Mathematical track:* Transfer matrix method; S-matrix; unitarity $T + R = 1$ from probability current conservation; analytic structure of $T(E)$ and its poles (resonances).

**Tier differentiation:** HS: "a quantum particle can pass through a wall — that's tunnelling." Beg. UG: rectangular barrier $T$ and $R$; tunnelling formula; step potential. Adv. UG: transfer matrix; resonance condition; Gamow factor; Hartman effect. MSc: S-matrix; analytic continuation; poles and resonances; time delay. PhD: Breit–Wigner resonances; complex scaling; multi-channel scattering; Feshbach resonances.

**Assessment:** Set A — $T$ and $R$ for rectangular barrier at given $E$ and $V_0$; verify $T+R = 1$; step potential. Set B — transfer matrix derivation; thin-barrier $T \approx e^{-2\kappa a}$; resonance condition. Projects: (simple) $T(E)$ plotter for rectangular barrier: resonances and tunnelling regime; (complex) STM tunnelling current vs. tip–sample distance: exponential decay and work function.

## Module I.4 / L11: Scattering from a Potential Well

**90-min pacing:** (0–20 min) Finite square well: $V(x) = -V_0 < 0$ for $|x| < a$, $V = 0$ outside; bound states ($E < 0$) vs. scattering states ($E > 0$); even and odd solutions from parity; (20–50 min) Bound states: inside $k = \sqrt{2m(E + V_0)}/\hbar$; outside $\kappa = \sqrt{-2mE}/\hbar$; transcendental equations: $k\tan(ka) = \kappa$ (even), $-k\cot(ka) = \kappa$ (odd); graphical solution; minimum well depth for first bound state $V_0^{\min} = \pi^2\hbar^2/8ma^2$; number of bound states $\approx \sqrt{2mV_0} \cdot 2a/\pi\hbar$; (50–70 min) Scattering states ($E > 0$): transmission resonances at $k_{\text{in}}a = n\pi/2$ (Ramsauer–Townsend effect); $T = 1$ at resonance (perfect transmission); physical interpretation: standing wave inside well; comparison with classical case; (70–90 min) Ramsauer–Townsend effect in rare-gas electron scattering: historical significance; comparison of 1D well with 3D spherically symmetric well; zero-energy scattering: scattering length $a_s$; $T \to 1$ for $a_s \to 0$; connection to Feshbach resonances in cold atom physics; unitarity limit.

**Core learning outcomes:**

- Solve the finite square well for both bound and scattering states.
- Derive the transcendental equations graphically and interpret their solutions.
- Explain transmission resonances and the Ramsauer–Townsend effect.
- Define scattering length and relate it to the Feshbach resonance.

**Dual-track content:**

- *Physical track:* Ramsauer–Townsend effect as the first evidence for wave nature of electrons; nuclear well models; cold-atom Feshbach resonances; unitarity limit in scattering.
- *Mathematical track:* Transcendental equations and graphical analysis; analytic structure of $T(E)$; scattering length from the zero-energy limit of the S-matrix; Levinson's theorem: $\delta(0) - \delta(\infty) = N_b\pi$.

**Tier differentiation:** HS: finite well as a better model than infinite well; why bound states

have limited energy. Beg. UG: even/odd bound states; graphical solution; transmission resonances. Adv. UG: minimum well depth; Ramsauer–Townsend; scattering length. MSc: Levinson's theorem; Feshbach resonances; connection to cold-atom physics. PhD: multi-channel scattering; effective range expansion; universality at unitarity.

**Assessment:** Set A — bound state count for given $V_0, a$; graphical transcendental solution; resonance $E$. Set B — derive transcendental equations from BCs; Levinson's theorem statement and example. Projects: (simple) bound state energies vs. $V_0$ for finite well: emergence of new states; (complex) Ramsauer–Townsend: $T(E)$ for Xe modelled as a finite spherical well.

## Module I.4 / L12: Delta-Function Potential and WKB Tunnelling

**90-min pacing:** (0–20 min) The delta-function potential $V(x) = -\alpha\delta(x)$; integration of Schrödinger equation across $x = 0$: discontinuity in $\psi'$, continuity of $\psi$; condition $\psi'(0^+) - \psi'(0^-) = -2m\alpha\psi(0)/\hbar^2$; (20–45 min) Bound state ($E < 0$): $\psi = Ae^{-\kappa|x|}$ with $\kappa = m\alpha/\hbar^2$; $E_b = -m\alpha^2/2\hbar^2$ (unique bound state for any $\alpha > 0$); scattering states: $T = 1/(1 + m\alpha^2/2\hbar^2 E)$; $R = 1 - T$; double delta-function: two wells at $\pm a$; bonding and antibonding states; connection to molecular bonding; (45–65 min) WKB approximation reviewed: $\psi \approx C/\sqrt{p(x)}\exp\left(\pm\frac{i}{\hbar}\int p(x)dx\right)$ in classically allowed regions; $p(x) = \sqrt{2m(E - V(x))}$; connection formulae across turning points; Bohr–Sommerfeld quantisation $\oint p\,dx = 2\pi\hbar(n + \frac{1}{2})$; (65–90 min) WKB tunnelling: barrier penetration factor $T \approx \exp\left(-\frac{2}{\hbar}\int_{x_1}^{x_2}\sqrt{2m(V - E)}\,dx\right)$; Gamow factor for Coulomb barrier: $G = \pi Z_1 Z_2 e^2/\hbar v$; alpha decay rate $\Gamma = \nu_0 T$ with attempt frequency $\nu_0$; Geiger–Nuttall law; WKB for bound states in a Morse potential; limitations at turning points.

**Core learning outcomes:**

- Solve the delta-function potential for bound and scattering states.
- Derive the WKB wavefunction and Bohr–Sommerfeld quantisation.
- Compute the WKB tunnelling factor for a general barrier.
- Apply the Gamow factor to alpha decay and derive the Geiger–Nuttall law.

**Dual-track content:**

- *Physical track:* Delta potential as a model for molecular bonding ($H_2^+$); WKB tunnelling in alpha decay, proton–proton fusion in the Sun; Geiger–Nuttall law as a triumph of WKB; enzyme catalysis via proton tunnelling.
- *Mathematical track:* Distributional derivative and delta-function BCs; WKB as the leading-order term in an $\hbar$ expansion; connection formulae from Airy functions; uniform approximation near turning points.

**Tier differentiation:** HS: the delta potential as the thinnest possible wall; WKB as "tunnelling with an exponential rule." Beg. UG: delta-function bound state and $T$; WKB quantisation for HO; Gamow factor formula. Adv. UG: double delta-function bonding/antibonding; WKB connection formulae; Geiger–Nuttall. MSc: Airy function connection formulae; Morse potential WKB levels; uniform approximation. PhD: Instanton as imaginary-time WKB; exact WKB; resurgence; Gamow vectors and complex scaling.

**Assessment:** Set A — delta-function $E_b$ and $T$; WKB levels for harmonic oscillator; Gamow factor for $^{238}$U. Set B — double delta-function bonding/antibonding states; WKB connection formula derivation. Projects: (simple) WKB energy levels for Morse potential vs. exact: anharmonicity; (complex) Geiger–Nuttall law: alpha decay rate vs. $Z$ and $E_\alpha$ for actinide series.

### Assessment Strategy

***Per-Lecture Assessment Bundle*** *(every lecture): concept check (10–15 min, 5–8 short conceptual questions); Problem Set A (simple, 6–10 computational exercises); Problem Set B (complex, 3–6 multi-step derivations and proofs); mini-project simple (1–3 hours); mini-project complex (5–10 hours, open-ended).*

**Course-level grading template:**

- Weekly homework (best 9/12): 40%
- System synthesis project (one chosen system, deep study): 35%
- Final oral/whiteboard exam: 25%

Mixed cohorts: HS/UG → Set A + simple projects; MSc/PhD → Set B + complex projects.

### Reading Strategy

**Track A — Core textbooks:** Sakurai & Napolitano, *Modern Quantum Mechanics* Ch. 2 (HO, hydrogen), 4 (Zeeman/Stark), 5 (PT); Griffiths, *Introduction to Quantum Mechanics* Ch. 2–6 (all systems).

**Track B — Support:** Cohen-Tannoudji, Diu & Laloë, *Quantum Mechanics* Vol. II (spin–orbit, Stark, scattering); Atkins & Friedman, *Molecular Quantum Mechanics* (Zeeman, hydrogen, spectroscopy).

**Track C — Specialised:** Allen & Eberly, *Optical Resonance and Two-Level Atoms* (Rabi model); Landau & Lifshitz, *Quantum Mechanics* §23 (delta potential), §50 (WKB), §113 (Stark); Holstein, Am. J. Phys. **57**, 714 (1989) — Thomas precession; Gamow, Z. Phys. **51**, 204 (1928) — original tunnelling calculation.

## 1.5 Module I.5 — Introductory Physical Systems (Lecture Menu)

**Series I / Module I.5** *Introductory Physical Systems — Lecture Menu*

**Format:** 12 stand-alone lecture topics at the level of a first quantum mechanics course. Each topic is self-contained and applies the formalism of Modules I.1–I.3 to a canonical physical system. Instructors may select any subset as worked-example lectures or problem classes. Full expansion to the standard 90-minute format (dual-track content, five-tier pedagogy, Problem Sets A & B, mini-projects) is available on request.

**Recommended audience:** Undergraduate students during or immediately after Module I.3. Prerequisites: Modules I.1–I.3.

### Module Overview

| Lec. | Title | Key Concepts / Theme |
|---|---|---|
| L01 | The Finite Square Well | Bound state counting; graphical transcendental; parity |
| L02 | The Quantum Harmonic Oscillator in 1D (Wave Mechanics) | Hermite polynomials; recursion; zero-point; wave functions |
| L03 | The Rigid Rotor | Spherical harmonics; moment of inertia; rotational spectra |
| L04 | The Hydrogen-Like Ion | $Z_{\text{eff}}$; scaling of $E_n$; radial densities; periodic table |
| L05 | The Particle on a Ring | $e^{im\phi}$; angular momentum quantisation; Aharonov–Bohm |
| L06 | The Double-Well Potential | Tunnelling splitting; symmetric/antisymmetric; inversion in $NH_3$ |
| L07 | The Kronig–Penney Model | Periodic $\delta$-function array; band structure; Bloch waves |
| L08 | The Helium Atom | Two-electron problem; variational $Z_{\text{eff}}$; exchange |
| L09 | The $H_2$ Molecule (Simple LCAO) | Bonding/antibonding; overlap integral; dissociation curve |
| L10 | The Square Barrier and Tunnel Diode | Resonant tunnelling; $T(E)$ oscillations; device physics |
| L11 | The Linear Stark Effect in Hydrogen | Degenerate PT; $n = 2$ level splitting; parabolic eigenstates |
| L12 | The 1D Dirac Comb and Band Gaps | Bloch's theorem applied; dispersion relation; effective mass |

### Lecture Titles and Scope

**L01 — The Finite Square Well**

The finite square well $V(x) = -V_0$ for $|x| < a$ as the prototype for quantum confinement with finite barriers. Even and odd bound-state solutions; transcendental equations and their graphical solution; minimum well depth for the first bound state; dependence of bound-state count on $V_0$ and $a$; penetration of the wavefunction into classically forbidden regions; comparison with the infinite square well; application to quantum dots, nuclear potential wells, and optical waveguides.

**L02 — The Quantum Harmonic Oscillator in 1D (Wave-Mechanics Approach)**

Schrödinger equation in dimensionless units; asymptotic behaviour at large $|x|$; power-series (Frobenius) solution; Hermite polynomials $H_n(\xi)$; termination condition and energy quantisation $E_n = (n + \frac{1}{2})\hbar\omega$; normalised wavefunctions $\psi_n(x) = N_n H_n(\alpha x) e^{-\alpha^2 x^2/2}$; parity; node theorem; zero-point energy; comparison with classical probability density; connection to the

ladder-operator treatment of Module I.4-L03.

**L03 — The Rigid Rotor**

Quantum rigid rotor as a particle constrained to a sphere of radius $R$; Hamiltonian $\hat{H} = \hat{\mathbf{L}}^2/2I$ with moment of inertia $I = \mu R^2$; eigenfunctions $Y_l^m(\theta, \phi)$ and eigenvalues $E_l = \hbar^2 l(l+1)/2I$; degeneracy $2l + 1$; rotational constant $B = \hbar^2/2I$; electric dipole selection rule $\Delta l = \pm 1$; microwave rotational spectrum: equally spaced lines at $\bar{\nu} = 2B(l+1)$; determination of bond lengths; centrifugal distortion correction; connection to diatomic molecular spectroscopy.

**L04 — Hydrogen-Like Ions**

Generalisation of hydrogen to nuclear charge $Z$: $\hat{H} = -\nabla^2/2 - Z/r$; scaling $E_{nl} \to Z^2 E_n^{\mathrm{H}}$ and $\langle r \rangle \to \langle r \rangle^{\mathrm{H}}/Z$; radial wavefunctions for $He^+$, $Li^{2+}$, and general $Z$; connection to Slater screening and $Z_{\mathrm{eff}}$ in many-electron atoms; ionisation energies across the periodic table; X-ray emission energies (Moseley's law); positronium ($\mu \to m_e/2$) and muonic hydrogen ($\mu \to m_\mu$); reduced mass correction and its significance.

**L05 — The Particle on a Ring**

Particle confined to a 1D ring of radius $R$; Hamiltonian $\hat{H} = -(\hbar^2/2mR^2)\partial^2/\partial\phi^2$; periodic boundary conditions; eigenfunctions $e^{im\phi}/\sqrt{2\pi}$; quantised angular momentum $\hbar m$, $m \in \mathbb{Z}$; doubly degenerate levels $E_m = \hbar^2 m^2/2I$ ($m \neq 0$); application to $\pi$-electrons in benzene (Hückel analogy); Aharonov–Bohm effect on a ring: flux-dependent energy levels $E_m \propto (m - \Phi/\Phi_0)^2$; persistent current in mesoscopic rings.

**L06 — The Double-Well Potential and Molecular Inversion**

Symmetric double well $V(x) = \frac{\lambda}{4}(x^2 - a^2)^2$; qualitative picture: two nearly degenerate levels split by tunnelling; symmetric (+) and antisymmetric (−) combinations of localised states; tunnelling splitting $\Delta E \propto e^{-\gamma}$; inversion of $NH_3$: experimental microwave spectrum at 23.79 GHz; application to maser principle; asymmetric double well and localisation; connection to instanton picture (Module II.2 L08); two-level system Hamiltonian $\hat{H} = -\frac{\Delta}{2}\hat{\sigma}_z - \frac{\epsilon}{2}\hat{\sigma}_x$.

**L07 — The Kronig–Penney Model**

1D periodic array of $\delta$-function barriers $V(x) = \sum_n V_0\,\delta(x - na)$; Bloch's theorem applied; transfer matrix for one period; transcendental dispersion relation $\cos(ka) = \cos(Ka) + \frac{mV_0}{\hbar^2 K}\sin(Ka)$; graphical solution: allowed bands where $|\cos ka| \leq 1$; band gaps at $ka = n\pi$; effective mass near band edges; weak- and strong-coupling limits; connection to the nearly-free electron and tight-binding models; relevance to semiconductor superlattices.

**L08 — The Helium Atom**

Two-electron Hamiltonian $\hat{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - 2/r_1 - 2/r_2 + 1/r_{12}$; zeroth-order product wavefunction; first-order perturbation theory: $\langle 1/r_{12} \rangle = 5Z/8$; variational treatment with $Z_{\mathrm{eff}}$; shielding constant; exchange integral; para- and ortho-helium (singlet and triplet); comparison of perturbative, variational, and experimental energies; connection to Slater–Condon rules and the many-body problem.

**L09 — The $H_2$ Molecule: Simple LCAO-MO**

$H_2$ as the simplest two-electron molecule; Born–Oppenheimer PES; LCAO-MO ansatz with $1s$ hydrogen orbitals; bonding $\sigma_g$ and antibonding $\sigma_u^*$ MOs; overlap $S$, Coulomb $\alpha$, and resonance $\beta$ integrals; variational MO energies; bond order; optimise bond length and compare to experiment; Heitler–London VB wavefunction; comparison of MO and VB predictions for dissociation; correlation energy.

**L10 — The Square Barrier and Resonant Tunnel Diode**

Double-barrier heterostructure $V(x)$: two rectangular barriers of height $V_0$ and width $b$ separated by a well of width $a$; transmission $T(E)$ via the transfer matrix; resonant transmission $T = 1$ when quasi-bound state energy $E_n$ lies below $V_0$; resonance width $\Gamma$; current–voltage ($I$–$V$) characteristic of a resonant tunnel diode; peak-to-valley ratio; negative differential resistance; GaAs/AlAs heterostructure parameters; applications in high-frequency oscillators.

**L11 — The Linear Stark Effect in Hydrogen**

Electric field perturbation $\hat{H}' = -eEz$ applied to the degenerate $n = 2$ level of hydrogen ($2s$, $2p_{-1}$, $2p_0$, $2p_1$); selection rules from parity and $\Delta m = 0$; only $\langle 2s|z|2p_0\rangle = -3ea_0$ is non-zero; degenerate perturbation theory: diagonalise the $4 \times 4$ perturbation matrix; four eigenstates with energies $\pm 3eEa_0$ and two unshifted states; physical picture: linear combinations with permanent electric dipole moment; parabolic eigenstates $|n_1, n_2, m\rangle$; comparison with the quadratic (second-order) Stark effect of the $n = 1$ ground state.

**L12 — The 1D Dirac Comb and Band Structure**

Dirac comb: infinite periodic array of $\delta$-function wells $V(x) = -\alpha \sum_n \delta(x - na)$; exact solution using Bloch's theorem and transfer-matrix matching; dispersion relation $\cos(ka) = \cos(Ka) - \frac{m\alpha}{\hbar^2 K}\sin(Ka)$; energy bands and gaps in the reduced zone scheme; effective mass $m^*$ at band edges; density of states in 1D; dependence of band gaps on $\alpha$: strong-coupling (tight-binding) and weak-coupling (nearly-free electron) limits; connection to the Kronig–Penney model (L07) and Bloch electron (Module I.4-L02).

***Module I.5 Note:*** *These lecture topics are pitched at the level of a standard first-year quantum mechanics course (advanced undergraduate). Each can be taught as a standalone 90-minute worked-example class or problem session. Full expansion to the standard syllabus format (90-min pacing, dual-track content, five-tier pedagogy, Problem Sets A & B, simple and complex mini-projects) is available on request.*

## 2 Series II — Advanced Quantum Mechanics

**Series II Overview**

**Prerequisite:** Completion of Series I or equivalent.
**Duration:** Each module comprises 12 lectures × 90 minutes.
**Core aim:** Systematically deepen the formalism and connect quantum mechanics to modern research areas.

| Module | Title | Lectures |
|---|---|---|
| II.1 | Ground-State Approximation Methods | 12 |
| II.2 | Path Integrals in Quantum Mechanics | 12 |
| II.3 | Theory of Angular Momentum | 12 |
| II.4 | Second Quantisation and Introductory Many-Body Theory | 12 |
| II.5 | Non-Equilibrium Quantum Many-Body Green's Functions | 12 |
| II.6 | Light–Matter Interaction, Linear Response, and Spectroscopy | 12 |
| II.7 | Quantum Statistical Mechanics in Equilibrium | 12 |
| II.8 | Advanced Physical Systems (Lecture Menu) | 12 |
| II.9 | Quantum Information and Quantum Logic | 12 |
| II.10 | Signature Systems of Quantum Information | 12 |
| II.11 | The Tenfold Way | 12 |
| II.12 | Signature Systems of the Tenfold Way | 12 |
| | **Total** | **144** |

## 2.1 Module II.1 — Ground-State Approximation Methods

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Variational Principle | Ritz method; Rayleigh quotient; upper bound; trial states |
| L02 | Linear Variational Method | Secular determinant; basis expansion; Hückel MO theory |
| L03 | Hartree and Hartree–Fock | Mean-field; Fock operator; self-consistency; Koopman |
| L04 | Coupled-Cluster Methods | $e^{\hat{T}}$ ansatz; CCSD(T); size consistency |
| L05 | Quantum Monte Carlo I | VMC; importance sampling; Metropolis algorithm |
| L06 | Quantum Monte Carlo II | DMC; fixed-node approximation; sign problem |
| L07 | DMRG | Entanglement area law; SVD; MPS; renormalisation group |
| L08 | Tensor Networks | MPS, MERA; contraction algorithms; bond dimension |
| L09 | Natural Orbitals | 1-RDM; NOONs; multi-configurational methods; PNOs |
| L10 | Multi-Reference Methods | CASSCF; MRCI; NEVPT2; active space selection |
| L11 | Benchmark Systems | $H_2$, LiH, $N_2$; accuracy hierarchy; error budget |
| L12 | Method Selection in Practice | Cost scaling; applicability regimes; crossover criteria |

### Five-Tier Layered Pedagogy

| Tier | Cohort | Characteristic content |
|---|---|---|
| Tier 1 | HS | Conceptual motivation; energy minimisation intuition; analogy with curve fitting |
| Tier 2 | Beg. UG | Definitions; matrix secular equations; basic HF; worked 2-electron examples |
| Tier 3 | Adv. UG | Rigorous derivations; convergence proofs; algorithm implementation; error analysis |
| Tier 4 | MSc | Advanced methods (CC, QMC, DMRG); diagrammatic techniques; active space design |
| Tier 5 | PhD | Frontiers: stochastic CC, FCIQMC, tensor hypercontraction, cost-scaling optimisation |

### Lecture-by-Lecture Syllabi

**Module II.1 / L01: The Variational Principle**

**90-min pacing:** (0–20 min) Motivation: exact diagonalisation is exponentially hard; variational methods as the first systematic route around this wall; statement of the variational theorem: for any normalised trial state $|\phi\rangle$, $\langle\phi|\hat{H}|\phi\rangle \geq E_0$, with equality iff $|\phi\rangle = |0\rangle$; (20–45 min) Proof from spectral decomposition; Rayleigh quotient $R[\phi] = \langle\phi|\hat{H}|\phi\rangle/\langle\phi|\phi\rangle$; optimisation as a constrained minimisation; Lagrange multiplier and the stationarity condition $\delta R = 0 \Leftrightarrow \hat{H}\,|\phi\rangle = E\,|\phi\rangle$; (45–70 min) Excited states: Hylleraas–Undheim–MacDonald theorem (eigenvalues of a truncated Hamiltonian interlace the exact eigenvalues); upper bounds on excited-state energies; Temple's lower bound; (70–90 min) Practical trial states: Gaussian-type orbitals; Padé approximants; neural-network wavefunctions (VMC preview); error analysis: how a variational error $\delta E_{\text{var}}$ in the energy corresponds to a first-order error in the wavefunction.

**Core learning outcomes:**

- State and prove the variational theorem from spectral decomposition.
- Compute the Rayleigh quotient for a trial wavefunction and minimise it analytically.
- State the Hylleraas–Undheim theorem and apply it to excited-state bounds.
- Explain why a second-order error in the wavefunction gives only a fourth-order energy error.

**Dual-track content:**

- *Physical/intuitive track:* Energy minimisation as "best guess in a restricted class of states"; analogy with curve-fitting subject to a norm constraint; Gaussian orbitals as computationally convenient trial states.
- *Mathematical track:* Ritz–Galerkin theory; min-max characterisation of eigenvalues; Temple's lower bound; error propagation from wavefunction to energy.

**Tier differentiation:** HS: "the ground-state energy is the lowest possible expectation value of the Hamiltonian" — variational principle as quantum minimisation. Beg. UG: compute $\langle\hat{H}\rangle$ for Gaussian trial state in a harmonic potential; optimise over the width parameter $\alpha$. Adv. UG: full proof of variational theorem; Hylleraas–Undheim theorem; Temple lower bound. MSc: error analysis; Padé and Hylleraas-type trial functions; multi-parameter optimisation with gradient and Hessian. PhD: neural-network wavefunctions; stochastic reconfiguration; natural gradient descent; connection to variational Monte Carlo.

**Assessment:** Set A — Rayleigh quotient for Gaussian trial in hydrogen; optimise $\alpha$; bound

on $E_0$. Set B — prove variational theorem; derive stationarity condition; Hylleraas–Undheim theorem proof. Projects: (simple) variational parameter scan for helium with $Z_{\text{eff}}$; (complex) multi-parameter Hylleraas wavefunction for helium and convergence to exact energy.

## Module II.1 / L02: The Linear Variational Method

**90-min pacing:** (0–20 min) Linear ansatz: $|\phi\rangle = \sum_{k=1}^{N} c_k |\chi_k\rangle$ for a fixed basis $\{|\chi_k\rangle\}$; Rayleigh quotient in matrix form; Hamiltonian matrix $H_{kl} = \langle\chi_k|\hat{H}|\chi_l\rangle$ and overlap matrix $S_{kl} = \langle\chi_k|\chi_l\rangle$; (20–45 min) Secular (generalised eigenvalue) equation $\mathbf{Hc} = E\mathbf{Sc}$; Cholesky decomposition of $\mathbf{S}$ to convert to standard eigenvalue problem; $N$ variational eigenvalues as upper bounds to exact eigenvalues (Hylleraas–Undheim); orthonormalisation of basis via Löwdin symmetric or canonical procedures; (45–65 min) Hückel molecular orbital theory as a paradigm application: $\pi$ system of ethylene, butadiene, benzene; Hückel parameters $\alpha$, $\beta$; secular determinant; MO energies and coefficients; aromaticity from Hückel rule $4n+2$; (65–90 min) Extended Hückel and limitations; basis set convergence in general; non-orthogonal basis subtleties; condition number of $\mathbf{S}$; linear dependence and near-linear dependence; canonical truncation.

**Core learning outcomes:**

- Set up and solve the secular equation for a given basis.
- Apply Hückel MO theory to simple $\pi$ systems and interpret results.
- Orthonormalise a non-orthogonal basis via Löwdin or canonical procedures.
- Diagnose and handle linear dependence in a basis set.

**Dual-track content:**

- *Chemical/applied track:* Hückel MO energies and their relation to reactivity; aromatic stabilisation energy; frontier MO theory (HOMO/LUMO); extended Hückel.
- *Mathematical track:* Generalised eigenvalue problem; Cholesky and Löwdin orthogonalisation; condition number; linear dependence detection via SVD of overlap matrix.

**Tier differentiation:** HS: Hückel MO theory for benzene — why $\pi$ electrons delocalise and stabilise the ring. Beg. UG: set up and solve $2 \times 2$ secular equation; Hückel for ethylene and allyl. Adv. UG: Hylleraas–Undheim theorem for variational eigenvalues; Löwdin orthogonalisation derivation. MSc: condition number analysis; canonical truncation; extended Hückel parameterisation. PhD: connection to finite-element Galerkin methods; adaptively contracted basis sets; DIIS convergence acceleration.

**Assessment:** Set A — secular equation for particle-in-a-box with polynomial basis; Hückel MO diagram for butadiene and benzene. Set B — prove Hylleraas–Undheim theorem for linear variational method; Löwdin orthogonalisation derivation; condition number of Gaussian overlap matrix. Projects: (simple) Hückel MO solver for arbitrary planar $\pi$ systems; (complex) Löwdin orthogonalisation vs. canonical truncation: numerical comparison for near-linearly-dependent Gaussian basis.

## Module II.1 / L03: Hartree and Hartree–Fock Theory

**90-min pacing:** (0–20 min) The $N$-electron problem; exponential wall; Hartree product ansatz $\Psi = \prod_i \phi_i(\mathbf{r}_i)$; variational optimisation gives Hartree equations $[\hat{h} + \hat{v}_i^{\text{H}}]\phi_i = \varepsilon_i\phi_i$; Hartree potential as classical mean field; self-consistency loop; (20–50 min) Antisymmetry requirement: Slater determinant; Hartree–Fock energy from variational principle on Slater determinant; Fock operator $\hat{F}_i = \hat{h}_i + \hat{J}_i - \hat{K}_i$; Coulomb $\hat{J}$ and exchange $\hat{K}$ operators

defined and compared; HF equations $\hat{F}\phi_i = \varepsilon_i\phi_i$; canonical vs. localised orbitals; (50–70 min) Self-consistent field (SCF) algorithm: Roothaan–Hall equations $\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}$ in an AO basis; density matrix $\mathbf{P} = 2\mathbf{C}_{\text{occ}}\mathbf{C}_{\text{occ}}^{\dagger}$; convergence and DIIS; Koopman's theorem ($-\varepsilon_i \approx$ ionisation energy); (70–90 min) Properties of HF: size consistency; variational; missing correlation; correlation energy $E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}$; restricted (RHF), unrestricted (UHF), and restricted-open-shell (ROHF); spin contamination in UHF.

**Core learning outcomes:**

- Derive the Hartree and Hartree–Fock equations from a variational principle.
- Define the Coulomb and exchange operators and state their physical meaning.
- Describe the SCF algorithm and the Roothaan–Hall equations.
- Apply Koopman's theorem and state the limitations of HF theory.

**Dual-track content:**

- *Physical/chemical track:* Mean-field picture; exchange as a purely quantum effect from antisymmetry; HF as the starting point for all post-HF methods; correlation hole; Fermi hole vs. Coulomb hole.
- *Mathematical/formal track:* Functional derivative of the HF energy; Brillouin's theorem; Koopman's theorem derivation; relation of HF to DFT (Kohn–Sham as an orbital-optimised density-functional analogue).

**Tier differentiation:** HS: "each electron moves in the average field of all others" — mean field as an approximation to avoid solving a coupled $3N$-dimensional problem exactly. Beg. UG: Hartree equations for helium; two-electron integrals; SCF iteration by hand. Adv. UG: HF equations from variational principle; Brillouin's theorem; Koopman's theorem proof. MSc: Roothaan–Hall in AO basis; DIIS convergence; symmetry breaking in UHF; RHF vs. UHF for $H_2$. PhD: Brueckner orbitals; natural orbitals from 1-RDM; HF as saddle point of full CI energy; orbital-optimised MP2.

**Assessment:** Set A — HF energy of helium by direct integration; Koopman's theorem for first ionisation; identify Coulomb vs. exchange terms in a given two-electron integral. Set B — derive HF equations from $\delta E_{\text{HF}} = 0$; prove Brillouin's theorem; show SCF density matrix idempotency $\mathbf{P}^2 = \mathbf{P}$ at convergence. Projects: (simple) HF SCF loop for $H_2$ in a minimal (STO-3G) basis with convergence plot; (complex) RHF vs. UHF dissociation curve for $H_2$: spin contamination and correlation energy.

## Module II.1 / L04: Coupled-Cluster Methods

**90-min pacing:** (0–20 min) Limitations of truncated CI: size inconsistency; Brillouin–Wigner vs. Rayleigh–Schrödinger CI; size consistency defined; the exponential ansatz: $|\Psi_{\text{CC}}\rangle = e^{\hat{T}}|\Phi_0\rangle$ with cluster operator $\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots$; why the exponential structure guarantees size consistency; (20–50 min) CC amplitude equations: project $\langle\Phi_i^a|\, e^{-\hat{T}}\hat{H}e^{\hat{T}}\,|\Phi_0\rangle = 0$ (doubles) and similarly for singles; Baker–Campbell–Hausdorff expansion of $e^{-\hat{T}}\hat{H}e^{\hat{T}}$ terminates at quartic in $\hat{T}$ (for two-body $\hat{H}$); CCSD: $\hat{T} = \hat{T}_1 + \hat{T}_2$; $\mathcal{O}(N^6)$ scaling; (50–70 min) Perturbative triples: CCSD(T) — the "gold standard"; $\mathcal{O}(N^7)$ scaling; relative cost and accuracy; equation-of-motion CC (EOM-CC) for excited states; CCSD(T)-F12 for basis set convergence acceleration; (70–90 min) Diagrammatic representation of CC equations: Goldstone diagrams; connected vs. disconnected; Wick's theorem applied to CC; local correlation methods: PNO-CCSD(T), DLPNO-CCSD(T) and linear scaling.

**Core learning outcomes:**

- Explain why the exponential ansatz ensures size consistency.
- Derive the CCSD amplitude equations via BCH expansion.
- State the computational scaling of CCSD and CCSD(T) and place them in the accuracy hierarchy.
- Describe PNO-based local correlation and its scaling advantage.

**Dual-track content:**

- *Computational chemistry track:* CCSD(T) as the de facto gold standard; benchmark accuracy for thermochemistry (sub-kJ/mol); EOM-CC for spectra; DLPNO-CCSD(T) for large molecules; software packages (ORCA, MOLPRO, CFOUR).
- *Many-body theory track:* Diagrammatic CC; linked-cluster theorem; CC as a resummation of MBPT; connection to coupled-cluster Green's functions (CCGF).

**Tier differentiation:** HS: "instead of writing the wavefunction as a sum, write it as an exponential" — why this small change guarantees that two well-separated molecules have independent energies. Beg. UG: CCD amplitude equations in schematic form; size consistency demonstration for He+He. Adv. UG: BCH expansion to quartic; derive CCSD equations; perturbative triples derivation. MSc: EOM-CCSD; CCSD(T)-F12; PNO-CCSD(T) and local correlation ideas. PhD: Diagrammatic CC; CCGF and ionisation potentials; transcorrelated CC; FCIQMC-tailored CC.

**Assessment:** Set A — verify size consistency of CCSD for He dimer; CCD energy in schematic notation; rank CCSD(T) in the accuracy hierarchy. Set B — derive BCH expansion of $e^{-\hat{T}}\hat{H}e^{\hat{T}}$ to quartic; linked-cluster theorem. Projects: (simple) CCSD vs. FCI energy curve for $H_2$ as function of bond length; (complex) EOM-CCSD excitation energies for ethylene vs. TDDFT and experiment.

## Module II.1 / L05: Quantum Monte Carlo I — Variational QMC

**90-min pacing:** (0–20 min) Monte Carlo integration: basic idea; central limit theorem; error $\propto 1/\sqrt{N_{\text{samples}}}$; importance sampling to reduce variance; application to high-dimensional integrals (the QMC motivation); (20–45 min) Variational QMC (VMC): evaluate $E_{\text{var}} = \langle\Psi_T|\hat{H}|\Psi_T\rangle/\langle\Psi_T|\Psi_T\rangle$ stochastically; local energy $E_L(\mathbf{R}) = \hat{H}\Psi_T/\Psi_T$; variance minimisation vs. energy minimisation; the Metropolis–Hastings algorithm: proposal, acceptance ratio $A = \min(1, |\Psi_T(\mathbf{R}')|^2/|\Psi_T(\mathbf{R})|^2)$; detailed balance and ergodicity; (45–70 min) Trial wavefunction design: Slater–Jastrow form $\Psi_T = \mathcal{A}[\phi]\exp(J)$; Jastrow factor: electron–electron, electron–nuclear, three-body terms; cusp conditions: $\partial_r E_L$ finite requires $(\partial_r\Psi_T/\Psi_T)|_{r=0}$ to cancel the $1/r$ divergence ($s$-wave cusp: $a_{ee} = 1/2$ for unlike spin, $1/4$ for like); (70–90 min) Parameter optimisation: stochastic reconfiguration (SR) and the natural gradient; ADAM and gradient descent in stochastic setting; correlated sampling; VMC scaling: $\mathcal{O}(N^3)$–$\mathcal{O}(N^4)$ for determinant evaluation.

**Core learning outcomes:**

- Explain importance sampling and the Metropolis algorithm.
- Compute the local energy for a Slater–Jastrow trial wavefunction.
- Derive and apply the electron–electron cusp condition.
- Compare energy minimisation and variance minimisation for parameter optimisation.

**Dual-track content:**

- *Computational/algorithmic track:* Metropolis algorithm implementation; autocorrelation time and statistical efficiency; blocking analysis for error estimation; GPU acceleration;

practical VMC workflows.

- *Mathematical/stochastic track:* Markov chain theory; detailed balance; ergodicity; central limit theorem for correlated samples; stochastic reconfiguration as natural gradient descent on the manifold of quantum states.

**Tier differentiation:** HS: "roll the dice millions of times to estimate an integral in $3N$ dimensions" — Monte Carlo as a solution to the curse of dimensionality. Beg. UG: Metropolis walk for a harmonic oscillator; blocking analysis; variance estimate. Adv. UG: Slater–Jastrow VMC; cusp conditions derivation; variance minimisation. MSc: stochastic reconfiguration; natural gradient; correlated sampling for forces. PhD: neural-network Jastrow and backflow; transcorrelated VMC; ab initio QMC for solids.

**Assessment:** Set A — Metropolis walk for $|\psi_0|^2$ in HO; local energy for Gaussian trial; cusp check. Set B — derive electron cusp condition; prove detailed balance for Metropolis; variance minimisation stationarity condition. Projects: (simple) VMC for helium with one-parameter Jastrow; convergence with $N_{\text{samples}}$; (complex) stochastic reconfiguration optimisation of multi-parameter Jastrow for LiH.

## Module II.1 / L06: Quantum Monte Carlo II — Diffusion QMC

**90-min pacing:** (0–20 min) From VMC to exact: the diffusion QMC (DMC) idea; imaginary-time Schrödinger equation $-\partial_\tau \Psi = (\hat{H} - E_T)\Psi$; large-$\tau$ projection onto ground state; analogy with diffusion + branching process; Green's function $G(\mathbf{R}', \mathbf{R}; \tau)$ as the propagator; (20–45 min) DMC algorithm: walkers (configurations) undergo diffusion steps $\Delta\tau$; branching (replication/death) with weight $\exp[-(E_L(\mathbf{R}) - E_T)\Delta\tau]$; importance sampling with trial function: modified Green's function; drift velocity $\mathbf{F} = \nabla \ln |\Psi_T|$; guided diffusion; (45–65 min) The fermion sign problem: positive and negative walkers annihilate; fixed-node approximation: constrain walker crossings of nodal surface of $\Psi_T$; fixed-node energy as an upper bound; quality of nodes determines accuracy; release-node (unphysical) and its failure for electrons; (65–90 min) Pseudopotentials in DMC: non-locality and T-moves; relativistic effects; lattice regularised DMC; twist-averaged boundary conditions for periodic systems; scaling: $\mathcal{O}(N^3)$; DMC for solids — computational workflow and accuracy benchmarks.

**Core learning outcomes:**

- Derive the imaginary-time Schrödinger equation and explain ground-state projection.
- Describe the DMC algorithm with importance sampling and branching.
- Explain the sign problem and the fixed-node approximation.
- Assess the accuracy of DMC for atoms and molecules relative to CCSD(T).

**Dual-track content:**

- *Physical/computational track:* DMC as the most accurate practical method for large systems; solid-state applications (transition metal oxides, ice polymorphs); comparison with CCSD(T) for benchmark sets; practical workflow with CASINO or QMCPack.
- *Mathematical track:* Feynman–Kac formula; Green's function Monte Carlo; variational fixed-node theorem; node optimisation; relation to path-integral MC.

**Tier differentiation:** HS: "walkers explore space like diffusing particles, and the ones in low-energy regions survive" — projection onto the ground state as a survival-of-the-fittest process. Beg. UG: one-dimensional DMC for a particle in a box; branching and population control. Adv. UG: fixed-node theorem proof; importance sampling derivation; timestep bias analysis. MSc: T-moves for non-local pseudopotentials; twist-averaged boundary conditions; node optimisation. PhD: Feynman–Kac formula; lattice regularisation; backflow nodes;

auxiliary field QMC.

**Assessment:** Set A — 1D DMC for HO; branching weight calculation; timestep extrapolation. Set B — prove fixed-node variational theorem; derive importance-sampled Green's function. Projects: (simple) DMC for helium: fixed-node energy vs. VMC and exact; (complex) node quality study — HF vs. CASSCF nodes for $N_2$ at various bond lengths.

## Module II.1 / L07: Density Matrix Renormalisation Group

**90-min pacing:** (0–20 min) Exponential wall revisited: why the Hilbert space dimension is $d^N$ for a spin-$\frac{1}{2}$ chain; entanglement as the key to efficient representation; area law vs. volume law for ground states of gapped local Hamiltonians; entanglement entropy $S = -\mathrm{Tr}(\hat{\rho}_A \ln \hat{\rho}_A)$ and its area law $S \leq c \cdot L^{d-1}$ in $d$ spatial dimensions; (20–45 min) Schmidt decomposition and its cost: $|\Psi\rangle = \sum_\alpha \lambda_\alpha |\alpha\rangle_L |\alpha\rangle_R$; singular value decomposition (SVD); truncation to bond dimension $m$: approximation error bounded by discarded weights $\epsilon = \sum_{\alpha>m} \lambda_\alpha^2$; the renormalisation group idea: keep the $m$ most relevant states; (45–70 min) Matrix Product State (MPS) representation: $|\Psi\rangle = \sum_{\{s_i\}} \mathrm{Tr}[A^{s_1} A^{s_2} \cdots A^{s_N}] |s_1 \cdots s_N\rangle$; bond dimension $m$ as the key parameter; DMRG as variational optimisation over MPS; two-site DMRG algorithm: superblock Hamiltonian; Lanczos diagonalisation; SVD truncation; (70–90 min) DMRG in quantum chemistry (DMRG-SCF / DMRG-CASSCF): orbital ordering and entanglement; mutual information $I_{ij}$; active space beyond CAS(18,18); scaling $\mathcal{O}(m^3 N + m^2 N d^2)$; convergence with $m$ and comparison with CASSCF/MRCI.

**Core learning outcomes:**

- State the area law and explain why it justifies MPS representations.
- Perform a Schmidt decomposition and estimate the SVD truncation error.
- Describe the MPS ansatz and the two-site DMRG algorithm.
- Explain DMRG-CASSCF and its advantage over conventional CASSCF.

**Dual-track content:**

- *Quantum chemistry/applications track:* DMRG as a large-active-space method; orbital selection via mutual information; applications to transition metal complexes, $f$-electron systems, and spin chains; software: ITensor, CheMPS2, Block2.
- *Quantum information/mathematical track:* MPS as a variational class; area-law proof sketch (Hastings); entanglement spectrum; connection to topological order.

**Tier differentiation:** HS: "instead of keeping all $2^N$ amplitudes, keep only the most entangled ones" — DMRG as smart truncation guided by quantum information. Beg. UG: Schmidt decomposition for a 4-site chain; SVD truncation and error. Adv. UG: MPS canonical forms; two-site DMRG algorithm; area law statement. MSc: DMRG-CASSCF; orbital ordering optimisation; mutual information; convergence with $m$. PhD: DMRG as tensor network; time-evolving block decimation (TEBD); finite-temperature DMRG; DMRG for periodic systems.

**Assessment:** Set A — Schmidt decomposition for a 4-qubit state; truncation error estimate; MPS representation of a product state. Set B — prove area-law scaling for 1D gapped systems (Hastings, sketch); derive the variational condition in two-site DMRG. Projects: (simple) DMRG energy vs. bond dimension $m$ for Heisenberg chain; (complex) DMRG-CASSCF active space selection via mutual information for $Cr_2$ dimer.

## Module II.1 / L08: Tensor Networks

**90-min pacing:** (0–20 min) MPS as a special case of a general tensor network (TN); graphical notation: tensors as nodes, contracted indices as edges; review of MPS canonical forms (left/right canonical); why generalising beyond 1D requires new network geometries; (20–45 min) Tree tensor networks (TTN): branching structure; efficient contraction; multi-scale entanglement renormalisation ansatz (MERA): disentanglers and isometries; causal cone; critical systems and logarithmic entanglement; Projected Entangled Pair States (PEPS) for 2D systems: exponentially hard contraction; practical approximations (boundary MPS, Monte Carlo sampling of PEPS); (45–65 min) Contraction algorithms: optimal contraction order ($\mathcal{NP}$-hard in general); DMRG as MPS contraction; tensor train decomposition; Tucker decomposition and its connection to HOSVD; tensor hypercontraction (THC) in quantum chemistry: $\mathcal{O}(N^4) \to \mathcal{O}(N^2)$ for two-electron integrals; (65–90 min) Tensor network methods in machine learning: tensor trains as efficient neural network layers; Born machine; quantum-inspired ML; connection to renormalisation group in field theory; overview of software: ITensor (Julia), TensorFlow Quantum, Quimb.

**Core learning outcomes:**

- Read and draw tensor network diagrams and identify contraction patterns.
- Distinguish MPS, TTN, MERA, and PEPS in terms of entanglement structure and cost.
- Explain tensor hypercontraction and its computational advantage.
- Connect tensor networks to renormalisation group and machine learning.

**Dual-track content:**

- *Quantum chemistry/algorithms track:* THC for two-electron integrals; PEPS for 2D correlated materials; tensor-network quantum embedding; software workflows.
- *Quantum information/mathematical track:* Entanglement structure of different TN geometries; holographic duality (AdS/MERA); MERA as a discrete realisation of AdS/CFT; complexity of TN contraction.

**Tier differentiation:** HS: "data stored as a network of multiplications rather than one giant array" — tensor networks as efficient compression of high-dimensional data. Beg. UG: matrix product of three matrices as a 3-tensor MPS; DMRG as 1D TN. Adv. UG: MERA causal cone; PEPS contraction complexity; Tucker decomposition. MSc: THC for electron repulsion integrals; Born machine; quantum-inspired ML. PhD: AdS/MERA connection; PEPS for frustrated magnets; real-space RG from TN perspective.

**Assessment:** Set A — draw and evaluate simple TN contractions (matrix chain, star network); MPS representation of a GHZ state; MERA causal cone width. Set B — derive SVD-based truncation error in TTN; THC factorisation of a $2\times2$ ERI block. Projects: (simple) MPS compression of a random state: truncation error vs. bond dimension; (complex) MERA ansatz for Ising critical point: scaling exponents from entanglement spectrum.

## Module II.1 / L09: Natural Orbitals and the One-Particle Density Matrix

**90-min pacing:** (0–20 min) The one-particle reduced density matrix (1-RDM): $\gamma(\mathbf{x};\mathbf{x}') = N\int \Psi^*(\mathbf{x}',\mathbf{x}_2,\ldots)\Psi(\mathbf{x},\mathbf{x}_2,\ldots)d\mathbf{x}_2\cdots d\mathbf{x}_N$; one- and two-electron properties from the 1-RDM and 2-RDM; $N$-representability of the 1-RDM: necessary conditions (eigenvalues in $[0,1]$, trace $= N$); (20–45 min) Natural orbitals (NOs): eigenfunctions of the 1-RDM; occupation numbers $n_i \in [0,1]$ (NOONs); $n_i = 0$ or 1 iff single Slater determinant; fractional occupations as a measure of correlation; natural orbital expansion of $\Psi$ converges faster than canonical MO expansion; analogy with PCA / SVD of the wavefunction; (45–65 min) Natu-

ral orbitals in practice: NOON-based active space selection; iterative NO-based CI (iterative MRCI); CASSCF natural orbitals; pair natural orbitals (PNO): biorthogonal pair functions defined from the 2-RDM; PNO-CCSD(T) and DLPNO-CCSD(T): local correlation with PNO truncation; (65–90 min) The 2-RDM and $N$-representability: P, Q, G conditions; direct variational optimisation of the 2-RDM (2-RDM methods); semidefinite programming (SDP) formulation; accuracy and scaling; antisymmetric geminal power (AGP) and generalised valence bond (GVB) as correlated pair wavefunctions.

**Core learning outcomes:**

- Define the 1-RDM and compute it for a two-electron wavefunction.
- Interpret NOONs as a measure of static correlation.
- Explain NOON-based active space selection and PNO local correlation.
- State the $N$-representability problem for the 2-RDM and the SDP approach.

**Dual-track content:**

- *Computational chemistry track:* NOONs as diagnostics for multireference character; DLPNO-CCSD(T) active space from PNOs; GVB for bond breaking; SDP-2RDM software (Maple QMC).
- *Mathematical/formal track:* 1-RDM as a Hermitian trace-class operator; Coleman's $N$-representability conditions; SDP duality; connection to quantum marginal problem.

**Tier differentiation:** HS: "occupation numbers tell you how many electrons are in each orbital" — NOONs of 1 or 0 mean no correlation; numbers in between signal strong correlation. Beg. UG: compute 1-RDM for He with two-configuration CI; diagonalise to get NOs; interpret NOONs. Adv. UG: proof that NOs diagonalise the 1-RDM; NOON-based active space selection criterion. MSc: PNO construction from 2-RDM; DLPNO truncation thresholds; $N$-representability P, Q, G conditions. PhD: SDP for 2-RDM; Coleman's theorem; connection to quantum marginal problem; v2RDM.

**Assessment:** Set A — compute NOONs for $H_2$ at stretched geometry; identify multireference character; compare natural orbital expansion rate with canonical MO expansion. Set B — prove that eigenfunctions of 1-RDM are NOs; derive the P condition for $N$-representability; PNO truncation error bound. Projects: (simple) NOON diagnostic: HF vs. FCI NOONs for $H_2$ dissociation; (complex) 2-RDM SDP for $H_2$: comparison with FCI and CCSD.

## Module II.1 / L10: Multi-Reference Methods

**90-min pacing:** (0–20 min) When single-reference fails: near-degeneracies, bond breaking, transition metal chemistry, conical intersections; static vs. dynamic correlation; multireference diagnostics: $T_1$ and $D_1$ amplitudes, NOONs, $\langle \hat{S}^2 \rangle$ deviation; the complete active space (CAS) concept: partitioning MOs into inactive (doubly occupied), active (variably occupied), and virtual (empty); (20–45 min) CASSCF: simultaneous optimisation of CI coefficients and orbitals; Newton–Raphson and DIIS for orbital rotations; super-CI and DIIS; second-order CASSCF (CASSCF-2) and RASSCF (restricted active space); active space selection heuristics: chemical intuition, NOON-based, automated (AVAS, DMET); (45–70 min) MRCI: configuration expansion from CASSCF reference; MRCI+Q (Davidson correction) for size-extensivity; NEVPT2 (strongly-contracted, partially-contracted): second-order perturbation theory on a CASSCF reference; CASPT2 and RASPT2; MRMP2 and QDPT; accuracy and applicability comparison; (70–90 min) Automated active space methods: AutoCAS; AVAS (atomic valence active space); embedding: DMET (density ma-

trix embedding theory); SEET (self-energy embedding theory); multireference methods in periodic systems: CASSCF for solids (preview); MR-EOM-CC for excited states; summary and method choice guidelines.

**Core learning outcomes:**

- Identify situations where multireference methods are necessary using diagnostic criteria.
- Describe the CASSCF algorithm and the role of orbital optimisation.
- Compare MRCI+Q, NEVPT2, and CASPT2 in accuracy and computational cost.
- Apply automated active space selection (AVAS, DMET) to a chemical problem.

**Dual-track content:**

- *Applied computational chemistry track:* Method selection for transition metals, photochemistry, and bond breaking; benchmark datasets (GMTKN55, MR-TDDFT comparison); active space size limitations and DMRG as the remedy.
- *Formal/mathematical track:* MRCI as a truncated CI from a multi-determinant reference; Davidson correction and its size-extensivity restoration; NEVPT2 from second-order Rayleigh–Schrödinger PT on a CAS Hamiltonian.

**Tier differentiation:** HS: "when one configuration is not enough, use a committee" — CASSCF as choosing the most important configurations from a controlled active space. Beg. UG: minimal active space for $H_2$ bond breaking; CASSCF(2,2) energy curve. Adv. UG: CASSCF orbital gradient and Hessian; Davidson correction derivation; NEVPT2 Hamiltonian. MSc: RASSCF partitioning; AutoCAS/AVAS; DMET embedding; multireference diagnostics. PhD: MRCI internal contraction; NEVPT2 strongly vs. partially contracted; MR-EOM-CC; CASSCF for solids.

**Assessment:** Set A — choose active space for $O_3$, $Fe(CO)_5$, and a Cr complex using NOONs; CASSCF(2,2) energy for $H_2$; $T_1$ diagnostic interpretation. Set B — derive Davidson correction; NEVPT2 second-order energy formula derivation; CASSCF orbital gradient equation. Projects: (simple) CASSCF vs. RHF potential energy surface for $N_2$ dissociation; (complex) NEVPT2 vs. CASPT2 excitation energies for a transition metal complex.

## Module II.1 / L11: Benchmark Systems and Accuracy Hierarchies

**90-min pacing:** (0–20 min) Philosophy of benchmarking: what does "accurate" mean? Comparison to experiment vs. to higher-level theory; error cancellation; basis set vs. method error; systematic and non-systematic errors; error statistics: MAE, RMSE, MAX; (20–45 min) Benchmark systems and datasets: $H_2$ potential energy curve — analytic benchmarks; helium atom: exact (Pekeris) vs. HF vs. MP2 vs. CCSD(T) vs. FCI; LiH: basis set convergence; $N_2$ dissociation: multireference character onset; GMTKN55 composite benchmark: thermochemistry (W4-11), reaction barriers (BH76), non-covalent interactions (S66, WATER27); (45–65 min) Accuracy hierarchy: HF → MP2 → CCSD → CCSD(T) → CCSDT → FCI; scaling: $N^4$, $N^5$, $N^6$, $N^7$, $N^8$, exponential; DFT comparison: relative accuracy and cost; QMC in the hierarchy: DMC vs. CCSD(T) and FCI; basis set convergence: CBS extrapolation schemes (Helgaker, Peterson); composite methods: G3, W4, HEAT; (65–90 min) Error analysis in practice: basis set incompleteness error (BSIE); core–valence correlation; scalar relativistic effects; spin–orbit coupling; diagonal Born–Oppenheimer correction; zero-point vibrational energy; thermochemical accuracy: sub-kJ/mol goal; error budget for atomisation energies.

**Core learning outcomes:**

- Rank electronic structure methods by accuracy and computational scaling.

- Apply CBS extrapolation to estimate the basis set limit.
- Decompose the total error budget for a thermochemical property.
- Interpret benchmark statistics (MAE, RMSE, MAX) for method comparison.

**Dual-track content:**

- *Applied track:* GMTKN55 and HEAT benchmarks as standards of chemical accuracy; composite methods for thermochemistry; relativistic and DBOC corrections; choosing method for a given system and accuracy target.
- *Formal/statistical track:* Error propagation; CBS extrapolation theory; Helgaker $n^{-3}$ correlation energy extrapolation; systematic error cancellation in isodesmic reactions.

**Tier differentiation:** HS: "higher-level methods are more accurate but cost more time on the computer" — accuracy ladder as a trade-off between exactness and feasibility. Beg. UG: compare HF, MP2, CCSD(T) for $H_2O$ geometry; CBS extrapolation for a simple molecule. Adv. UG: full error budget for $N_2$ atomisation energy; W4 methodology components. MSc: GMTKN55 analysis; relativistic corrections; DBOC evaluation; composite method design. PhD: complete basis set limit theory; stochastic error in QMC benchmarks; renormalised MBPT.

**Assessment:** Set A — rank methods by accuracy and cost; CBS extrapolation for He and $H_2O$; compute MAE for a set of 5 atomisation energies. Set B — derive Helgaker CBS formula; error budget for $N_2$ dissociation energy; design a composite method for sub-kJ/mol thermochemistry. Projects: (simple) accuracy ladder plot for $H_2O$: HF through CCSD(T)/CBS with error bars; (complex) full W4-style error budget for $C_2H_2$: CCSD(T), core–valence, relativity, DBOC, ZPE.

## Module II.1 / L12: Method Selection in Practice

**90-min pacing:** (0–20 min) The method selection problem: given a molecule, property, and accuracy target, which method is optimal? Decision tree: system size $N$; required accuracy; multireference character; available computational resources; time constraints; (20–45 min) Cost-scaling summary: HF $\mathcal{O}(N^3)$–$\mathcal{O}(N^4)$ (conventional); MP2 $\mathcal{O}(N^5)$; CCSD $\mathcal{O}(N^6)$; CCSD(T) $\mathcal{O}(N^7)$; CASSCF $\mathcal{O}(e^{N_{\text{act}}})$; DMRG $\mathcal{O}(m^3N)$; DMC $\mathcal{O}(N^3)$; DFT $\mathcal{O}(N^3)$ (hybrid); linear-scaling DFT $\mathcal{O}(N)$; crossover system sizes for practical use cases; (45–65 min) Applicability regimes: single-reference systems: CCSD(T)/CBS as gold standard; DFT for large single-reference; multireference: CASSCF/NEVPT2 for moderate active spaces; DMRG for large active spaces; solids and extended systems: DFT, periodic HF, embedding (DMET, QMMM); excited states: EOM-CC, TDDFT, MR-EOM-CC, ADC; non-equilibrium and dynamics: TDDFT, MCTDH, ML potentials; (65–90 min) Practical workflow design: choosing basis set, method, software, and hardware; black-box vs. expert use; automation and high-throughput screening; machine learning potentials (MLPs): how DFT or CCSD(T) data trains MLPs; future outlook: quantum computing approaches (VQE, QITE); fault-tolerant quantum chemistry.

**Core learning outcomes:**

- Apply the decision tree to select an appropriate method for a given system and target.
- Compare formal scaling with practical crossover sizes for common methods.
- Design a complete computational workflow from structure to property.
- Describe the role of ML potentials and quantum computing in the future of the field.

**Dual-track content:**

- *Practical/industrial track:* High-throughput screening; automation with AiiDA or ASE;

ORCA and MOLPRO workflows; ML potential training pipelines; QM/MM for enzymes.
- *Foundational track:* Formal complexity and why crossovers occur; locality of correlation; quantum computing resources for quantum chemistry; resource estimation for fault-tolerant quantum chemistry.

**Tier differentiation:** HS: "match the tool to the job" — a pocket guide to choosing a method; analogy with choosing the right level of map detail for your journey. Beg. UG: method decision tree for 5 representative molecules; identify bottleneck. Adv. UG: crossover analysis; formal scaling vs. prefactor discussion; workflow design. MSc: high-throughput screening design; ML potential training workflow; embedding strategy. PhD: quantum computing resource estimation for FeMoco; fault-tolerant vs. NISQ; future of the field: where will the next major accuracy gains come from?

**Assessment:** Set A — method selection for: (i) drug-like molecule geometry, (ii) transition metal reaction barrier, (iii) solid-state band gap, (iv) photochemical conical intersection. Set B — derive crossover size between CCSD(T) and DMC; design an error budget and workflow for sub-kJ/mol prediction of the $N_2$ dissociation energy. Projects: (simple) decision-tree web app for method selection; (complex) full computational study of a reaction: method convergence, basis set extrapolation, and comparison to experiment.

### Assessment Strategy

***Per-Lecture Assessment Bundle*** *(every lecture): concept check (10–15 min, 5–8 short conceptual questions); Problem Set A (simple, 6–10 computational exercises); Problem Set B (complex, 3–6 multi-step derivations and proofs); mini-project simple (1–3 hours); mini-project complex (5–10 hours, open-ended); research questions well-defined (2–4 scoped prompts); research questions fully open (1–2 frontier-style prompts).*

**Course-level grading template:**

- Weekly homework (best 10/12): 40%
- Method comparison project (after L06): 20%
- Computational research project: 25%
- Final exam (take-home or whiteboard): 15%

Mixed cohorts: Beg. UG → Set A + simple projects; MSc/PhD → Set B + complex projects.

### Reading Strategy

**Track A — Core textbooks:** Helgaker, Jørgensen & Olsen, *Molecular Electronic-Structure Theory* (Wiley, 2000) — the definitive reference for Chapters 3–14; Szabo & Ostlund, *Modern Quantum Chemistry* (Dover) — HF and post-HF primer.

**Track B — Methods references:** Jensen, *Introduction to Computational Chemistry* (Wiley) — accessible survey; Cramer, *Essentials of Computational Chemistry* (Wiley); Nightingale & Umrigar (eds.), *Quantum Monte Carlo Methods in Physics and Chemistry* (Kluwer).

**Track C — Advanced and frontier:** Schollwöck, Rev. Mod. Phys. **77** (2005) and **82** (2011) — DMRG reviews; Orús, Ann. Phys. **349** (2014) — tensor networks; Needs *et al.*, J. Chem. Phys. **152** (2020) — CASINO DMC review.

## 2.2 Module II.2 — Path Integrals in Quantum Mechanics

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Feynman's Postulate | Sum over histories; action; classical limit; stationary phase |
| L02 | Derivation of the PI | Trotter; discretisation; continuum limit; measure |
| L03 | Gaussian Path Integrals | HO propagator; fluctuation determinant; van Vleck–Morette |
| L04 | Saddle Point and WKB | Semiclassical expansion; Bohr–Sommerfeld; tunnelling |
| L05 | Imaginary-Time PIs | Wick rotation; Euclidean action; Matsubara frequencies |
| L06 | Statistical Mechanics Connection | $Z$ as periodic PI; PIMC; exchange cycles |
| L07 | Coherent State PI | Bosonic/fermionic coherent states; Grassmann; Berry term |
| L08 | Instantons and Tunnelling | Bounce; dilute gas; splitting; resurgence |
| L09 | PI in Field Theory | $\phi^4$; generating functional; Feynman rules; one-loop |
| L10 | Gauge Fields and Topology | Aharonov–Bohm; Wilson loop; $\theta$-vacuum; Chern–Simons |
| L11 | Numerical Path Integrals | PIMC; ring polymer; RPMD; worm algorithm |
| L12 | Berry Phase from PIs | Geometric phase; Chern number; TKNN; non-Abelian |

### Lecture-by-Lecture Syllabi

**Module II.2 / L01: Feynman's Postulate**

**90-min pacing:** (0–20 min) Three formulations of QM: Schrödinger, Heisenberg, Feynman; Feynman's postulate — amplitude from $x_a$ to $x_b$ is sum over all paths weighted by $e^{iS/\hbar}$; (20–45 min) Classical action $S = \int L\,dt$; stationary phase $\hbar \to 0$: only classical path survives; Fermat's principle analogy; double-slit from two-path sum; (45–70 min) From PI to Schrödinger equation: Trotter factorisation $e^{-i\hat{H}\varepsilon/\hbar} \approx e^{-i\hat{T}\varepsilon/\hbar}e^{-i\hat{V}\varepsilon/\hbar}$; recover $i\hbar\partial_t\psi = \hat{H}\psi$ as $\varepsilon \to 0$; (70–90 min) Relation to propagator $K(x_b, x_a; T) = \langle x_b| e^{-i\hat{H}T/\hbar} |x_a\rangle$; Dirac's precursor idea; historical context. **Core learning outcomes:**

- State Feynman's postulate and relate it to the propagator.
- Derive the classical limit from the stationary-phase argument.
- Recover the Schrödinger equation from the short-time propagator.
- Compute the double-slit amplitude from a two-path sum.

**Tier differentiation:** HS: "nature tries all paths; at human scales only the classical path matters." Beg. UG: stationary phase; free-particle PI; double-slit amplitude. Adv. UG: Trotter factorisation; recover Schrödinger; momentum-space PI. MSc: functional measure; Wiener measure; Feynman–Kac theorem. PhD: mathematical rigour of PI; lattice regularisation; stochastic PDE connection. **Assessment:** Set A — stationary phase for free particle; double-slit amplitude from two paths. Set B — derive Schrödinger from short-time propagator; Trotter error analysis. Projects: (simple) path interference visualiser for double slit; (complex) Trotter error for HO propagator vs. number of slices.

## Module II.2 / L02: Derivation of the Path Integral

**90-min pacing:** (0–20 min) Time-slicing: insert $N-1$ resolutions of identity $\int dx_k \, |x_k\rangle \langle x_k|$; $\varepsilon = T/N$; (20–45 min) Momentum Gaussian integral: $\int dp \, e^{ip(x_{k+1}-x_k)/\hbar - i\varepsilon p^2/2m\hbar}$; Lagrangian form; Riemann sum $\to \int L \, dt$; measure $\mathcal{D}x(t)$; (45–65 min) Free particle: exact Gaussian; $K_0 = (m/2\pi i\hbar T)^{1/2} e^{im(x_b-x_a)^2/2\hbar T}$; check at $T \to 0$; (65–90 min) Ordering ambiguities: midpoint vs. endpoint; Weyl ordering; lattice regularisation as rigorous definition; continuum limit. **Core learning outcomes:**

- Derive the PI by time-slicing and integrating over momenta.
- Obtain the Lagrangian form and define the PI measure.
- Compute the free-particle propagator exactly.
- Explain ordering ambiguities and the Weyl prescription.

**Tier differentiation:** HS: "slice time into many steps; combine them to get the full amplitude." Beg. UG: time-sliced derivation; momentum Gaussian; free-particle result. Adv. UG: ordering ambiguity; Weyl prescription; linear-potential PI. MSc: Wiener measure; lattice regularisation; Feynman–Kac. PhD: cylindrical measures; stochastic analysis; connection to QFT PI. **Assessment:** Set A — derive $K_0$ from PI; check normalisation; linear-potential PI.

Set B — ordering ambiguity derivation; Weyl prescription; lattice-PI convergence. Projects: (simple) numerical time-sliced propagator vs. analytic for HO; (complex) ordering ambiguity: midpoint vs. pre-point for anharmonic oscillator.

## Module II.2 / L03: Gaussian Path Integrals

**90-min pacing:** (0–20 min) Quadratic actions give exact results; split $S = S_{\text{cl}} + S[\eta]$; cross terms vanish by EOM; (20–50 min) HO: classical solution; $S_{\text{cl}}$; fluctuation determinant $\det\left(-\partial_t^2 - \omega^2\right)^{-1/2}$; $\zeta$-function regularisation; Gel'fand–Yaglom; $K_{\text{HO}} = \left(\frac{m\omega}{2\pi i\hbar \sin \omega T}\right)^{1/2} e^{iS_{\text{cl}}/\hbar}$; (50–70 min) Stationary-phase approximation: expand around classical; leading Gaussian correction; van Vleck–Morette determinant; validity $S_{\text{cl}} \gg \hbar$; (70–90 min) Multiple saddles; Morse theory; conjugate points; Maslov index; multi-dimensional generalisation. **Core learning outcomes:**

- Evaluate Gaussian PIs by separating classical and fluctuation parts.
- Compute the HO propagator exactly via the fluctuation determinant.
- Apply stationary-phase approximation for non-Gaussian actions.
- Define the van Vleck–Morette determinant and Maslov index.

**Tier differentiation:** HS: HO PI as "the exactly solvable model" of path integrals. Beg. UG: $S_{\text{cl}}$ for HO; verify propagator normalisation. Adv. UG: fluctuation determinant; stationary-phase derivation. MSc: van Vleck–Morette; Maslov index; multi-dimensional saddle. PhD: $\zeta$-function; Gel'fand–Yaglom method; functional determinants in QFT. **Assessment:** Set A — $S_{\text{cl}}$ for HO; propagator at $T = \pi/\omega$; coherent-state check. Set B — fluctuation determinant via $\zeta$-function; Maslov index for caustic crossing. Projects: (simple) HO propagator: PI vs. operator method; (complex) semiclassical approximation for quartic potential vs. exact.

## Module II.2 / L04: Saddle-Point Approximation and WKB

**90-min pacing:** (0–20 min) WKB ansatz $\psi = Ae^{iS/\hbar}$; Hamilton–Jacobi from $\hbar^0$; transport equation for $A$; (20–45 min) WKB in allowed/forbidden regions; connection formulae from Airy functions; Bohr–Sommerfeld $\oint p\,dx = 2\pi\hbar(n + \frac{1}{2})$; Maslov index gives $\frac{1}{2}$; (45–65 min) Tunnelling: $\gamma = \hbar^{-1}\int |p|dx$; Gamow factor for alpha decay; field emission; (65–90 min) Beyond WKB: uniform approximation; complex turning points; Stokes lines; exact WKB and resurgence; Gutzwiller trace formula (sum over periodic orbits). **Core learning outcomes:**

- Derive WKB wavefunctions and connection formulae.
- Apply Bohr–Sommerfeld and compute energy levels.
- Calculate tunnelling amplitudes via the barrier integral.
- Connect WKB to PI saddle-point approximation.

**Tier differentiation:** HS: "particles tunnel through barriers in imaginary time." Beg. UG: WKB for HO; Bohr–Sommerfeld; simple tunnelling rate. Adv. UG: connection formulae from Airy; Gamow factor derivation. MSc: uniform approximation; complex turning points; Stokes lines; resurgence. PhD: exact WKB; Gutzwiller trace formula; quantum chaos. **Assessment:** Set A — Bohr–Sommerfeld for HO and square well; Gamow factor for Coulomb barrier. Set B — connection formulae from Airy; Bohr–Sommerfeld with Maslov. Projects: (simple) WKB energy levels vs. exact for Morse potential; (complex) Gamow factor for alpha decay vs. experimental data.

## Module II.2 / L05: Imaginary-Time Path Integrals

**90-min pacing:** (0–20 min) Wick rotation $t \to -i\tau$; Euclidean action $S_E = \int(\frac{m}{2}\dot{x}^2 + V)d\tau$; damped integrand $e^{iS_M} \to e^{-S_E}$; (20–45 min) $Z = \mathrm{Tr}\, e^{-\beta\hat{H}} = \oint \mathcal{D}x\, e^{-S_E}$ over periodic paths $x(0) = x(\beta\hbar)$; HO via Matsubara modes $\omega_n = 2\pi n/\beta\hbar$; $Z_{\mathrm{HO}} = [\sinh(\beta\hbar\omega/2)]^{-1}$; (45–65 min) Imaginary-time GF $G(\tau) = -\langle T_\tau x(\tau)x(0)\rangle$; spectral function via inverse Laplace; (65–90 min) Thermal averages as PI with operator insertion; analytic continuation $G(\tau) \to G^R(\omega)$; ill-posedness and MEM. **Core learning outcomes:**

- Perform the Wick rotation and construct the Euclidean PI.
- Derive $Z$ as a periodic Euclidean PI.
- Compute the HO partition function via Matsubara modes.
- Define the imaginary-time GF and relate it to the spectral function.

**Tier differentiation:** HS: "imaginary time turns quantum oscillations into statistical weights." Beg. UG: Wick rotation; Euclidean action; $Z_{\mathrm{HO}}$. Adv. UG: Matsubara modes; imaginary-time correlator; analytic continuation. MSc: spectral function from $G(\tau)$; MEM; ill-posedness. PhD: Wiener measure in Euclidean time; stochastic quantisation; lattice field theory. **Assessment:** Set A — Wick-rotate HO action; $Z_{\mathrm{HO}}$ via Gaussian; ground-state energy. Set B — Matsubara product formula; imaginary-time correlator for HO. Projects: (simple) $Z_{\mathrm{HO}}(\beta)$ and specific heat; (complex) analytic continuation of $G(\tau)$ to $G^R(\omega)$: stability and MEM.

## Module II.2 / L06: Statistical Mechanics Connection

**90-min pacing:** (0–20 min) $Z = \mathrm{Tr}\, e^{-\beta\hat{H}}$ as Euclidean PI with periodic BCs; $F = -k_B T \ln Z$; $U, S$ from $Z$; (20–45 min) Quantum ideal gases from PI: fermionic anti-

periodic BCs; bosonic periodic; BE/FD distributions derived from PI; grand canonical $\Xi = \mathrm{Tr}\, e^{-\beta(\hat{H}-\mu\hat{N})}$; (45–65 min) Interacting systems: linked-cluster theorem for $\ln Z$; virial coefficients; $\hbar^2$ Wigner–Kirkwood corrections; (65–90 min) Path-integral Monte Carlo: ring-polymer representation; Metropolis sampling; exchange cycles; bosonic permutations and superfluidity onset. **Core learning outcomes:**

- Derive quantum statistical mechanics from the Euclidean PI.
- Show how Bose/Fermi statistics arise from boundary conditions.
- Apply linked-cluster theorem to compute $\ln Z$ perturbatively.
- Describe path-integral Monte Carlo and exchange cycles.

**Tier differentiation:** HS: "quantum particles can swap paths in imaginary time — that's exchange." Beg. UG: $Z$ for free boson/fermion from PI; Bose/Fermi distributions from BCs. Adv. UG: linked-cluster theorem; second virial coefficient from PI. MSc: Wigner–Kirkwood expansion; PIMC algorithm; exchange cycles. PhD: worldline MC; sign problem for fermions; superfluid description from PI. **Assessment:** Set A — partition function for two-state system; Bose/Fermi statistics from BCs. Set B — linked-cluster theorem; second virial coefficient for hard-sphere gas. Projects: (simple) PIMC for HO: energy vs. $\beta$; (complex) PIMC exchange cycles: bosonic statistics and superfluid onset for 4 particles.

## Module II.2 / L07: Coherent-State Path Integrals

**90-min pacing:** (0–20 min) Bosonic coherent states $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$; over-completeness $\int d^2\alpha\,|\alpha\rangle\langle\alpha|/\pi = \hat{1}$; (20–45 min) Bosonic PI: insert over-completeness; action $S = \int(\alpha^*\partial_\tau\alpha + H(\alpha^*,\alpha))d\tau$; Berry connection term; $Z$ as closed-path integral; (45–65 min) Grassmann algebra: $\psi\psi' = -\psi'\psi$; Berezin integration $\int d\psi\,\psi = 1$; fermionic coherent states; anti-periodic BCs; Grassmann Gaussian $\int\prod d\bar{\psi}d\psi\, e^{-\bar{\psi}M\psi} = \det M$; (65–90 min) Applications: free boson/fermion $Z$; Matsubara frequencies from BCs; Hubbard–Stratonovich decoupling preview. **Core learning outcomes:**

- Construct the bosonic coherent-state PI and identify the Berry term.
- Define Grassmann algebra and perform Berezin integration.
- Derive the fermionic PI with anti-periodic BCs.
- Evaluate Grassmann Gaussian integrals.

**Tier differentiation:** HS: coherent states as "most classical" quantum states. Beg. UG: bosonic coherent-state PI; $Z$ for free boson. Adv. UG: Grassmann algebra; fermionic PI; determinant formula. MSc: Hubbard–Stratonovich; mean-field from saddle point. PhD: supersymmetric QM; Witten index; localisation. **Assessment:** Set A — over-completeness; Berry term; free boson $Z$ from PI. Set B — Berezin integral identities; Grassmann Gaussian $= \det M$; fermionic $Z$. Projects: (simple) bosonic PI for driven HO: Berry phase from cyclic evolution; (complex) Hubbard–Stratonovich decoupling and mean-field saddle point.

## Module II.2 / L08: Instantons and Quantum Tunnelling

**90-min pacing:** (0–20 min) Tunnelling as imaginary-time classical solution; double-well $V = \frac{\lambda}{4}(x^2 - v^2)^2$; instanton $x_{\text{inst}}(\tau) = v\tanh[\omega(\tau-\tau_0)/2]$; (20–45 min) Instanton action $S_{\text{inst}} = \frac{4}{3\lambda}(m\omega)^3/\hbar$; zero mode from translational invariance; collective coordinate $\tau_0$; dilute instanton gas; splitting $\Delta E = 2\hbar\omega C e^{-S_{\text{inst}}/\hbar}$; (45–65 min) Bounce for metastable decay; imaginary part of energy; Affleck's formula; vacuum decay in cosmology; (65–90 min) Multi-

instanton interactions; resurgence; Borel resummation; trans-series in QM. **Core learning outcomes:**

- Derive the instanton solution and compute its action.
- Treat the zero mode via collective coordinates.
- Compute the ground-state splitting from the dilute instanton gas.
- Explain the bounce solution and metastable decay.

**Tier differentiation:** HS: "the instanton is the imaginary-time classical path for tunnelling." Beg. UG: double-well instanton; action; splitting formula. Adv. UG: zero mode; dilute gas; derive $\Delta E$. MSc: bounce; Affleck's formula; metastable decay rate. PhD: resurgence; instanton–anti-instanton; Borel resummation. **Assessment:** Set A — instanton profile and action; splitting formula; compare to exact. Set B — zero mode collective coordinate; fluctuation determinant with zero mode. Projects: (simple) double-well splitting vs. barrier height: instanton vs. exact; (complex) dilute gas: multi-instanton partition function.

## Module II.2 / L09: Path Integrals in Field Theory

**90-min pacing:** (0–20 min) QM to QFT: $x(t) \to \phi(x,t)$; generating functional $Z[J] = \int \mathcal{D}\phi\, e^{i(S+\int J\phi)}$; $W[J] = \ln Z[J]$; 1PI effective action $\Gamma$; (20–45 min) $\phi^4$ theory; Feynman rules from PI: propagator $(k^2+m^2)^{-1}$; vertex $-\lambda$; loop expansion $= \hbar$ expansion; (45–65 min) One-loop effective potential; UV divergences; dimensional regularisation; renormalisation group equation; (65–90 min) Gauge theories: Yang–Mills; Faddeev–Popov ghosts; BRST symmetry; QED Feynman rules from PI. **Core learning outcomes:**

- Construct the generating functional and derive Feynman rules for $\phi^4$.
- Perform the loop expansion and compute the one-loop effective potential.
- Explain UV divergences and dimensional regularisation.
- Outline the Faddeev–Popov procedure for gauge theories.

**Tier differentiation:** HS: "Feynman diagrams are pictures of particle interactions derived from the PI." Beg. UG: generating functional; $\phi^4$ Feynman rules; connected Green's functions. Adv. UG: one-loop effective potential; UV divergences; dimensional regularisation. MSc: Coleman–Weinberg; Faddeev–Popov; BRST. PhD: Wilsonian effective action; FRG; Zinn-Justin. **Assessment:** Set A — tree-level Feynman diagrams for $\phi^4$ 4-point function; propagator from $Z_0[J]$. Set B — one-loop effective potential; dimensional regularisation of mass divergence. Projects: (simple) Feynman diagram generator for $\phi^4$ at two loops; (complex) Coleman–Weinberg mechanism: symmetry restoration vs. breaking as function of $\lambda/m^2$.

## Module II.2 / L10: Gauge Fields and Topological Effects

**90-min pacing:** (0–20 min) $U(1)$ gauge invariance; Aharonov–Bohm phase $e^{ie\oint \mathbf{A}\cdot d\mathbf{l}/\hbar}$; physical despite $\mathbf{B} = 0$; (20–45 min) PI derivation of AB phase; holonomy; non-Abelian gauge fields; Wilson loop $W_C = \mathrm{Tr}\,\mathcal{P}e^{ig\oint A_\mu dx^\mu}$; area law and confinement; (45–65 min) Lattice gauge theory: plaquette action; strong-coupling expansion; Monte Carlo for $SU(2)$; (65–90 min) Topological field theories: Chern–Simons in 2+1D; $\theta$-vacuum of QCD; topological charge $Q = \frac{g^2}{32\pi^2}\int F\tilde{F}$; instanton number; axion. **Core learning outcomes:**

- Derive the Aharonov–Bohm phase from the PI.

- Define the Wilson loop and relate its area law to confinement.
- Describe lattice gauge theory and Monte Carlo.
- Explain the topological charge and $\theta$-vacuum.

**Tier differentiation:** HS: "a particle can feel a magnetic field it never touches" — AB effect. Beg. UG: AB phase; Wilson loop; topological charge. Adv. UG: area law from strong coupling; topological charge calculation. MSc: lattice gauge MC; plaquette action; $SU(2)$. PhD: Chern–Simons TQFT; $\theta$-vacuum; axion; instanton number. **Assessment:** Set A — AB phase for given flux; Wilson loop; topological charge for instanton. Set B — area law from strong-coupling expansion; Chern–Simons action. Projects: (simple) AB interference pattern vs. flux; (complex) $SU(2)$ lattice: plaquette energy vs. coupling by MC.

## Module II.2 / L11: Numerical Path Integrals

**90-min pacing:** (0–20 min) Why numerical PI: non-Gaussian integrals; PIMC via ring-polymer representation; bead-and-spring; (20–45 min) PIMC algorithm: Metropolis on ring polymer; staging moves for efficiency; energy estimator; Trotter error vs. $P$; (45–65 min) PIMD: ring polymer MD; thermostat (PILE); RPMD for rate theory; quantum effects in proton transfer; (65–90 min) Worm algorithm for bosons: worldlines; permutation cycles; BEC from winding number; sign problem for fermions; restricted PIMC. **Core learning outcomes:**

- Set up ring-polymer representation and describe PIMC.
- Explain staging moves and their role in decorrelation.
- Describe RPMD and its application to rate theory.
- Outline worm algorithm and connection to BEC.

**Tier differentiation:** HS: "each quantum particle becomes a ring of classical beads." Beg. UG: ring polymer; energy estimator; Trotter error. Adv. UG: staging moves; RPMD; Trotter error analysis. MSc: worm algorithm; exchange cycles; superfluid density. PhD: sign problem; restricted PIMC; fixed-node PIMC for fermions. **Assessment:** Set A — ring polymer for HO; Trotter error vs. $P$; energy estimator. Set B — staging move derivation; winding number and superfluidity estimator. Projects: (simple) PIMC for 1D hydrogen: ground-state energy vs. temperature; (complex) RPMD rate constant for symmetric double well vs. exact quantum.

## Module II.2 / L12: Berry Phase from Path Integrals

**90-min pacing:** (0–20 min) Adiabatic evolution; Berry phase $\gamma_n = i \oint \langle n | \nabla_R | n \rangle \cdot dR$; Berry connection and curvature; (20–45 min) PI derivation: coherent-state PI; Berry term $i \oint \alpha^* \dot{\alpha}\, d\tau$ as geometric phase; adiabatic limit; (45–65 min) Chern number: $C = \frac{1}{2\pi} \int_{\mathrm{BZ}} \mathcal{F}_{xy} d^2k \in \mathbb{Z}$; TKNN formula $\sigma_{xy} = Ce^2/h$; bulk–boundary correspondence; Haldane model; (65–90 min) Non-Abelian Berry phase: degenerate states; Wilczek–Zee holonomy; topological quantum computing; experimental observations. **Core learning outcomes:**

- Derive the Berry phase from the adiabatic coherent-state PI.
- Define Berry curvature and Chern number.
- State the TKNN formula and explain bulk-boundary correspondence.
- Describe non-Abelian Berry phase and topological quantum computing.

**Tier differentiation:** HS: "some quantum phases cannot be removed by any gauge choice" — topological invariants. Beg. UG: Berry phase for spin-$\frac{1}{2}$; solid-angle formula. Adv. UG: Chern number; TKNN; Haldane model. MSc: non-Abelian Berry phase; Wilczek–Zee; topological insulator. PhD: anyons in FQH; topological quantum error correction; holonomy in QFT. **Assessment:** Set A — Berry phase for adiabatic spin cycle; solid-angle formula. Set B — Chern number for Haldane model; TKNN from Kubo formula. Projects: (simple) Berry curvature map for two-band model; (complex) Haldane model topological phase diagram.

***Module II.2 Reading:*** *Zinn-Justin, QFT and Critical Phenomena; Schulman, Techniques and Applications of Path Integration; Kleinert, Path Integrals; Nakahara, Geometry, Topology and Physics.*

## 2.3 Module II.3 — Theory of Angular Momentum

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | $\mathfrak{su}(2)$ Algebra | Commutation relations; Casimir; irreps; Schurś lemma |
| L02 | Ladder Operators & Spectrum | $\hat{J}_\pm$; quantisation of $j, m$; matrix elements; Schwinger bosons |
| L03 | Orbital Angular Momentum | $\hat{\mathbf{L}}$ in spherical coords; spherical harmonics; addition theorem |
| L04 | Spin Angular Momentum | Pauli matrices; spinors; $SU(2)$ double cover; spin coherent states |
| L05 | Addition of Angular Momenta | Triangle rule; Clebsch–Gordan coefficients; singlet/triplet |
| L06 | Wigner $3j$-Symbols & Racah | $3j, 6j, 9j$ symbols; selection rules; line strengths |
| L07 | Wigner–Eckart Theorem | Spherical tensor operators; reduced matrix elements; Landé $g$ |
| L08 | Hydrogen Atom & $SO(4)$ | Runge–Lenz vector; accidental degeneracy; parabolic coordinates |
| L09 | Fine and Hyperfine Structure | Spin–orbit; Darwin; Fermi contact; 21 cm; Breit–Rabi |
| L10 | Wigner $D$-matrices | Euler angles; $D^{(j)}$; symmetric top; NMR pulses |
| L11 | Angular Momentum Many-Body | Term symbols; Hundś rules; shell model; $LS$ vs. $jj$ coupling |
| L12 | Rotational Spectroscopy | Rigid rotor; symmetric top; Raman; nuclear spin statistics |

### Lecture-by-Lecture Syllabi

**Module II.3 / L01: The $\mathfrak{su}(2)$ Algebra**

**90-min pacing:** (0–20 min) Rotation symmetry generators $\hat{J}_x, \hat{J}_y, \hat{J}_z$; algebra $[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k$; Casimir $\hat{\mathbf{J}}^2$ commutes with all $\hat{J}_i$; (20–45 min) Irreducible representations: $j \in \{0, \frac{1}{2}, 1, \ldots\}$; simultaneous eigenstates $|j, m\rangle$; $(2j+1)$-dimensional; (45–70 min) $SU(2)$ vs. $SO(3)$: $2\pi$ rotation gives $-1$ for half-integer $j$; Schur's lemma; (70–90 min) Physical realisations: orbital AM ($j$ integer), spin ($j$ half-integer); fundamental vs. adjoint representation. **Core learning outcomes:**

- State $\mathfrak{su}(2)$ commutation relations and identify the Casimir.
- Label irreps by $j, m$ and state their ranges.
- Distinguish $SU(2)$ from $SO(3)$ and explain double-valued representations.

- Apply Schur's lemma.

**Tier differentiation:** HS: "angular momentum comes in units of $\hbar$." Beg. UG: Casimir; $2 \times 2$ and $3 \times 3$ representations. Adv. UG: all irreps from Schur; $SU(2)$ vs. $SO(3)$. MSc: universal enveloping algebra; Casimir from Killing form. PhD: highest-weight theory; Verma modules; quantum groups $SU_q(2)$. **Assessment:** Set A — verify Casimir commutes with $\hat{J}_i$; eigenvalues for $j = 1, \frac{3}{2}$. Set B — prove irreps labelled by $j \geq 0$; $SU(2)$ double cover. Projects: (simple) angular momentum algebra visualiser; (complex) character table for $j = 0, \frac{1}{2}, 1, \frac{3}{2}$.

## Module II.3 / L02: Ladder Operators and Spectrum

**90-min pacing:** (0–20 min) Ladder operators $\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y$; $[\hat{J}_z, \hat{J}_\pm] = \pm\hbar\hat{J}_\pm$; (20–50 min) Matrix elements $\hat{J}_+ |j, m\rangle = \hbar\sqrt{j(j+1) - m(m+1)}\, |j, m+1\rangle$; termination at $m = \pm j$; quantisation of $j$; (50–70 min) Explicit matrices $j = \frac{1}{2}$ (Pauli) and $j = 1$; (70–90 min) Schwinger boson representation: $\hat{J}_+ = \hat{a}^\dagger\hat{b}$, $\hat{J}_- = \hat{b}^\dagger\hat{a}$; spin coherent states. **Core learning outcomes:**

- Derive spectrum and matrix elements via ladder operators.
- Explain quantisation of $j$ from ladder termination.
- Write explicit matrices for $j = \frac{1}{2}, 1$.
- Construct the Schwinger boson representation.

**Assessment:** Set A — matrix elements $\langle j, m'| \hat{J}_x |j, m\rangle$ for $j = 1$; $\hat{\mathbf{J}}^2$ check. Set B — quantisation from termination; Schwinger boson algebra. Projects: (simple) Bloch sphere visualiser; (complex) Schwinger bosons: generate all $j$ from two HO modes.

## Module II.3 / L03: Orbital Angular Momentum and Spherical Harmonics

**90-min pacing:** (0–20 min) $\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla)$; $\hat{L}_z = -i\hbar\partial_\phi$; $\hat{\mathbf{L}}^2$ in spherical coordinates; (20–50 min) Spherical harmonics $Y_l^m(\theta, \phi)$; associated Legendre polynomials; orthonormality; parity $(-1)^l$; (50–70 min) Addition theorem; solid harmonics; multipole expansion; (70–90 min) Rigid rotor $\hat{H} = \hat{\mathbf{L}}^2/2I$; spectrum $E_l = \hbar^2 l(l+1)/2I$; selection rule $\Delta l = \pm 1$. **Core learning outcomes:**

- Express $\hat{L}_z$ and $\hat{\mathbf{L}}^2$ in spherical coordinates.
- State properties of spherical harmonics and use the addition theorem.
- Solve the rigid rotor and identify selection rules.
- Apply multipole expansion.

**Assessment:** Set A — $Y_l^m$ for $l = 0, 1$; addition theorem; rigid rotor spectrum. Set B — derive $\hat{\mathbf{L}}^2$ in spherical coordinates; prove addition theorem. Projects: (simple) 3D orbital visualiser; (complex) multipole expansion of arbitrary charge distribution.

## Module II.3 / L04: Spin Angular Momentum

**90-min pacing:** (0–20 min) Spin: $\hat{\mathbf{S}} = \frac{\hbar}{2}\hat{\boldsymbol{\sigma}}$; Pauli matrices; spinors; (20–45 min) $SU(2)$ double cover; $2\pi$ rotation $\to -1$; neutron interferometry; (45–65 min) Spin in magnetic field: Larmor precession; Rabi formula; (65–90 min) Spin coherent states; Bloch sphere; over-completeness; spin-coherent-state PI. **Core learning outcomes:**

- Compute Pauli matrix algebra and spin precession.
- Explain the $SU(2)$ double cover physically.
- Solve spin in a magnetic field exactly.
- Describe spin coherent states and the Bloch sphere.

**Assessment:** Set A — Pauli algebra; Larmor frequency; $\langle \hat{S}_z \rangle(t)$. Set B — $SU(2)$ double cover; spin coherent state over-completeness. Projects: (simple) Bloch sphere trajectory for Rabi problem; (complex) spin-coherent-state PI: Berry phase.

### Module II.3 / L05: Addition of Angular Momenta

**90-min pacing:** (0–20 min) $\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2$; uncoupled and coupled bases; triangle rule; (20–50 min) Clebsch–Gordan coefficients; symmetry properties; ladder-operator derivation; tables for $\frac{1}{2}\otimes\frac{1}{2}$, $\frac{1}{2}\otimes 1$; (50–70 min) Singlet and triplet states; entanglement; (70–90 min) Spin–orbit states for hydrogen; $6j$-symbols as recoupling coefficients. **Core learning outcomes:**

- Apply the triangle rule and compute CG coefficients.
- Construct singlet and triplet states.
- Apply CG to spin–orbit coupling in hydrogen.
- Define $6j$-symbols and explain recoupling.

**Assessment:** Set A — CG for $\frac{1}{2} \otimes 1$; triplet/singlet states; spin–orbit $m_j$. Set B — derive CG by ladders; verify orthogonality of coupled basis. Projects: (simple) CG coefficient calculator; (complex) spin–orbit spectrum via CG recoupling.

### Module II.3 / L06: Wigner $3j$-Symbols and Racah Algebra

**90-min pacing:** (0–20 min) Wigner $3j$-symbols; symmetry properties; relation to CG; (20–45 min) Racah formula; selection rules $m_1 + m_2 + m_3 = 0$ and triangle rule; (45–70 min) $6j$-symbols as recoupling; $9j$-symbols; orthogonality; (70–90 min) Applications: atomic $E1$ selection rules; Laporte rule; astrophysical line strengths. **Core learning outcomes:**

- Define $3j$-symbol and state symmetry properties.
- Derive selection rules from $3j$-symbols.
- Define $6j$-symbols and explain recoupling.
- Apply to atomic electric dipole selection rules.

**Assessment:** Set A — $3j$-symbols for given $j, m$; identify forbidden transitions. Set B — $3j$ symmetry derivation; $6j$ orthogonality. Projects: (simple) atomic line-strength calculator; (complex) Racah algebra for hyperfine matrix elements.

### Module II.3 / L07: The Wigner–Eckart Theorem

**90-min pacing:** (0–20 min) Spherical tensor operators $\hat{T}_q^{(k)}$; transformation under $\hat{J}$; rank-0 (scalar), rank-1 (vector), rank-2 (quadrupole); (20–45 min) Wigner–Eckart theorem: $\langle j', m' | \hat{T}_q^{(k)} | j, m \rangle = \langle j, m; k, q | j', m' \rangle \frac{\langle j' \| \hat{T}^{(k)} \| j \rangle}{\sqrt{2j'+1}}$; reduced matrix element; (45–65 min) Applications: $E1$ selection rules; dipole matrix elements; Landé $g$-factor; (65–90 min) Composite systems; $6j$-symbols; effective operators. **Core learning outcomes:**

- Define spherical tensor operators and verify transformation properties.
- State W-E theorem and identify reduced matrix element.

- Apply W-E to selection rules and Landé $g$-factor.

**Assessment:** Set A — identify ranks; $E1, M1, E2$ selection rules; Landé $g$ for $^3P_1$. Set B — W-E proof from Schur; reduced matrix element for HO ladder. Projects: (simple) line-strength calculator; (complex) quadrupole moments for nuclear models.

### Module II.3 / L08: Hydrogen Atom and $SO(4)$ Symmetry

**90-min pacing:** (0–20 min) Hydrogen Hamiltonian; separation; radial equation; (20–45 min) Radial wavefunctions; Laguerre polynomials; $E_n = -13.6/n^2$ eV; degeneracy $n^2$; (45–65 min) $SO(4)$: Runge–Lenz vector $\hat{\mathbf{A}}$; $[\hat{H}, \hat{\mathbf{A}}] = 0$; $\mathfrak{so}(4)$ from $\hat{\mathbf{L}}$ and rescaled $\hat{\mathbf{A}}$; algebraic derivation of $1/n^2$ spectrum; (65–90 min) Parabolic coordinates; $n_1, n_2, m$; Stark effect. **Core learning outcomes:**

- Solve the hydrogen radial equation and identify quantum numbers.
- Define Runge–Lenz and show $[\hat{H}, \hat{\mathbf{A}}] = 0$.
- Derive the spectrum algebraically from $\mathfrak{so}(4)$.
- Describe parabolic coordinates and Stark effect.

**Assessment:** Set A — $R_{nl}$ for $2s, 2p$; $n = 2$ degeneracy count. Set B — verify Runge–Lenz commutator; $n^2$ degeneracy from $\mathfrak{so}(4)$. Projects: (simple) hydrogen orbital density visualiser; (complex) algebraic spectrum without radial equation.

### Module II.3 / L09: Fine and Hyperfine Structure

**90-min pacing:** (0–20 min) Three fine-structure corrections: relativistic KE, spin–orbit, Darwin; (20–45 min) First-order PT: $\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = \frac{1}{2}(\hat{\mathbf{J}}^2 - \hat{\mathbf{L}}^2 - \hat{\mathbf{S}}^2)$; $\langle r^{-3} \rangle_{nl}$; energy $E^{(1)}_{nj}$; (45–65 min) Hyperfine: Fermi contact + orbital + dipole–dipole; $\hat{\mathbf{F}} = \hat{\mathbf{J}} + \hat{\mathbf{I}}$; 21 cm line; (65–90 min) Zeeman: weak (anomalous) and strong (Paschen–Back); Breit–Rabi diagram. **Core learning outcomes:**

- Evaluate the three fine-structure corrections.
- Derive spin–orbit splitting via PT in the $j$-basis.
- Describe hyperfine Hamiltonian and 21 cm line.
- Analyse Zeeman effect in weak and strong limits.

**Assessment:** Set A — fine-structure splitting of $2p$; 21 cm energy. Set B — $\langle r^{-3} \rangle_{nl}$; Darwin term; Breit–Rabi diagram. Projects: (simple) hydrogen energy-level diagram with fine and hyperfine; (complex) Zeeman splitting of hyperfine levels: Breit–Rabi formula.

### Module II.3 / L10: Wigner $D$-matrices

**90-min pacing:** (0–20 min) Euler angles; rotation operator $\hat{R}(\alpha, \beta, \gamma) = e^{-i\alpha \hat{J}_z/\hbar} e^{-i\beta \hat{J}_y/\hbar} e^{-i\gamma \hat{J}_z/\hbar}$; (20–45 min) $D^{(j)}_{m'm}(\alpha, \beta, \gamma)$; $d^{(j)}(\beta)$ real matrices; explicit forms for $j = \frac{1}{2}, 1$; (45–65 min) Properties: unitarity; orthogonality $\int D^{(j)*} D^{(j')} d\Omega \propto \delta_{jj'}$; connection $D^{(l)*}_{m0} \propto Y^m_l$; (65–90 min) Applications: rotation of $Y^m_l$; symmetric top; NMR pulse sequences; single-qubit gates. **Core learning outcomes:**

- Parametrise $SO(3)$ and compute $D$-matrices for $j = \frac{1}{2}, 1$.
- Apply orthogonality and connection to spherical harmonics.

- Rotate $Y_l^m$ and analyse symmetric top.

**Assessment:** Set A — $D^{(1/2)}(\alpha, \beta, \gamma)$; rotate $Y_1^0$; orthogonality. Set B — group multiplication law; symmetric top selection rules. Projects: (simple) $D$-matrix rotation visualiser; (complex) symmetric top levels and NMR pulses.

### Module II.3 / L11: Angular Momentum in Many-Body Systems

**90-min pacing:** (0–20 min) Multi-electron atoms: $LS$ vs. $jj$ coupling; term symbols ${}^{2S+1}L_J$; Hund's rules; (20–45 min) $LS$ coupling: spin–orbit via W-E; Lande interval rule $E(J) - E(J-1) = \frac{A}{2}J$; (45–65 min) Nuclear shell model: magic numbers; spin–orbit splitting; $nlj$ states; (65–90 min) $jj$ coupling for heavy nuclei; intermediate coupling; Racah for many-body. **Core learning outcomes:**

- Construct term symbols and apply Hund's rules.
- Derive the Lande interval rule.
- Explain magic numbers from the shell model.
- Distinguish $LS$ and $jj$ coupling.

**Assessment:** Set A — term symbols for $p^2, d^3$; Hund's ground state; interval rule for ${}^3P$. Set B — Lande interval rule; $LS$ vs. $jj$ for Pb vs. C. Projects: (simple) atomic term symbol calculator; (complex) shell model: magic numbers from spin–orbit.

### Module II.3 / L12: Rotational Spectroscopy

**90-min pacing:** (0–20 min) Rigid rotor: $E_J = \hbar^2 J(J+1)/2I$; $B = \hbar/4\pi cI$; (20–45 min) Selection rules $\Delta J = \pm 1$; microwave spectrum; bond lengths; centrifugal distortion; (45–65 min) Symmetric top: prolate/oblate; $E_{J,K} = \hbar^2[BJ(J+1) + (A-B)K^2]$; (65–90 min) Raman: $\Delta J = 0, \pm 2$; nuclear spin statistics; ortho/para-$H_2$; alternating intensity. **Core learning outcomes:**

- Derive rigid rotor spectrum and selection rules.
- Extract $B$ and bond length from rotational spectrum.
- Solve symmetric top and identify quantum numbers.
- Explain nuclear spin statistics and intensity alternation.

**Assessment:** Set A — rotational spectrum of HCl; $B$ and bond length; centrifugal distortion. Set B — symmetric top energy levels; ortho/para-$H_2$ statistics. Projects: (simple) simulated rotational spectrum of a diatomic; (complex) ortho/para-$H_2$ ratio vs. temperature.

***Module II.3 Reading:*** *Sakurai & Napolitano, Ch. 3; Edmonds, Angular Momentum in QM; Varshalovich, Moskalev & Khersonskii, Quantum Theory of Angular Momentum; Bunker & Jensen, Molecular Symmetry and Spectroscopy.*

## 2.4 Module II.4 — Second Quantisation and Introductory Many-Body Theory

## Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Fock Space | Occupation number; creation/annihilation; Jordan–Wigner; field operators |
| L02 | Operators in 2nd Quantisation | One- and two-body operators; normal ordering; Wicks theorem |
| L03 | Homogeneous Electron Gas | Jellium; Fermi sea; HF exchange; Lindhard; Friedel; plasmon |
| L04 | HF in 2nd Quantisation | Mean-field decoupling; Brillouin; Koopman; HFB; Nambu |
| L05 | Beyond Mean Field | CI; MP2; Goldstone diagrams; linked cluster; GW; RPA |
| L06 | Green's Functions I | Retarded GF; Lehmann representation; spectral function; Galitskii–Migdal |
| L07 | Green's Functions II | Dyson equation; HF and GW self-energies; quasiparticle weight |
| L08 | Interacting Bosons: BEC | Bogoliubov transformation; Goldstone; Landau criterion; vortices |
| L09 | Interacting Fermions: BCS | Cooper instability; gap equation; Bogoliubov–Valatin; BCS–BEC |
| L10 | The Hubbard Model | Tight-binding; Mott insulator; superexchange; DMFT |
| L11 | Quantum Magnetism | Heisenberg model; Holstein–Primakoff; magnons; frustrated magnets |
| L12 | Towards Non-Equilibrium | Keldysh motivation; Floquet theory; driven-dissipative balance |

## Lecture-by-Lecture Syllabi

### Module II.4 / L01: Fock Space and Second Quantisation

**90-min pacing:** (0–20 min) Fock space $\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}^{(N)}$; vacuum; (20–50 min) Bosons $[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}$; fermions $\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij}$; Pauli exclusion from $\hat{c}^2 = 0$; occupation basis; (50–70 min) Many-body states from creation operators; inner products; comparison with Slater determinants; Jordan–Wigner string; (70–90 min) Field operators $\hat{\psi}(\mathbf{r}) = \sum_k \phi_k \hat{c}_k$; density operator $\hat{n}(\mathbf{r}) = \hat{\psi}^\dagger \hat{\psi}$. **Core learning outcomes:**

- Construct Fock space and define creation/annihilation operators.
- Show Pauli exclusion from fermionic anti-commutation.
- Write many-body states as products of creation operators.
- Define field operators and their (anti-)commutation relations.

**Assessment:** Set A — $\hat{a}\hat{a}^\dagger |n\rangle$; verify Pauli from $\hat{c}^2 = 0$; write $|1, 0, 1, 0\rangle$. Set B — Jordan–Wigner derivation; field operator commutation. Projects: (simple) boson vs. fermion occupation statistics; (complex) Jordan–Wigner for 1D spin chain: map to free fermions.

### Module II.4 / L02: Operators in Second Quantisation

**90-min pacing:** (0–20 min) One-body: $\hat{T} = \sum_{ij} t_{ij} \hat{c}_i^\dagger \hat{c}_j$; kinetic energy; external potential; (20–45 min) Two-body: $\hat{V} = \frac{1}{2} \sum_{ijkl} V_{ijkl} \hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_l \hat{c}_k$; Coulomb interaction; normal ordering; Wick's theorem; (45–65 min) Contractions; compute $\langle \hat{T} \rangle, \langle \hat{V} \rangle$ in Slater determinant; (65–90 min) Generalised Wick; Isserlis theorem; Feynman diagrams as Wick contractions. **Core learning outcomes:**

- Write one- and two-body operators in second quantisation.

- Apply Wick's theorem to evaluate expectation values.
- Draw Feynman diagrams as Wick contractions.

**Assessment:** Set A — one-body Hamiltonian; $\langle \hat{V} \rangle$ in HF by Wick. Set B — prove Wick for 4-operator product; generalised Wick for BCS state. Projects: (simple) Wick contraction calculator; (complex) Wick-based HF energy: all diagrams.

## Module II.4 / L03: The Homogeneous Electron Gas

**90-min pacing:** (0–20 min) Jellium model; Hamiltonian components; (20–45 min) Fermi sea; $k_F = (3\pi^2 n)^{1/3}$; kinetic energy; (45–65 min) HF exchange energy $E_{\text{ex}} = -\frac{3e^2 k_F}{4\pi}$; correlation energy; $r_s$ parameter; Gell-Mann–Brueckner high-density limit; (65–90 min) Lindhard dielectric function; Thomas–Fermi screening; plasmon; Friedel oscillations. **Core learning outcomes:**

- Set up jellium model and compute HF exchange energy.
- Define correlation energy and describe its $r_s$ dependence.
- Derive Thomas–Fermi screening and identify the plasmon.

**Assessment:** Set A — $k_F, E_F$; HF exchange energy; Gell-Mann–Brueckner high-density limit. Set B — Lindhard function at $q \to 0$; plasmon dispersion. Projects: (simple) HEG energy vs. $r_s$; (complex) Lindhard dielectric function and Friedel oscillations.

## Module II.4 / L04: Hartree–Fock in Second Quantisation

**90-min pacing:** (0–20 min) HF ansatz; variational principle; orbital rotations; (20–45 min) Mean-field decoupling; Fock matrix; Brillouin's theorem; Koopman's theorem; (45–65 min) SCF density matrix idempotency; symmetry breaking; (65–90 min) HFB generalisation; pairing; BCS as HFB; Bogoliubov transformation in Nambu space. **Core learning outcomes:**

- Derive HF mean-field decoupling and prove Brillouin's theorem.
- Describe the HFB generalisation and BCS connection.
- Perform a Bogoliubov transformation in Nambu space.

**Assessment:** Set A — mean-field decoupling; Brillouin theorem check. Set B — Thouless stability; Bogoliubov transformation: diagonalise HFB matrix. Projects: (simple) HF for Hubbard dimer; (complex) HFB pairing: gap equation.

## Module II.4 / L05: Beyond Mean Field — MP2, Diagrams, and Resummations

**90-min pacing:** (0–20 min) Correlation energy; CI; size inconsistency; (20–45 min) MBPT: MP2 $E^{(2)} = \sum_{ia,jb} \frac{|\langle \Phi_{ij}^{ab} | \hat{V} | \Phi_0 \rangle|^2}{E_0 - E_{ij}^{ab}}$; Goldstone diagrams; (45–65 min) Linked-cluster theorem; MP2 as two Goldstone diagrams; MP3; Dyson equation; (65–90 min) GW: screened interaction $W = v/(1 - vP)$; self-consistent GW; RPA as ring-diagram sum; connection to TDDFT. **Core learning outcomes:**

- Compute MP2 energy and draw its Goldstone diagrams.
- State the linked-cluster theorem.
- Describe GW and RPA as diagrammatic resummations.

**Assessment:** Set A — MP2 for $H_2$; two Goldstone diagrams; size-consistency check. Set B — linked-cluster derivation; Dyson equation from self-energy. Projects: (simple) MP2 correlation energy vs. basis size; (complex) RPA correlation energy for HEG.

### Module II.4 / L06: Green's Functions I

**90-min pacing:** (0–20 min) Retarded GF $G^R(t) = -i\theta(t)\langle\{\hat{c}(t), \hat{c}^\dagger(0)\}\rangle$; (20–50 min) Lehmann representation; spectral function $A(k,\omega) = -2\mathrm{Im}G^R$; poles at ionisation/affinity energies; (50–70 min) Non-interacting GF; quasiparticle peak; incoherent background; $Z_k$; lifetime; (70–90 min) Observables: Galitskii–Migdal; DOS; momentum distribution. **Core learning outcomes:**

- Define retarded GF and derive the Lehmann representation.
- Interpret poles as ionisation/affinity energies.
- Compute spectral function and DOS for a non-interacting system.

**Assessment:** Set A — non-interacting GF; spectral function; DOS for HEG. Set B — Lehmann representation; Galitskii–Migdal energy. Projects: (simple) spectral function: non-interacting to quasiparticle; (complex) Lehmann from CI eigenvalues of a cluster.

### Module II.4 / L07: Green's Functions II — Dyson Equation

**90-min pacing:** (0–20 min) Self-energy $\Sigma$ via Dyson $G = G_0 + G_0\Sigma G$; (20–45 min) HF self-energy diagrams; GW: $\Sigma^{GW} = iGW$; $W = v/(1 - vP)$; (45–65 min) Quasiparticle equation; $Z_k = [1 - \partial_\omega \mathrm{Re}\Sigma]^{-1}$; lifetime $\tau^{-1} = -2Z_k\mathrm{Im}\Sigma$; (65–90 min) $G_0W_0$ band gaps; ARPES; self-consistent GW. **Core learning outcomes:**

- Derive Dyson equation and compute HF and GW self-energies.
- Solve quasiparticle equation; compute $Z_k$ and lifetime.
- Describe $G_0W_0$ accuracy for band gaps.

**Assessment:** Set A — HF and GW diagrams; $Z_k$ for model self-energy; DOS from $G$. Set B — Dyson derivation; prove $Z_k \leq 1$; lifetime from $\mathrm{Im}\Sigma$. Projects: (simple) model spectral function: QP peak + satellite; (complex) $G_0W_0$ band gap for Si.

### Module II.4 / L08: Interacting Bosons and BEC

**90-min pacing:** (0–20 min) Weakly interacting Bose gas; BEC: $\hat{a}_0 \approx \sqrt{N_0}$; (20–50 min) Bogoliubov transformation; dispersion $E_k = \sqrt{\varepsilon_k(\varepsilon_k + 2gn_0)}$; phonon mode; Goldstone; (50–70 min) Depletion; Lee–Huang–Yang; Landau criterion; two-fluid model; (70–90 min) Quantised vortices; BKT in 2D; GP equation; experimental realisations. **Core learning outcomes:**

- Perform Bogoliubov transformation and derive dispersion.
- Identify Goldstone mode and explain Landau criterion.
- Describe vortices and GP equation.

**Assessment:** Set A — Bogoliubov spectrum; depletion; Landau critical velocity. Set B — Bogoliubov derivation; Goldstone: $E_k \propto k$. Projects: (simple) Bogoliubov dispersion vs. free particle; (complex) GP vortex solution and energy.

## Module II.4 / L09: Interacting Fermions and BCS Theory

**90-min pacing:** (0–20 min) Cooper instability; exponential binding energy $E_B \sim -2\hbar\omega_D e^{-1/\lambda}$; (20–50 min) BCS wavefunction; gap equation $\Delta_k = -\sum_{k'} V_{kk'} \frac{\Delta_{k'}}{2E_{k'}}$; quasiparticle energy $E_k = \sqrt{\xi_k^2 + |\Delta_k|^2}$; (50–70 min) Bogoliubov–Valatin transformation; condensation energy; Meissner effect; (70–90 min) BCS–BEC crossover; Feshbach resonance; unitary Fermi gas; cold atoms. **Core learning outcomes:**

- Derive Cooper pairing and the gap equation.
- Perform the Bogoliubov–Valatin transformation.
- Describe BCS–BEC crossover.

**Assessment:** Set A — Cooper binding energy; gap equation at $T = 0$; quasiparticle spectrum. Set B — derive gap equation; Bogoliubov–Valatin derivation. Projects: (simple) BCS gap vs. coupling $\lambda$; (complex) BCS–BEC crossover: quasiparticle spectrum vs. $1/k_F a_s$.

## Module II.4 / L10: The Hubbard Model

**90-min pacing:** (0–20 min) Tight-binding + Hubbard $U$; Hamiltonian; (20–45 min) Half-filling: non-interacting metal; Mott insulator at large $U$; spectral weight transfer; (45–65 min) Limits: $U \to 0$ (perturbation theory); $U \to \infty$ (Heisenberg/$t$–$J$); magnetic phase diagram; Stoner criterion; (65–90 min) Exact 1D solution (Bethe ansatz); spinon–holon separation; DMFT; high-$T_c$ cuprates. **Core learning outcomes:**

- Write the Hubbard Hamiltonian and describe its two limiting cases.
- Explain the Mott insulator and derive the Heisenberg model.
- Describe DMFT and connections to high-$T_c$.

**Assessment:** Set A — tight-binding band; Mott gap; Stoner criterion. Set B — superexchange derivation ($t$–$J$ model); Mott transition. Projects: (simple) Hubbard dimer: exact vs. HF; (complex) DMFT self-consistency for Bethe lattice.

## Module II.4 / L11: Quantum Magnetism

**90-min pacing:** (0–20 min) Heisenberg model $J \sum \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j$; mean-field; (20–45 min) Holstein–Primakoff: $\hat{S}^+ \approx \sqrt{2S}\hat{a}$; magnon dispersion $\omega_k = 4JS(1 - \gamma_k)$; Goldstone at $k = 0$; (45–65 min) AFM: two-sublattice; Bogoliubov transformation; zero-point fluctuations; Mermin–Wagner in 1D; (65–90 min) Frustrated magnets; quantum spin liquids; Kitaev honeycomb: exact solution via Majorana fermions. **Core learning outcomes:**

- Apply Holstein–Primakoff and derive magnon dispersion.
- Explain zero-point fluctuations in AFM and Mermin–Wagner.
- Describe frustrated magnets and quantum spin liquids.

**Assessment:** Set A — magnon dispersion for FM; mean-field $T_c$; AFM zero-point fluctuation. Set B — AFM Bogoliubov transformation; Mermin–Wagner sketch. Projects: (simple) magnon dispersion for 1D/2D FM; (complex) AFM spin-wave theory: zero-point reduction vs. dimension.

**Module II.4 / L12: Towards Non-Equilibrium**

**90-min pacing:** (0–20 min) Limitations of equilibrium: laser-driven systems, biased junctions, periodic drives; (20–45 min) Keldysh contour motivation; closed-time-path; contour-ordered GF; (45–65 min) Floquet theory: $|\psi(t)\rangle = e^{-i\varepsilon t/\hbar}|u(t)\rangle$; quasi-energies; effective Floquet Hamiltonian from Magnus; (65–90 min) Driven-dissipative steady states; Lindblad from Keldysh limit; Floquet topological insulators; time-crystalline order. **Core learning outcomes:**

- Explain the Keldysh contour and its relation to real-time GFs.
- State Floquet's theorem and derive the effective Hamiltonian.
- Describe driven-dissipative steady states.

**Assessment:** Set A — Keldysh contour GF relations; Floquet quasi-energies for two-level system. Set B — Magnus expansion for Floquet Hamiltonian; Lindblad steady-state derivation. Projects: (simple) Floquet quasi-energy spectrum for driven TLS; (complex) Floquet engineering of Haldane model from driven graphene.

***Module II.4 Reading:*** *Fetter & Walecka, Quantum Theory of Many-Particle Systems; Bruus & Flensberg, Introduction to Many-Body QT in Condensed Matter; Altland & Simons, Condensed Matter Field Theory; Auerbach, Interacting Electrons and Quantum Magnetism.*

## 2.5 Module II.5 — Non-Equilibrium Quantum Many-Body Green's Functions

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Motivation and Scope | Transport; ultrafast; driven systems; failure of equilibrium MBPT |
| L02 | The Keldysh Contour | CTP; forward/backward branches; contour ordering; generating functional |
| L03 | Contour Green's Functions | $G^c, G^<, G^>, G^R, G^A, G^K$; LO matrix; FDT; Matsubara limit |
| L04 | Langreth Rules | Analytic continuation; convolution identities; GW and 2B self-energies |
| L05 | Kadanoff–Baym Equations | EOM; two-time structure; $\Phi$-derivable; conserving approximations |
| L06 | Self-Energies | HF; GW; second Born; T-matrix; KB conservation theorem |
| L07 | Quantum Boltzmann Equation | Wigner transform; gradient expansion; collision integral; Drude |
| L08 | Steady-State Transport | Meir–Wingreen; Landauer; Kondo; IETS; shot noise |
| L09 | Electron–Phonon Coupling | Fröhlich; Fan–Migdal; polaron; Eliashberg; $T_c$ |
| L10 | Pump–Probe Dynamics | Time-resolved spectral function; Peierls; tr-ARPES; thermalisation |
| L11 | Driven-Dissipative Systems | Lindblad from NEGF; Floquet–Keldysh; NESS; polaritons |
| L12 | Numerical NEGF | Time-stepping KB; GKBA; NESSi; ML surrogate self-energies |

**Lecture-by-Lecture Syllabi**

### Module II.5 / L01: Motivation and Scope of NEGF

**90-min pacing:** (0–20 min) Paradigmatic non-equilibrium situations: molecular junctions under bias; laser-driven solids; open quantum systems; (20–45 min) Why equilibrium MBPT fails: adiabatic switching; no transient dynamics; (45–65 min) Survey of formalisms: Redfield/Lindblad (Markovian); TDCI (exponential); NEGF (systematically improvable); (65–90 min) Key observables: $I(t)$; non-equilibrium spectral function; tr-ARPES; entanglement entropy from NEGF. **Core learning outcomes:**

- Identify physical situations requiring non-equilibrium treatment.
- Explain why Matsubara MBPT fails for driven systems.
- Compare Lindblad, TDCI, and NEGF in accuracy and cost.
- List observables accessible from the NEGF.

**Assessment:** Set A — identify equilibrium vs. non-equilibrium scenarios; match formalism to problem. Set B — limitations of Matsubara: why adiabatic switching fails. Projects: (simple) scenario classification exercise; (complex) NEGF vs. TDDFT comparison for driven quantum dot.

### Module II.5 / L02: The Keldysh Contour

**90-min pacing:** (0–20 min) CTP $\mathcal{C} = \mathcal{C}^+ \cup \mathcal{C}^-$; motivation from $\langle \hat{O}(t) \rangle = \mathrm{Tr}[\hat{U}(t_0,t)\hat{O}\hat{U}(t,t_0)\hat{\rho}_0]$; (20–45 min) Contour-ordering $T_{\mathcal{C}}$; contour-ordered GF $G^c = -i\langle T_{\mathcal{C}}\hat{c}(t)\hat{c}^\dagger(t')\rangle$; (45–65 min) Perturbation theory on contour; Wick's theorem; (65–90 min) Keldysh generating functional; Schwinger–Dyson equations. **Core learning outcomes:**

- Construct the CTP contour and justify its necessity.
- Define contour ordering and the contour GF.
- Apply Wick's theorem on the Keldysh contour.
- Derive GFs as functional derivatives of the generating functional.

**Assessment:** Set A — contour ordering examples; $G^c$ for non-interacting system. Set B — Wick on contour; Keldysh action for free fermions. Projects: (simple) diagrammatic free-fermion GF on Keldysh contour; (complex) Keldysh generating functional: $G^<$ and $G^>$ as derivatives.

### Module II.5 / L03: Contour-Ordered Green's Functions

**90-min pacing:** (0–20 min) Six real-time GFs from contour decomposition: $G^c, G^{\bar{c}}, G^<, G^>, G^R, G^A$; (20–45 min) Keldysh GF $G^K = G^< + G^>$; Larkin–Ovchinnikov (LO) matrix $\hat{G} = \begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix}$; causality; (45–65 min) FDT at equilibrium: $G^K = (1-2f)(G^R - G^A)$; Lehmann for $G^R$ and $G^<$; (65–90 min) Analytic continuation Keldysh→Matsubara; connection at equilibrium. **Core learning outcomes:**

- Define all six real-time GFs and state their relations.
- Construct the LO matrix representation.
- Derive the FDT from Keldysh formalism.
- Connect Keldysh GFs to Matsubara GF at equilibrium.

**Assessment:** Set A — $G^R, G^<, G^>$ for non-interacting fermion; verify FDT. Set B —

LO matrix algebra; analytic continuation to Matsubara. Projects: (simple) GF component visualiser at various temperatures; (complex) LO Dyson equation for model self-energy.

### Module II.5 / L04: Langreth Rules

**90-min pacing:** (0–20 min) Need for Langreth rules: contour convolutions in real time; (20–50 min) Basic rules: $C^R = A^R B^R$; $C^< = A^R B^< + A^< B^A$; $C^K = A^R B^K + A^K B^A$; $n$-fold generalisations; (50–70 min) Mixed products for vertex corrections; (70–90 min) GW self-energy $\Sigma^R = iG^R W^< + iG^< W^R$; second-Born $\Sigma^< = i^2 G^< G^< G^>$; T-matrix. **Core learning outcomes:**

- State and derive Langreth rules for contour convolutions.
- Apply rules to convert contour diagrams to real-time expressions.
- Derive real-time GW and second-Born self-energies.

**Assessment:** Set A — apply Langreth rules to product of two GFs; HF self-energy in real time. Set B — GW $\Sigma^{R,<}$ from Langreth; verify equilibrium FDT. Projects: (simple) Langreth rule calculator; (complex) second-Born $\Sigma^{R,<}$: verify FDT at equilibrium.

### Module II.5 / L05: Kadanoff–Baym Equations

**90-min pacing:** (0–20 min) EOM for $G^c(t,t')$: apply $i\partial_t$; (20–50 min) KB equations for $G^<$ and $G^R$; two-time integro-differential structure; initial conditions; (50–65 min) Matsubara track for correlated initial states; (65–90 min) $\Phi$-derivable conserving approximations: $\Sigma = \delta\Phi/\delta G$; KB conservation theorem. **Core learning outcomes:**

- Write the KB equations for $G^R$ and $G^<$.
- Explain two-time structure and initial conditions.
- Define conserving approximations via the $\Phi$-functional.

**Assessment:** Set A — EOM for $G^<$; particle conservation for HF self-energy. Set B — derive KB from $\Phi$-functional; conserving approximation check. Projects: (simple) KB for two-level system: exact vs. HF; (complex) energy conservation in KB with second-Born for quantum dot.

### Module II.5 / L06: Self-Energies and Conserving Approximations

**90-min pacing:** (0–20 min) Self-energy hierarchy from $\Phi$-functional; HF $\Phi_{\mathrm{HF}}$; (20–45 min) GW: $\Phi_{\mathrm{GW}} = -\frac{i}{2}\,\mathrm{Tr}[\ln(1 - vP)]$; $W = v + vPW$; $\Sigma^{\mathrm{GW}} = iGW$; $\mathcal{O}(N^3)$/step; (45–65 min) Second Born: $\Sigma^{\mathrm{2B}} = i^2 VGGV$ + exchange; T-matrix: ladder resummation; (65–90 min) Comparison: accuracy vs. cost; benchmark vs. TDCI; KB conservation theorem. **Core learning outcomes:**

- Derive GW, second-Born, and T-matrix from $\Phi$-functionals.
- State the KB conservation theorem.
- Compare self-energy approximations in accuracy and scaling.

**Assessment:** Set A — HF and GW diagrams; identify conserving vs. non-conserving. Set B — $\Phi$ derivation of GW; KB conservation theorem proof. Projects: (simple) GW vs. HF spectral function for Hubbard dimer; (complex) benchmark GW vs. 2B vs. exact.

## Module II.5 / L07: Quantum Boltzmann Equation

**90-min pacing:** (0–20 min) Wigner transform: $T = (t + t')/2$, $\omega \leftrightarrow t - t'$; $G(T, \omega)$; (20–45 min) Gradient expansion of KB; QBE: $\partial_T f + \mathbf{v}_k \cdot \nabla f + \mathbf{F} \cdot \partial_k f = I_{\text{coll}}[f]$; Wigner distribution $f(T, \mathbf{k}) = G^</(2\pi)$; (45–65 min) Collision integral from self-energy; Born approximation; H-theorem; (65–90 min) Drude conductivity; electron–phonon scattering; relaxation-time approximation; beyond Born. **Core learning outcomes:**

- Perform Wigner transform and gradient expansion of KB equations.
- Derive the QBE and identify the collision integral.
- Compute Drude conductivity from the QBE.
- State the H-theorem and entropy production.

**Assessment:** Set A — Wigner transform of $G^<$; Drude from RTA. Set B — gradient expansion of KB; collision integral from Born self-energy. Projects: (simple) Boltzmann relaxation to equilibrium; (complex) electron–phonon QBE: $T$-dependent resistivity.

## Module II.5 / L08: Steady-State Transport

**90-min pacing:** (0–20 min) Nanojunction: central region + leads L,R at $\mu_L, \mu_R$; (20–50 min) Meir–Wingreen formula; embedding self-energy $\Sigma^R_{L/R}$; broadening $\Gamma_{L/R}$; Landauer non-interacting limit: $T(\omega) = \text{Tr}[\Gamma_L G^R \Gamma_R G^A]$; (50–70 min) Kondo effect; IETS; shot noise; Fano factor; (70–90 min) Experimental comparison: STM break junction; mechanically controlled break junction. **Core learning outcomes:**

- Derive the Meir–Wingreen formula.
- Obtain the Landauer formula in the non-interacting limit.
- Compute transmission for a molecular junction.
- Describe Kondo effect and IETS.

**Assessment:** Set A — Landauer for single-channel wire; transmission for resonant level. Set B — Meir–Wingreen derivation; Kondo temperature estimate. Projects: (simple) Landauer transmission: conductance vs. gate voltage; (complex) Meir–Wingreen for Anderson model: interacting vs. non-interacting.

## Module II.5 / L09: Electron–Phonon Coupling

**90-min pacing:** (0–20 min) Fröhlich Hamiltonian $\hat{H}_{\text{el-ph}} = \sum_{kq} g_{kq} \hat{c}^\dagger_{k+q} \hat{c}_k (\hat{b}_q + \hat{b}^\dagger_{-q})$; (20–45 min) Fan–Migdal electron self-energy; phonon self-energy from electrons; (45–65 min) Polaron: Fröhlich effective mass; Holstein small polaron; Lang–Firsov; (65–90 min) Eliashberg theory: $\alpha^2 F(\omega)$; $\lambda = 2 \int d\omega \alpha^2 F/\omega$; McMillan $T_c$; comparison to BCS. **Core learning outcomes:**

- Write the Fröhlich Hamiltonian and derive Fan–Migdal self-energy.
- Describe the polaron and compute its effective mass.
- Explain Eliashberg theory and the phonon contribution to $T_c$.

**Assessment:** Set A — Fröhlich coupling; electron lifetime; Drude resistivity. Set B — Fan–Migdal self-energy derivation; large polaron effective mass. Projects: (simple) electron spectral function renormalisation from phonons; (complex) Eliashberg $T_c$ from ab initio $\alpha^2 F$.

## Module II.5 / L10: Pump–Probe Dynamics

**90-min pacing:** (0–20 min) Femtosecond pump–probe; time-resolved ARPES; transient absorption; (20–45 min) Non-equilibrium spectral function $A(T, \omega)$; Wigner-transformed $G^<$; time-energy uncertainty; (45–65 min) KB equations for pump-driven system; Peierls substitution; photo-excited $f(T, \mathbf{k})$; (65–90 min) Relaxation: electron–electron thermalisation; electron–phonon cooling; Auger recombination; comparison to experiment. **Core learning outcomes:**

- Define time-resolved spectral function and relate to tr-ARPES.
- Set up KB equations for a laser-driven solid.
- Describe the hierarchy of relaxation processes after photo-excitation.

**Assessment:** Set A — non-equilibrium $f(\mathbf{k}, T)$ after Gaussian pulse; energy conservation check. Set B — Peierls substitution derivation; KB for driven two-band model. Projects: (simple) two-band model: population dynamics after laser pulse; (complex) tr-ARPES simulation for driven graphene.

## Module II.5 / L11: Driven-Dissipative Systems

**90-min pacing:** (0–20 min) Open quantum systems: Markovian bath; Lindblad from NEGF by tracing out bath; (20–45 min) Bath self-energy gives imaginary contribution; jump operators from bath spectral function; (45–65 min) Floquet–Keldysh: periodic drive + bath; NESS from Dyson equation; (65–90 min) Polariton condensation; quantum synchronisation; Floquet topological states; time-crystalline order. **Core learning outcomes:**

- Derive Lindblad by integrating out a Markovian bath in NEGF.
- Set up Floquet–Keldysh GF for a periodically driven open system.
- Compute NESS from the Dyson equation.

**Assessment:** Set A — Lindblad from bath self-energy; NESS for two-level system. Set B — Floquet–Keldysh GF structure; NESS for driven resonant level. Projects: (simple) NESS occupation for driven-dissipative resonant level; (complex) Floquet–Keldysh NESS for driven Haldane model.

## Module II.5 / L12: Numerical Methods for NEGF

**90-min pacing:** (0–20 min) Computational challenge: 2D time grid $\mathcal{O}(N_T^2 N_k^2)$ memory; $\mathcal{O}(N_T^3 N_k^3)$ cost; (20–45 min) Predictor–corrector (Adams–Bashforth–Moulton); stability; parallelisation; (45–65 min) GKBA: $G^<(t,t') \approx G^R(t,t')\rho(t') - \rho(t)G^A(t,t')$; $\mathcal{O}(N_T N_k^3)$; accuracy vs. full KB; (65–90 min) NESSi library; GPU acceleration; ML surrogate self-energies; benchmark vs. TDCI. **Core learning outcomes:**

- Explain the computational cost of KB equations on a 2D time grid.
- Describe the predictor–corrector algorithm for KB.
- Derive the GKBA and state its accuracy and cost advantage.
- Outline NESSi software and ML-based acceleration.

**Assessment:** Set A — scaling analysis; GKBA for driven two-level system. Set B — predictor–corrector derivation; stability analysis. Projects: (simple) full KB vs. GKBA for driven HO: accuracy and cost; (complex) GPU-accelerated KB for Hubbard dimer: scaling benchmarks.

***Module II.5 Reading:*** *Stefanucci & van Leeuwen, Nonequilibrium Many-Body Theory of Quantum Systems (Cambridge, 2013); Kadanoff & Baym, Quantum Statistical Mechanics; Haug & Jauho, Quantum Kinetics in Transport and Optics of Semiconductors.*

## 2.6 Module II.6 — Light–Matter Interaction, Linear Response, and Spectroscopy

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Classical and Quantum EM Fields | Coulomb gauge; mode expansion; photon; vacuum fluctuations |
| L02 | Light–Matter Coupling | Minimal coupling; dipole approximation; PZW; RWA; strong field |
| L03 | Linear Response & Kubo Formula | Retarded susceptibility; causality; Kramers–Kronig |
| L04 | Fluctuation–Dissipation Theorem | FDT; Johnson–Nyquist; quantum noise; effective temperature |
| L05 | Optical Conductivity | Drude from Kubo; $f$-sum rule; KK analysis; superfluid weight |
| L06 | Absorption and Emission | Fermiś golden rule; Einstein coefficients; line shapes |
| L07 | Selection Rules | $E1, M1, E2$; Laporte; molecular symmetry; IR vs. Raman |
| L08 | Non-Linear Response | $\chi^{(2)}, \chi^{(3)}$; SHG; TPA; FWM; 2DES |
| L09 | Time-Domain Spectroscopy | FID; photon echo; $R^{(3)}$ correlator; 2D IR/ES |
| L10 | Raman and Resonance Raman | Kramers–Heisenberg; Franck–Condon; SERS; TERS; CARS |
| L11 | Photoelectron Spectroscopy | Sudden approximation; Dyson orbital; ARPES; shake-up |
| L12 | Cavity QED & Strong Coupling | Jaynes–Cummings; vacuum Rabi splitting; polaritons; circuit QED |

### Lecture-by-Lecture Syllabi

**Module II.6 / L01: Classical and Quantum Electromagnetic Fields**

**90-min pacing:** (0–20 min) Maxwell; Coulomb gauge $\nabla \cdot \mathbf{A} = 0$; transverse/longitudinal; (20–45 min) Mode expansion; quantisation $[\hat{a}_{\mathbf{k}\lambda}, \hat{a}^\dagger_{\mathbf{k}'\lambda'}] = \delta_{\mathbf{k}\mathbf{k}'}\delta_{\lambda\lambda'}$; photon; (45–65 min) Vacuum fluctuations $\langle 0|\hat{\mathbf{E}}^2|0\rangle \neq 0$; Casimir effect; (65–90 min) Coherent states; Fock vs. coherent vs. thermal; Glauber–Sudarshan $P$-function; $g^{(1)}, g^{(2)}$. **Core learning outcomes:**

- Quantise the EM field in Coulomb gauge.
- Define photon operators and number states.
- Explain vacuum fluctuations and the Casimir effect.
- Distinguish coherent, Fock, and thermal states.

**Assessment:** Set A — EM energy in photon numbers; vacuum $\langle E^2\rangle$; Casimir force. Set B — commutation relations from canonical quantisation; $g^{(2)}$ for coherent vs. Fock. Projects: (simple) Casimir force vs. plate separation; (complex) Wigner function of single-photon Fock state.

## Module II.6 / L02: Light–Matter Coupling

**90-min pacing:** (0–20 min) Minimal coupling $\hat{H} = \frac{1}{2m}(\hat{\mathbf{p}} - e\mathbf{A})^2 + V$; expand to linear in $\mathbf{A}$; (20–45 min) Dipole approximation; length and velocity gauges; gauge equivalence via unitary transformation; (45–65 min) Power–Zienau–Woolley transformation; multipolar Hamiltonian; $E1, M1, E2$; (65–90 min) Strong field: beyond dipole; Floquet; RWA; driven two-level system. **Core learning outcomes:**

- Derive light–matter interaction from minimal coupling.
- Apply dipole approximation and verify gauge equivalence.
- Describe PZW transformation and multipolar Hamiltonian.
- Apply RWA to a driven two-level system.

**Assessment:** Set A — expand minimal coupling; dipole approximation; RWA for TLS. Set B — gauge equivalence proof; PZW transformation. Projects: (simple) Rabi oscillations: exact vs. RWA; (complex) Bloch–Siegert shift beyond RWA.

## Module II.6 / L03: Linear Response Theory and the Kubo Formula

**90-min pacing:** (0–20 min) Linear response: $\langle \delta\hat{B}(t)\rangle = \int \chi_{BA}(t-t')F(t')dt'$; (20–50 min) Kubo formula: $\chi_{BA}(t) = -i\theta(t)\langle[\hat{B}(t), \hat{A}(0)]\rangle_0$; causality $\Rightarrow$ upper-half analyticity; Kramers–Kronig relations; (50–70 min) Physical examples: dielectric, magnetic, optical conductivity; density–density response; (70–90 min) Wiener–Khinchin: $\mathrm{Im}\,\chi^R$ from equilibrium fluctuations. **Core learning outcomes:**

- Derive the Kubo formula for retarded susceptibility.
- Apply causality to derive Kramers–Kronig.
- Compute optical conductivity from current–current Kubo.
- Connect $\mathrm{Im}\,\chi^R$ to fluctuations via Wiener–Khinchin.

**Assessment:** Set A — $\chi^R(\omega)$ for damped HO; Kramers–Kronig check. Set B — Kubo derivation from TDPT; $f$-sum rule from commutator. Projects: (simple) KK verification for Lorentzian; (complex) optical conductivity for tight-binding chain from Kubo.

## Module II.6 / L04: The Fluctuation–Dissipation Theorem

**90-min pacing:** (0–20 min) FDT: $S_{BA}(\omega) = 2\hbar n_B(\omega)\mathrm{Im}\,\chi^R_{BA}(\omega)$; (20–45 min) Derivation from Lehmann: detailed balance; classical limit $\hbar\omega \ll k_BT$; (45–65 min) Johnson–Nyquist noise $S_V = 4k_BTR$; dissipation from noise; (65–90 min) Quantum noise; standard quantum limit; back-action; quantum optomechanics; FDT in non-equilibrium: effective temperature. **Core learning outcomes:**

- State the FDT and derive it from the Lehmann representation.
- Apply FDT to derive Johnson–Nyquist noise.
- Discuss FDT in non-equilibrium: effective temperature.

**Assessment:** Set A — FDT for RC circuit; Johnson–Nyquist noise; classical limit. Set B — Lehmann derivation; FDT for non-interacting fermion. Projects: (simple) FDT verification: numerical dynamics vs. $\mathrm{Im}\,\chi^R$; (complex) effective temperature from FDT in non-equilibrium electron gas.

## Module II.6 / L05: Optical Conductivity and the $f$-Sum Rule

**90-min pacing:** (0–20 min) $\sigma(\omega)$ from current–current Kubo; $\mathrm{Re}\,\sigma(\omega)$=power absorbed; (20–45 min) Drude from Kubo; $\sigma_{\mathrm{Drude}} = \frac{ne^2}{m}\frac{1}{\gamma - i\omega}$; Drude–Lorentz; (45–65 min) $f$-sum rule: $\int_0^\infty \mathrm{Re}\,\sigma\, d\omega = \frac{\pi n e^2}{2m}$; derivation from $[\hat{H}, \hat{j}]$; (65–90 min) Ellipsometry; KK analysis; Drude weight; superfluid density; Mott insulator gap. **Core learning outcomes:**

- Derive optical conductivity from current–current Kubo.
- Obtain Drude and Drude–Lorentz models.
- Prove and apply the $f$-sum rule.
- Interpret reflectance spectra via KK analysis.

**Assessment:** Set A — Drude parameters from spectrum; $f$-sum rule numerical check. Set B — $f$-sum derivation from Kubo; KK analysis of reflectance. Projects: (simple) Drude–Lorentz fit to metal optical conductivity; (complex) superfluid density from spectral weight: BCS vs. cuprate.

## Module II.6 / L06: Absorption, Emission, and Einstein Coefficients

**90-min pacing:** (0–20 min) Fermi's golden rule: $\Gamma_{i\to f} = \frac{2\pi}{\hbar}|\langle f|\hat{H}'|i\rangle|^2\rho(E_f)$; (20–45 min) Einstein: $A_{21}, B_{12}, B_{21}$; relations $A_{21} = \frac{\hbar\omega^3}{\pi^2 c^3}B_{21}$; Planck from detailed balance; (45–65 min) Spontaneous emission from QED: $\Gamma_{\mathrm{sp}} = \frac{e^2\omega^3}{3\pi\epsilon_0\hbar c^3}|d_{fi}|^2$; natural linewidth; (65–90 min) Line shapes: Lorentzian (lifetime); Gaussian (Doppler); Voigt; power broadening. **Core learning outcomes:**

- Apply Fermi's golden rule to compute absorption cross sections.
- Define Einstein coefficients and derive their relation.
- Compute spontaneous emission rate from first principles.
- Describe physical origins of line shapes.

**Assessment:** Set A — Einstein $A$ for H Lyman-$\alpha$; $B_{12}$ from $A_{21}$; natural linewidth. Set B — spontaneous emission from minimal coupling; Wigner–Weisskopf sketch. Projects: (simple) absorption spectrum vs. line shape parameters; (complex) Purcell factor: cavity modification of $A_{21}$.

## Module II.6 / L07: Selection Rules

**90-min pacing:** (0–20 min) Selection rules from AM conservation; $\Delta m = 0, \pm 1$; (20–45 min) $E1$ from W-E: $\Delta l = \pm 1$; parity change; Laporte rule; $\Delta j = 0, \pm 1$ (not $0 \to 0$); (45–65 min) $M1$: $\Delta l = 0$, parity unchanged; $E2$: $\Delta l = 0, \pm 2$; competition; (65–90 min) Molecular selection rules: symmetry group; $\langle\psi_f|\hat{\mu}|\psi_i\rangle \neq 0$ iff $\Gamma_f \otimes \Gamma_\mu \otimes \Gamma_i \supset A_1$; IR vs. Raman; mutual exclusion. **Core learning outcomes:**

- State and apply $E1$ selection rules from W-E.
- Identify Laporte-forbidden transitions and higher-order mechanisms.
- Apply molecular symmetry to determine selection rules.
- Distinguish IR-active from Raman-active modes.

**Assessment:** Set A — allowed/forbidden for H $1s \to 2s, 2p, 3d$; $CO_2$ IR modes. Set B — $E2$ matrix element; group theory for $C_{2v}$ molecule. Projects: (simple) atomic selection rule table; (complex) symmetry analysis of benzene normal modes: IR vs. Raman.

### Module II.6 / L08: Non-Linear Response

**90-min pacing:** (0–20 min) $\chi^{(2)}_{ijk}(\omega_1, \omega_2)$; $\chi^{(3)}$; (20–45 min) $\chi^{(2)}$: SHG; sum/difference frequency; optical rectification; phase matching; $\chi^{(2)} = 0$ centrosymmetric; (45–65 min) $\chi^{(3)}$: TPA; FWM; self-phase modulation; solitons; (65–90 min) Ultrafast: transient absorption; 2DES; double-sided Feynman diagrams; rephasing vs. non-rephasing. **Core learning outcomes:**

- Define $\chi^{(2)}$ and $\chi^{(3)}$ and identify associated phenomena.
- Explain phase matching for SHG.
- Describe TPA, FWM, and transient absorption.
- Interpret double-sided Feynman diagrams for 2DES.

**Assessment:** Set A — SHG phase matching; $\chi^{(2)} = 0$ for centrosymmetric; TPA cross-section. Set B — $\chi^{(3)}$ from PT; Liouville pathway for TA. Projects: (simple) SHG spectrum from phase-matching bandwidth; (complex) 2DES for coupled TLS: cross-peak identification.

### Module II.6 / L09: Time-Domain Spectroscopy

**90-min pacing:** (0–20 min) FID; NMR analogy; Fourier transform spectroscopy; (20–45 min) Photon echo: $\pi/2$–$\tau$–$\pi$; rephasing; inhomogeneous broadening removal; (45–65 min) Response from correlators: $\chi^R(t) \propto \theta(t)\langle[\hat{\mu}(t), \hat{\mu}(0)]\rangle$; $R^{(3)}$ as four-point correlator; (65–90 min) 2D IR; 2DES; waiting-time dynamics; quantum biology; energy transfer. **Core learning outcomes:**

- Explain FID and photon echo in terms of rephasing.
- Express linear and third-order response as correlators.
- Describe 2DES and interpret waiting-time dynamics.

**Assessment:** Set A — FID for TLS; photon echo timing; Fourier transform to spectrum. Set B — derive $R^{(3)}$ as four-point correlator; rephasing vs. non-rephasing pathway. Projects: (simple) 2DES for coupled TLS; (complex) 2D IR for Morse oscillator.

### Module II.6 / L10: Raman and Resonance Raman Spectroscopy

**90-min pacing:** (0–20 min) Raman: Stokes/anti-Stokes; cross-section $\propto (\omega_i \pm \Omega)^4|\alpha|^2$; (20–45 min) Kramers–Heisenberg formula; resonance enhancement; (45–65 min) Vibronic coupling; Condon; Herzberg–Teller; Franck–Condon overlaps; (65–90 min) SERS; TERS; single-molecule SERS; graphene $G$ and $D$ bands; CARS microscopy. **Core learning outcomes:**

- Derive the Kramers–Heisenberg formula and identify resonance condition.
- Compute Raman intensities from Franck–Condon factors.
- Describe SERS, TERS, and CARS.

**Assessment:** Set A — Raman spectrum for diatomic; Stokes/anti-Stokes ratio; selection rule. Set B — Kramers–Heisenberg derivation; Franck–Condon for displaced HO. Projects: (simple) Raman spectrum of $CO_2$; (complex) resonance Raman intensity vs. excitation frequency.

**Module II.6 / L11: Photoelectron Spectroscopy**

**90-min pacing:** (0–20 min) XPS/UPS: $E_B = h\nu - E_{\text{kin}} - \phi$; Koopman's theorem; (20–45 min) Sudden approximation; spectral function $A(\omega)$ directly measured; shake-up/shake-off satellites; (45–65 min) Dyson orbital; pole strength = quasiparticle weight $Z$; (65–90 min) ARPES: $E_{\text{kin}}(\theta, \phi)$ maps band structure; **k**-resolved spectral function; matrix element effects; tr-ARPES. **Core learning outcomes:**

- Explain the sudden approximation and relate PES to spectral function.
- Define the Dyson orbital and quasiparticle weight.
- Interpret ARPES as **k**-resolved spectral function.
- Identify shake-up satellites.

**Assessment:** Set A — binding energies from XPS; Koopman vs. ΔSCF; satellite identification. Set B — Dyson orbital derivation; quasiparticle weight from pole strength. Projects: (simple) simulated ARPES for tight-binding band; (complex) Fermi liquid spectral function: QP peak and incoherent background.

**Module II.6 / L12: Cavity QED and Strong Light–Matter Coupling**

**90-min pacing:** (0–20 min) Cavity QED: three regimes $(\kappa, g, \gamma)$; (20–45 min) Jaynes–Cummings: $\hat{H}_{\text{JC}} = \hbar\omega_c \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_0}{2}\hat{\sigma}_z + \hbar g(\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-)$; dressed states; vacuum Rabi splitting $2g$; (45–65 min) Polaritons: upper/lower branches; Hopfield coefficients; polariton condensation; (65–90 min) Ultra-strong coupling; photon blockade; circuit QED; quantum information applications. **Core learning outcomes:**

- Solve JC model and identify dressed states.
- Derive vacuum Rabi splitting and explain strong coupling.
- Describe polariton branches and Hopfield coefficients.
- Explain photon blockade and circuit QED.

**Assessment:** Set A — JC dressed states; Rabi splitting; photon blockade condition. Set B — JC dynamics: collapse and revival; Hopfield coefficients. Projects: (simple) JC dynamics: Fock vs. coherent state Rabi oscillations; (complex) polariton dispersion: upper/lower branches and Hopfield coefficients.

***Module II.6 Reading:*** *Mukamel, Principles of Nonlinear Optical Spectroscopy; Mandel & Wolf, Optical Coherence and Quantum Optics; Walls & Milburn, Quantum Optics; Atkins & Friedman, Molecular Quantum Mechanics.*

## 2.7 Module II.7 — Quantum Statistical Mechanics in Equilibrium

## Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Density Operators in Thermodynamics | Gibbs state; von Neumann entropy; max-entropy principle |
| L02 | Canonical and Grand-Canonical | Partition function; $\mu$; grand potential; ensemble equivalence |
| L03 | Quantum Ideal Gases | BE/FD distributions; Sommerfeld expansion; degeneracy pressure |
| L04 | BEC and Superfluidity | $T_c$; ODLRO; $^4$He; BKT transition |
| L05 | Fermi Liquid Theory | Quasiparticles; Landau parameters; non-Fermi liquid |
| L06 | Matsubara Greens Functions | Imaginary-time GF; Matsubara frequencies; analytic continuation |
| L07 | Perturbation Theory at Finite $T$ | Diagrammatics; Dyson; free energy; linked-cluster theorem |
| L08 | Phase Transitions I | Landau theory; order parameter; SSB; Ginzburg criterion |
| L09 | Phase Transitions II | Ising model; RG; universality; quantum phase transitions |
| L10 | Quantum Criticality | QCP; $\omega/T$ scaling; non-Fermi liquid; strange metal |
| L11 | Entanglement Many-Body | Area/volume law; entanglement entropy; topological order |
| L12 | Open Problems & Frontiers | High-$T_c$; FQH; quantum simulation; outlook |

## Lecture-by-Lecture Syllabi

### Module II.7 / L01: Density Operators in Thermodynamics

**90-min pacing:** (0–20 min) Statistical quantum mechanics: mixed states as ensembles; density operator $\hat{\rho}$; purity; thermal equilibrium state $\hat{\rho} = e^{-\beta\hat{H}}/Z$; (20–45 min) Partition function $Z = \mathrm{Tr}\, e^{-\beta\hat{H}}$; Helmholtz free energy $F = -k_B T \ln Z$; internal energy $U = -\partial_\beta \ln Z$; entropy $S = -k_B \mathrm{Tr}(\hat{\rho}\ln\hat{\rho})$ (von Neumann); heat capacity $C_V$; (45–65 min) Maximum entropy principle: Jaynes' derivation — maximise $S$ subject to $\mathrm{Tr}\,\hat{\rho} = 1$ and $\mathrm{Tr}(\hat{\rho}\hat{H}) = U$; Lagrange multipliers give Gibbs state; (65–90 min) Quantum information perspective: von Neumann entropy as entanglement entropy at zero temperature; relative entropy $S(\hat{\rho}\|\hat{\sigma}) = \mathrm{Tr}(\hat{\rho}\ln\hat{\rho} - \hat{\rho}\ln\hat{\sigma}) \geq 0$; trace distance; quantum thermodynamics preview.

**Core learning outcomes:**

- Write the Gibbs state and compute $F, U, S, C_V$ from $Z$.
- Derive the Gibbs state from the maximum entropy principle.
- Define von Neumann entropy and compare with thermodynamic entropy.
- Evaluate relative entropy and trace distance for simple density matrices.

**Dual-track content:**

- *Physical track:* Two-level system, harmonic oscillator, and spin-$\frac{1}{2}$ as paradigmatic examples; specific heat anomalies; Schottky anomaly.
- *Quantum information track:* Von Neumann entropy as information content; Jaynes' maximum entropy as Bayesian inference; relative entropy as statistical distinguishability.

**Tier differentiation:** HS: "thermal equilibrium is the state of maximum ignorance given the average energy." Beg. UG: $Z$ for two-level system; $F, U, S$; specific heat. Adv. UG: maximum entropy derivation; von Neumann entropy; Gibbs state uniqueness. MSc: relative entropy; trace distance; quantum thermodynamics (work extraction). PhD: quantum fluctuation theorems (Jarzynski, Crooks); quantum heat engines; resource theory of thermodynamics.

**Assessment:** Set A — $Z$, $F$, $U$, $S$, $C_V$ for two-level system and HO; Schottky anomaly temperature. Set B — Jaynes derivation of Gibbs state; von Neumann entropy of Gibbs state; relative entropy positivity proof. Projects: (simple) $C_V(T)$ for two-level vs. HO: Schottky peak vs. Planck; (complex) quantum Otto heat engine: efficiency from $Z$ of working fluid.

## Module II.7 / L02: Canonical and Grand-Canonical Ensembles

**90-min pacing:** (0–20 min) Canonical ensemble: system in thermal contact with a reservoir; $Z_N = \mathrm{Tr}_N\, e^{-\beta\hat{H}}$; grand canonical: additionally allows particle exchange; grand partition function $\Xi = \sum_{N=0}^{\infty} e^{\beta\mu N} Z_N = \mathrm{Tr}\, e^{-\beta(\hat{H}-\mu\hat{N})}$; (20–45 min) Grand potential $\Omega = -k_B T \ln \Xi$; mean particle number $\langle \hat{N} \rangle = -\partial\Omega/\partial\mu$; number fluctuations $\langle (\Delta N)^2 \rangle = k_B T \partial\langle N \rangle/\partial\mu$; pressure $P = -\partial\Omega/\partial V$; Gibbs–Duhem relation $SdT + VdP = Nd\mu$; (45–65 min) Deriving quantum statistics: single-mode grand canonical; Bose–Einstein $n_k = (e^{\beta(\varepsilon_k-\mu)} - 1)^{-1}$; Fermi–Dirac $n_k = (e^{\beta(\varepsilon_k-\mu)} + 1)^{-1}$; fugacity expansion; (65–90 min) Ensemble equivalence in the thermodynamic limit; corrections for finite systems; Yang–Lee theorem: zeros of $\Xi$ in the complex $z$-plane signal phase transitions; microcanonical saddle-point approximation.

**Core learning outcomes:**

- Set up the grand canonical ensemble and derive thermodynamic relations.
- Derive BE and FD distributions from the single-mode grand canonical.
- State ensemble equivalence and identify its breakdown.
- Apply the Yang–Lee theorem to phase transitions.

**Tier differentiation:** HS: "the chemical potential is the energy cost of adding one more particle." Beg. UG: grand canonical; BE/FD from single-mode; $\langle N \rangle$ vs. $\mu$. Adv. UG: Gibbs–Duhem; ensemble equivalence; Yang–Lee zeros. MSc: fugacity expansion; virial coefficients; grand canonical for interacting systems. PhD: Yang–Lee zeros and phase transitions; Lee–Yang theorem; complex fugacity in QFT.

**Assessment:** Set A — $\langle N \rangle, \Omega, P$ for non-interacting fermion gas; BE/FD distributions. Set B — Yang–Lee theorem statement; ensemble equivalence derivation via saddle point. Projects: (simple) $\langle N \rangle$ vs. $\mu$ for Bose and Fermi gas at various temperatures; (complex) virial expansion to second order: quantum corrections to classical ideal gas.

## Module II.7 / L03: Quantum Ideal Gases

**90-min pacing:** (0–20 min) Density of states in 3D: $D(\varepsilon) = \frac{V}{2\pi^2}(\frac{2m}{\hbar^2})^{3/2}\varepsilon^{1/2}$; grand potential $\Omega = \mp k_B T \int D(\varepsilon) \ln\big(1 \pm e^{-\beta(\varepsilon-\mu)}\big) d\varepsilon$; (20–45 min) Ideal Fermi gas: Fermi energy $E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$; Sommerfeld expansion: $\langle f(\varepsilon) \rangle = \int_0^\infty f(\varepsilon) n_F d\varepsilon = \int_0^\mu f\, d\varepsilon + \frac{\pi^2}{6}(k_B T)^2 f'(\mu) + \ldots$; specific heat $C_V = \frac{\pi^2}{3} k_B^2 T D(E_F)$; Pauli susceptibility; (45–65 min) Ideal Bose gas: $\mu < 0$ at all $T > T_c$; $\mu \to 0^-$ signals BEC; $\Omega$ for bosons; pressure and density in terms of Bose functions $g_\nu(z)$; (65–90 min) Low-$T$ thermodynamics: $T^3$ law for bosons; Fermi gas degeneracy pressure; applications: He-3 specific heat; white dwarf and neutron star equations of state; cold atoms.

**Core learning outcomes:**

- Compute thermodynamic quantities for ideal Bose and Fermi gases.
- Derive the Sommerfeld expansion and apply it to specific heat and susceptibility.
- Explain degeneracy pressure and its role in stellar physics.

- Describe the approach to BEC as $\mu \to 0^-$ in the ideal Bose gas.

**Tier differentiation:** HS: FD and BE distributions; electronic specific heat linear in $T$; degeneracy pressure in white dwarfs. Beg. UG: Fermi energy; Sommerfeld linear $C_V$; Pauli susceptibility; BEC $T_c$. Adv. UG: Sommerfeld expansion derivation; Bose functions; degeneracy pressure for neutron star. MSc: Landau diamagnetism; de Haas–van Alphen; Bose gas thermodynamics below $T_c$. PhD: strongly interacting Fermi gas at unitarity; polaron in Fermi sea; quantum critical Fermi gas.

**Assessment:** Set A — Fermi energy for Cu; $C_V$ from Sommerfeld; Pauli susceptibility; BEC $T_c$ for $^{87}$Rb. Set B — Sommerfeld expansion derivation; pressure of degenerate neutron star; Bose functions. Projects: (simple) $C_V(T)$ for Fermi gas: crossover from linear to Dulong–Petit; (complex) equation of state for white dwarf: degeneracy pressure vs. gravitational collapse.

## Module II.7 / L04: Bose–Einstein Condensation and Superfluidity

**90-min pacing:** (0–20 min) BEC in 3D ideal gas: $\mu \to 0^-$ at $T_c$; condensate fraction $N_0/N = 1 - (T/T_c)^{3/2}$; thermodynamics below $T_c$: pressure independent of $N$; (20–45 min) Off-diagonal long-range order (ODLRO): $\lim_{|\mathbf{r}-\mathbf{r}'|\to\infty}\langle\hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r}')\rangle = n_0 \neq 0$; broken $U(1)$ symmetry; Goldstone (phonon) mode; superfluidity from ODLRO; Landau criterion $v < c_s$; (45–65 min) Superfluid $^4$He: lambda transition; two-fluid model (normal + superfluid fractions); quantised vortices $\kappa = h/m$; vortex energy and critical rotation speed; (65–90 min) 2D systems: Mermin–Wagner theorem forbids true LRO in 2D; algebraic ODLRO $\sim r^{-\eta}$ below BKT; Kosterlitz–Thouless transition: vortex–anti-vortex unbinding at $T_{\mathrm{BKT}}$; superfluid density jumps at $T_{\mathrm{BKT}}$; experimental observation in thin He films and cold atoms.

**Core learning outcomes:**

- Derive BEC critical temperature and condensate fraction.
- Define ODLRO and connect to broken symmetry and Goldstone mode.
- Describe quantised vortices and the Landau criterion for superfluidity.
- State the BKT transition and its experimental signatures.

**Tier differentiation:** HS: "below $T_c$ a macroscopic number of bosons occupy the same state." Beg. UG: $T_c$; condensate fraction; Landau criterion; two-fluid model. Adv. UG: ODLRO; vortex energy; BKT transition. MSc: algebraic order in 2D; superfluid density jump; BKT in ultracold atoms. PhD: BKT from renormalisation group; topological defects; superfluid density from Kubo formula.

**Assessment:** Set A — $T_c$ and $N_0/N$ for $^4$He and $^{87}$Rb; Landau critical velocity from Bogoliubov spectrum. Set B — ODLRO derivation; Mermin–Wagner theorem sketch; BKT vortex unbinding energy. Projects: (simple) condensate fraction vs. $T/T_c$ and specific heat anomaly; (complex) BKT transition: vortex–anti-vortex density and correlation function decay.

## Module II.7 / L05: Fermi Liquid Theory

**90-min pacing:** (0–20 min) Landau's postulate: adiabatic continuity from non-interacting to interacting Fermi gas; quasiparticle concept; one-to-one correspondence of states; sharp Fermi surface; (20–45 min) Quasiparticle energy: $\varepsilon_k^\star = \varepsilon_k^0 + \sum_{k'} f_{kk'}\delta n_{k'}$; Landau interaction function $f_{kk'}$; Landau parameters $F_l^s, F_l^a$; effective mass $m^* = m(1 + F_1^s/3)^{-1}$; (45–65 min)

Thermodynamics from quasiparticles: specific heat $C_V = \frac{\pi^2}{3} k_B^2 T D^*(E_F)$ with renormalised DOS; compressibility $\kappa = \kappa_0/(1+F_0^s)$; Pauli susceptibility $\chi = \chi_0/(1+F_0^a)$; Wilson ratio $R_W = \chi/C_V$; (65–90 min) Quasiparticle lifetime: $\tau^{-1} \propto (\varepsilon - E_F)^2 + (\pi k_B T)^2$ from Fermi's golden rule and phase space; Fermi liquid $\Leftrightarrow \operatorname{Im}\Sigma \propto \omega^2$; breakdown: Tomonaga–Luttinger in 1D; non-Fermi liquid at quantum critical points.

**Core learning outcomes:**

- State Landau's postulate and explain the quasiparticle concept.
- Compute renormalised thermodynamic quantities from Landau parameters.
- Derive quasiparticle lifetime from phase-space arguments.
- Identify signatures of non-Fermi liquid behaviour.

**Tier differentiation:** HS: "interacting electrons still behave like free electrons but with a different mass." Beg. UG: quasiparticle concept; effective mass; Sommerfeld linear $C_V$ from FL. Adv. UG: Landau parameters; compressibility; susceptibility; Wilson ratio. MSc: quasiparticle lifetime derivation; Luttinger liquid in 1D. PhD: Luttinger's theorem; microscopic derivation of FL from self-energy; quantum critical non-FL.

**Assessment:** Set A — effective mass for $^3$He from $C_V$; Pauli susceptibility from $F_0^a$; Wilson ratio. Set B — quasiparticle lifetime derivation; Luttinger theorem statement and use. Projects: (simple) FL thermodynamics: $C_V, \chi, \kappa$ as function of $F_0^s, F_0^a, F_1^s$; (complex) marginal Fermi liquid self-energy: $\operatorname{Im}\Sigma \propto \omega$ and its experimental signatures.

## Module II.7 / L06: Imaginary-Time Formalism and Matsubara Green's Functions

**90-min pacing:** (0–20 min) Imaginary-time evolution $\hat{O}(\tau) = e^{\hat{H}\tau/\hbar}\hat{O}e^{-\hat{H}\tau/\hbar}$; thermal averages as Euclidean path integrals (recap); Matsubara GF $\mathcal{G}(\mathbf{k},\tau) = -\langle T_\tau \hat{c}_k(\tau)\hat{c}_k^\dagger(0)\rangle_\beta$; (anti-)periodicity in $\tau \in [0, \beta\hbar]$; (20–45 min) Fourier decomposition: fermionic $\omega_n = (2n+1)\pi/\beta\hbar$; bosonic $\omega_n = 2n\pi/\beta\hbar$; non-interacting Matsubara GF $\mathcal{G}_0(\mathbf{k}, i\omega_n) = (i\omega_n - \varepsilon_k)^{-1}$; Dyson equation $\mathcal{G} = \mathcal{G}_0 + \mathcal{G}_0\Sigma\mathcal{G}$; (45–65 min) Matsubara frequency sums: contour integration; $\sum_n f(i\omega_n) = -\operatorname{Res}[n_F(\omega)f(\omega)]$; evaluation of one-loop self-energy; (65–90 min) Analytic continuation $i\omega_n \to \omega + i\eta$: spectral representation; Padé approximants for numerical continuation; maximum entropy method (MEM); ill-posedness and regularisation; connection between Matsubara and real-time GFs.

**Core learning outcomes:**

- Define the Matsubara GF and perform Matsubara Fourier decomposition.
- Evaluate Matsubara frequency sums via contour integration.
- Apply the Dyson equation to find the interacting Matsubara GF.
- Perform analytic continuation and describe numerical methods (Padé, MEM).

**Tier differentiation:** HS: "imaginary-time GFs encode thermal equilibrium information." Beg. UG: Matsubara frequencies; $\mathcal{G}_0$; Dyson equation in Matsubara. Adv. UG: frequency sums via contour; one-loop self-energy; analytic continuation. MSc: Padé approximants; MEM; spectral function reconstruction. PhD: Nevanlinna analytic continuation; stochastic MEM; sparse-modelling approach.

**Assessment:** Set A — $\mathcal{G}_0$ for fermion and boson; Dyson equation for HF self-energy. Set B — Matsubara sum via contour; analytic continuation of model self-energy. Projects: (simple) Matsubara GF for Hubbard dimer: spectral function via analytic continuation; (complex) MEM analytic continuation: spectral function from QMC-generated $\mathcal{G}(\tau)$.

## Module II.7 / L07: Perturbation Theory at Finite Temperature

**90-min pacing:** (0–20 min) Finite-$T$ perturbation theory: split $\hat{H} = \hat{H}_0 + \hat{V}$; interaction picture in imaginary time; $\mathcal{G} = \mathcal{G}_0 + \mathcal{G}_0 \Sigma \mathcal{G}$; Wick's theorem in imaginary time; (20–45 min) Diagrammatic rules at finite $T$: Feynman diagrams identical to $T = 0$ but with Matsubara frequencies; propagator $\mathcal{G}_0(k, i\omega_n)$; vertices $V$; frequency conservation; sums over $i\omega_n$ instead of integrals; (45–65 min) Free energy from diagrams: $F = F_0 - k_B T \ln\langle T_\tau e^{-\int_0^\beta \hat{V} d\tau}\rangle_0$; linked-cluster theorem: $\ln Z = \ln Z_0 +$ (linked diagrams only); first-order: $F^{(1)} = \langle \hat{V}\rangle_0$; second-order: $F^{(2)}$; (65–90 min) Self-consistent Dyson at finite $T$: HF at finite $T$ (free energy minimisation); GW at finite $T$; finite-$T$ spectral function; Matsubara self-energy and quasiparticle renormalisation; phase boundary from self-energy divergence.

**Core learning outcomes:**

- Apply Wick's theorem in imaginary time to derive Matsubara diagrammatic rules.
- Compute the free energy perturbatively using the linked-cluster theorem.
- Apply Dyson's equation at finite $T$ for the HF and GW self-energies.
- Identify phase boundaries from divergences in the self-energy.

**Tier differentiation:** HS: "quantum perturbation theory at finite temperature uses oscillating imaginary-time diagrams." Beg. UG: linked-cluster theorem; first-order free energy; HF at finite $T$. Adv. UG: full diagrammatic rules; second-order $F$; Dyson at finite $T$. MSc: GW at finite $T$; finite-$T$ spectral function; Mott transition from self-energy. PhD: $\Phi$-derivable approximations at finite $T$; Luttinger–Ward functional; DMFT at finite $T$.

**Assessment:** Set A — first-order $F$ for Fermi gas with contact interaction; HF self-energy at finite $T$. Set B — linked-cluster theorem at finite $T$; second-order free energy for electron gas. Projects: (simple) HF free energy vs. $T$: paramagnetic to ferromagnetic transition (Stoner); (complex) GW spectral function at finite $T$: quasiparticle peak broadening.

## Module II.7 / L08: Phase Transitions I — Landau Theory

**90-min pacing:** (0–20 min) Order parameter concept: zero above $T_c$, non-zero below; examples: magnetisation ($\hat{M}$), condensate ($\psi_0$), sublattice magnetisation; continuous (second-order) vs. first-order transitions; (20–45 min) Landau free energy: $F(\phi, T) = a(T)\phi^2 + b\phi^4 + c\phi^6 + \ldots$ (symmetry $\phi \to -\phi$); $a(T) = a_0(T - T_c)$; minimisation: $\phi^* = 0$ for $T > T_c$, $\phi^* = \sqrt{-a/2b}$ for $T < T_c$; second-order: $b > 0$; first-order: $b < 0$ with $c > 0$; (45–65 min) Thermodynamic consequences: specific heat jump $\Delta C = a_0^2 T_c/2b$; susceptibility $\chi = (2|a|)^{-1}$ above $T_c$; critical exponents $\alpha, \beta, \gamma, \delta$ from Landau; (65–90 min) Ginzburg criterion: Landau breaks down near $T_c$; $\xi = \xi_0(T - T_c)^{-\nu}$; fluctuations $\langle(\delta\phi)^2\rangle \sim k_B T/\xi^d$; Ginzburg–Landau theory (spatial order parameter); coherence length; domain walls; upper critical dimension $d_c^+ = 4$.

**Core learning outcomes:**

- Construct the Landau free energy from symmetry and compute the order parameter.
- Distinguish first- and second-order transitions from the sign of $b$.
- Derive critical exponents $\alpha, \beta, \gamma, \delta$ from Landau theory.
- State the Ginzburg criterion and identify the upper critical dimension.

**Tier differentiation:** HS: "order parameter measures how ordered the system is; it grows continuously at a second-order transition." Beg. UG: Landau free energy; minimisation; $T_c$; $\Delta C$. Adv. UG: critical exponents; susceptibility; first- vs. second-order. MSc: Ginzburg criterion; Ginzburg–Landau theory; coherence length; domain walls. PhD: renormalisation

group improvement of Landau; $\epsilon$-expansion; tricritical points.

**Assessment:** Set A — minimise Landau $F$; $T_c$; $\phi^*(T)$; $\Delta C$; $\chi$ above $T_c$. Set B — first-order transition: discontinuous $\phi$; latent heat; Ginzburg criterion derivation. Projects: (simple) Landau free energy landscape: second-order and first-order as function of temperature; (complex) Ginzburg–Landau order parameter profile: domain wall solution and energy.

## Module II.7 / L09: Phase Transitions II — Ising Model and Renormalisation Group

**90-min pacing:** (0–20 min) Ising model: $\hat{H} = -J\sum_{\langle ij\rangle}\sigma_i\sigma_j - h\sum_i \sigma_i$; exact solution in 1D (no phase transition); exact solution in 2D (Onsager); (20–45 min) Mean-field theory: $\sigma_i \approx m + \delta\sigma_i$; self-consistency $m = \tanh(\beta z J m + \beta h)$; $T_c = zJ/k_B$; critical exponents $\beta = \frac{1}{2}, \gamma = 1, \delta = 3$; mean-field fails in low $d$; (45–65 min) Kadanoff block spin and renormalisation group (RG) idea: coarse-grain by blocks of $b$ spins; effective Hamiltonian; flow of couplings; fixed points and universality; Wilson's RG in $d = 4 - \epsilon$; (65–90 min) Universality classes: Ising, Heisenberg, XY, $O(N)$; critical exponents from RG; scaling hypothesis; quantum phase transitions: $\hat{H} = \hat{H}_0 + g\hat{V}$ at $T = 0$; quantum critical point (QCP); dimensional crossover; quantum Ising model (TFIM) as paradigm.

**Core learning outcomes:**

- Apply mean-field theory to the Ising model and compute critical exponents.
- Describe Kadanoff block-spin RG and explain universality.
- Identify universality classes and their experimental realisations.
- Distinguish classical from quantum phase transitions and describe the TFIM.

**Tier differentiation:** HS: "all second-order phase transitions near their critical point behave the same way — universality." Beg. UG: mean-field Ising; $T_c$; critical exponents; universality concept. Adv. UG: Kadanoff block spin; RG flow; fixed points; universality classes. MSc: Wilson RG in $4-\epsilon$ dimensions; $\epsilon$-expansion; quantum Ising model. PhD: conformal field theory at the critical point; operator product expansion; exact critical exponents in 2D.

**Assessment:** Set A — mean-field $T_c$ for 3D Ising; critical exponents; Ising exact solution in 1D. Set B — Kadanoff block-spin RG; fixed point analysis; $\epsilon$-expansion for $\beta$. Projects: (simple) Monte Carlo simulation of 2D Ising model: $M(T)$ and $T_c$; (complex) RG flow for the 1D quantum Ising model: quantum phase transition.

## Module II.7 / L10: Quantum Criticality

**90-min pacing:** (0–20 min) Quantum phase transition at $T = 0$: varying a non-thermal parameter $g$ (pressure, field, doping); quantum critical point (QCP) at $g = g_c$; diverging correlation length $\xi \propto |g - g_c|^{-\nu}$; diverging correlation time $\xi_\tau \propto \xi^z$ (dynamical exponent $z$); (20–45 min) Quantum critical region: $T > 0$ near QCP; fan-shaped quantum critical region in $(g, T)$ plane; $\omega/T$ scaling: correlators $C(\omega, T) = T^{2\Delta/z} f(\omega/T)$ from conformal invariance; "strange metal" regime: $\rho \propto T$ linear resistivity; (45–65 min) Non-Fermi liquid signatures: self-energy $\mathrm{Im}\,\Sigma \propto \omega$ or $T$ (vs. $\omega^2, T^2$ in FL); cuprate phase diagram: pseudogap, strange metal, FL; QCP under the superconducting dome; (65–90 min) Theoretical approaches: $(d+1)$-dimensional classical field theory at QCP; $1/N$ expansion; holographic duality (AdS/CFT) for strongly coupled QCPs; quantum spin liquid as frustrated QCP; experimental signatures: NMR, transport, neutron scattering.

**Core learning outcomes:**

- Define a QCP and its characteristic scales $\xi$ and $\xi_\tau$.
- State the $\omega/T$ scaling hypothesis and its physical origin.
- Identify non-Fermi liquid signatures in transport and spectral functions.
- Describe the cuprate phase diagram and the role of the putative QCP.

**Tier differentiation:** HS: "near a quantum phase transition, quantum fluctuations dominate over thermal ones." Beg. UG: QCP definition; diverging $\xi$; quantum critical fan; linear resistivity. Adv. UG: $\omega/T$ scaling; dynamical exponent $z$; non-FL self-energy. MSc: holographic duality; $1/N$ expansion; cuprate phase diagram. PhD: conformal field theory at QCP; Hertz–Millis theory; non-Fermi liquid from FL instabilities.

**Assessment:** Set A — quantum critical scaling: $\xi(g)$, $\xi_\tau(g)$; quantum critical fan width vs. $T$. Set B — $\omega/T$ scaling derivation from conformal invariance; non-FL self-energy and resistivity. Projects: (simple) quantum critical fan diagram for transverse-field Ising model; (complex) $\omega/T$ scaling: numerical correlation function for quantum Ising chain at QCP.

## Module II.7 / L11: Entanglement in Many-Body Systems

**90-min pacing:** (0–20 min) Entanglement entropy as a probe of many-body physics: $S_A = -\mathrm{Tr}(\hat{\rho}_A \ln \hat{\rho}_A)$ where $\hat{\rho}_A = \mathrm{Tr}_B\, \hat{\rho}$; area law: $S_A \propto L^{d-1}$ for gapped systems; volume law for generic excited states and thermal states; (20–45 min) 1D gapped systems: area law $S_A = \text{const}$; MPS as efficient representation (connection to DMRG); 1D critical systems: $S_A = \frac{c}{3} \ln L$ where $c$ is the central charge (conformal field theory); Calabrese–Cardy formula; (45–65 min) Higher dimensions: area law in 2D and 3D gapped phases; subleading corrections: topological entanglement entropy $S_A^{\mathrm{top}} = -\ln \mathcal{D}$ where $\mathcal{D}$ is the total quantum dimension; detection of topological order (Kitaev–Preskill, Levin–Wen); (65–90 min) Entanglement spectrum: eigenvalues $\lambda_i$ of $\hat{\rho}_A$; entanglement Hamiltonian $H_E = -\ln \hat{\rho}_A$; bulk–boundary correspondence; entanglement negativity for mixed states; multipartite entanglement measures.

**Core learning outcomes:**

- Define entanglement entropy and compute it for simple bipartitions.
- State the area law and the Calabrese–Cardy formula for critical systems.
- Define topological entanglement entropy and describe how it detects topological order.
- Describe the entanglement spectrum and its relation to the entanglement Hamiltonian.

**Tier differentiation:** HS: "entanglement tells us how quantum-correlated two parts of a system are." Beg. UG: entanglement entropy for Bell states; area law; volume law for thermal states. Adv. UG: Calabrese–Cardy formula; central charge; topological entanglement entropy. MSc: entanglement spectrum; entanglement Hamiltonian; Kitaev–Preskill construction. PhD: modular Hamiltonian in QFT; entanglement in holography (Ryu–Takayanagi); multipartite entanglement.

**Assessment:** Set A — $S_A$ for Bell state, GHZ, and product state; area law check for 1D gapped chain. Set B — Calabrese–Cardy formula for XX chain; topological entropy for toric code. Projects: (simple) $S_A$ vs. subsystem size $L$ for critical vs. gapped 1D chain (DMRG); (complex) topological entanglement entropy from DMRG for Kitaev honeycomb model.

## Module II.7 / L12: Open Problems and Frontiers

**90-min pacing:** (0–20 min) Overview of open problems: high-$T_c$ superconductivity; fractional quantum Hall (FQH) effect; quantum spin liquids; non-equilibrium many-body physics; quantum simulation; (20–40 min) High-$T_c$ superconductivity: cuprate phase diagram (underdoped, optimally doped, overdoped); pseudogap; strange metal; pairing mechanism (not phonon-mediated); competing orders; $d$-wave symmetry of the gap; key experiments (ARPES, neutron, NMR); (40–55 min) Fractional quantum Hall effect: Landau levels; fractional filling $\nu = \frac{1}{3}, \frac{2}{5}, \ldots$; Laughlin wavefunction; fractional charge and statistics (anyons); non-Abelian FQH states ($\nu = \frac{5}{2}$) and topological quantum computing; (55–70 min) Quantum simulation: analog quantum simulators (cold atoms, trapped ions); digital quantum simulation (VQE, QPE); resource estimation for quantum chemistry on fault-tolerant hardware; (70–90 min) Outlook: machine learning for many-body physics (neural network quantum states, ML force fields); tensor network methods for open systems; non-equilibrium topological phases; quantum advantage in many-body simulations.

**Core learning outcomes:**

- Describe the cuprate phase diagram and the open questions in high-$T_c$.
- Explain the FQH effect and the role of Laughlin's wavefunction.
- Outline quantum simulation approaches (analog and digital).
- Identify frontier methods: neural network states, ML, non-equilibrium topology.

**Tier differentiation:** HS: "these are the unsolved problems that the next generation of physicists will crack." Beg. UG: cuprate phenomenology; FQH basics; quantum simulation concept. Adv. UG: Laughlin wavefunction; fractional statistics; VQE algorithm. MSc: non-Abelian anyons; topological quantum computing; analog simulators. PhD: current research frontiers; neural network quantum states; quantum error correction for many-body.

**Assessment:** Set A — sketch cuprate phase diagram; Laughlin wavefunction for $\nu = \frac{1}{3}$; VQE ansatz design. Set B — fractional charge from Laughlin; anyonic braiding statistics; QPE resource estimation for FeMoco. Projects: (simple) cuprate phase diagram: literature survey of key experiments and their interpretation; (complex) Laughlin wavefunction: Monte Carlo evaluation of density and pair correlation.

***Module II.7 Reading:*** *Altland & Simons, Condensed Matter Field Theory (Cambridge); Sachdev, Quantum Phase Transitions, 2nd ed. (Cambridge); Goldenfeld, Lectures on Phase Transitions and the Renormalization Group; Wen, Quantum Field Theory of Many-Body Systems (Oxford).*

## 2.8 Module II.8 — Advanced Physical Systems (Lecture Menu)

**Series II / Module II.8** *Advanced Physical Systems — Lecture Menu*

**Format:** 12 stand-alone lecture topics, each self-contained. Instructors select any subset as optional enrichment lectures following Modules I.1–I.4 and the preceding Series II modules and Series II. Full expansion (90-min pacing, dual-track content, five-tier pedagogy, assessment bundles) to be developed per instructor need. Prerequisites for each topic listed in parentheses.

**Recommended audience:** Advanced UG, MSc, and PhD students who have completed Modules I.1–I.4 and Series II.

### Module Overview

| Lec. | Title | Key Concepts / Theme |
|---|---|---|
| L01 | 3D Spherically Symmetric Potentials | I.4-L09; spherical harmonics; effective potential; 3D scattering |
| L02 | The Hydrogen Molecular Ion ($H_2^+$) | I.4-L09; LCAO; Born–Oppenheimer; bonding/antibonding MOs |
| L03 | Identical Particles & Quantum Statistics | I.3-L10; exchange symmetry; bosons; fermions; Gibbs paradox |
| L04 | The 3D Isotropic Harmonic Oscillator | I.4-L03; degeneracy; $\mathfrak{su}(3)$ symmetry; shell model |
| L05 | Magnetic Resonance (NMR/ESR) | I.4-L04/L06; rotating frame; Bloch equations; relaxation $T_1, T_2$ |
| L06 | Coherence, Decoherence, & the Quantum Eraser | I.3-L01; Mach–Zehnder; which-path; environment-induced decoherence |
| L07 | Path Integrals: First Steps | I.3-L05; Feynman's sum over histories; free particle; harmonic oscillator |
| L08 | Adiabatic Theorem & Berry Phase | I.3-L04/L05; slow parameter change; geometric phase; adiabatic approximation |
| L09 | Quantum Entanglement & Bell Inequalities | I.3-L01; EPR; CHSH; loophole-free tests; quantum cryptography |
| L10 | The WKB Method in 3D | I.4-L12; radial WKB; Langer correction; 3D Bohr–Sommerfeld; barrier |
| L11 | Supersymmetric Quantum Mechanics | I.4-L03/L09; SUSY partner potentials; factorisation; shape invariance |
| L12 | Exactly Solvable Models beyond the HO | I.4-L03; Pöschl–Teller; Morse; Pöschl–Teller II; reflectionless potentials |

### Lecture Abstracts

**L01 — 3D Spherically Symmetric Potentials**

Generalisation from 1D to 3D for any central potential $V(r)$; separation into radial and angular parts; effective potential including the centrifugal barrier $\hbar^2 l(l+1)/2mr^2$; radial Schrödinger equation for $u_{nl} = rR_{nl}$; general behaviour at $r \to 0$ (regularity) and $r \to \infty$ (bound vs. scattering); 3D infinite spherical well (spherical Bessel functions); 3D scattering: partial-wave expansion; phase shifts $\delta_l$; optical theorem; application to neutron–proton scattering and nuclear potential models.

**L02 — The Hydrogen Molecular Ion ($H_2^+$)**

Simplest molecular system: one electron, two protons; Born–Oppenheimer PES; LCAO-MO ansatz with $1s$ atomic orbitals; bonding ($\sigma_g$) and antibonding ($\sigma_u^*$) MOs; overlap integral $S$, Coulomb integral $\alpha$, resonance integral $\beta$; variational optimisation of $Z_{\text{eff}}$; exact solution

in prolate spheroidal coordinates $(\xi, \eta, \phi)$; comparison of LCAO vs. exact; equilibrium bond length and dissociation energy; connection to Hückel and VB theories.

**L03 — Identical Particles and Quantum Statistics**
The identity postulate for indistinguishable particles; exchange operator $\hat{P}_{12}$; symmetric $(+1)$ and antisymmetric $(-1)$ subspaces; fermions (half-integer spin, antisymmetric) and bosons (integer spin, symmetric); spin–statistics theorem (statement); many-particle wavefunctions; permanents and determinants; Gibbs paradox resolved; exchange interaction as a purely quantum effect; emergence of Bose–Einstein and Fermi–Dirac statistics; brief discussion of anyons and parastatistics.

**L04 — The 3D Isotropic Harmonic Oscillator**
Hamiltonian $\hat{H} = \hat{p}^2/2m + m\omega^2 r^2/2$; three approaches: Cartesian (product of 1D HOs), spherical (radial equation + spherical harmonics), and algebraic ($\mathfrak{su}(3)$ creation operators); energy levels $E_N = (N + 3/2)\hbar\omega$ with $N = 2n_r + l$; degeneracy $\frac{1}{2}(N+1)(N+2)$; connection to nuclear shell model (magic numbers 2, 8, 20 from single-particle filling); $\mathfrak{su}(3)$ algebra and its physical interpretation.

**L05 — Magnetic Resonance (NMR and ESR)**
Larmor precession of a spin in a static field $B_0$; resonance condition $\omega_0 = \gamma B_0$; oscillating transverse field $B_1 \cos(\omega t)$; rotating frame transformation; Rabi formula for spin flip; Bloch equations including phenomenological relaxation: longitudinal $T_1$ (spin–lattice) and transverse $T_2$ (spin–spin); free induction decay; Fourier transform to spectrum; pulsed NMR: $\pi/2$ pulse, FID, spin echo; chemical shift and $J$-coupling; MRI gradient encoding.

**L06 — Coherence, Decoherence, and the Quantum Eraser**
Quantum coherence as off-diagonal elements of $\hat{\rho}$; Mach–Zehnder interferometer with a "which-path" detector; loss of interference when path information is available; quantum eraser: restoring interference by erasing which-path information; decoherence from environment entanglement: $\hat{\rho}_{\text{sys}} = \text{Tr}_E(\hat{\rho}_{\text{total}})$; pointer states; decoherence time scale $t_D \sim \hbar/\Delta E_{\text{env}}$; connection to the measurement problem; Zurek's einselection; experimental realisations.

**L07 — Path Integrals: First Steps**
Feynman's sum over histories as the third formulation of QM; transition amplitude $K(x_b, t_b; x_a, t_a) = \int \mathcal{D}x(t)\, e^{iS[x]/\hbar}$; derivation from the short-time propagator; free particle: exact Gaussian integral; harmonic oscillator: classical action + fluctuation determinant; stationary phase and the classical limit; imaginary-time path integral and the partition function; brief discussion of field-theory generalisation; connection to PIMC (Module II.2).

**L08 — Adiabatic Theorem and Berry Phase**
Adiabatic theorem: slow variation of $\hat{H}(\mathbf{R}(t))$ keeps the system in the instantaneous eigenstate if the gap remains non-zero; quantum adiabatic condition $|\langle m|\dot{\hat{H}}|n\rangle|/(E_n - E_m)^2 \ll 1$; dynamical phase vs. geometric phase; Berry's phase $\gamma_n = i \oint \langle n(\mathbf{R})| \nabla_{\mathbf{R}} |n(\mathbf{R})\rangle \cdot d\mathbf{R}$; Berry connection and curvature; examples: spin-$\frac{1}{2}$ in a rotating field (solid-angle formula); Aharonov–Bohm as Berry phase; non-Abelian Berry phase; connection to topology.

**L09 — Quantum Entanglement and Bell Inequalities**
EPR paradox and local hidden variables; Bell's theorem; CHSH inequality $|S| \leq 2$ (classical) vs. $|S| \leq 2\sqrt{2}$ (quantum); derivation for singlet state; loophole-free experimental tests (2015 onwards); implications for non-locality; quantum cryptography: BB84 protocol; E91 entanglement-based QKD; quantum teleportation; entanglement as a resource; entanglement measures: concurrence, entanglement of formation; monogamy of entanglement.

**L10 — The WKB Method in 3D**

Extension of 1D WKB to 3D spherically symmetric potentials; radial WKB wavefunction; Langer correction $l(l+1) \to (l+\frac{1}{2})^2$ for centrifugal barrier near origin; 3D Bohr–Sommerfeld quantisation $\int_{r_1}^{r_2} p_r\, dr = \pi\hbar(n_r+\frac{1}{2})$; application to hydrogen: recover $E_n = -Z^2/2n^2$ a.u.; 3D tunnelling: barrier-penetration in a central field; alpha-decay Gamow factor in 3D; connection to quasi-classical limit and semiclassical mechanics.

**L11 — Supersymmetric Quantum Mechanics**

Factorisation of $\hat{H} = \hat{A}^\dagger\hat{A} + E_0$; supercharge operators $\hat{Q} = \hat{A}$, $\hat{Q}^\dagger = \hat{A}^\dagger$; SUSY partner potentials $V_\pm = W^2 \pm W'$ (superpotential $W$); isospectrality: $V_-$ and $V_+$ have identical spectra except ground state; shape invariance: hierarchy of solvable potentials; exactly solvable potentials from SUSY: infinite square well, harmonic oscillator, hydrogen; broken vs. unbroken SUSY; Witten index; connection to integrability and exactly solvable models.

**L12 — Exactly Solvable Models beyond the Harmonic Oscillator**

Criteria for exact solvability; Pöschl–Teller potential $V(x) = -V_0/\cosh^2(x/a)$: hypergeometric equation; $N$ bound states for $V_0 = \hbar^2 N(N+1)/2ma^2$; reflectionless property; Morse potential $V(r) = D_e(1-e^{-a(r-r_e)})^2$: exact vibrational levels $E_n = \hbar\omega_e(n+\frac{1}{2}) - x_e\hbar\omega_e(n+\frac{1}{2})^2$ (finite anharmonicity); Pöschl–Teller II (symmetric Rosen–Morse); inverse square potential; connection to SUSY hierarchy (L11); experimental relevance: diatomic molecules, optical solitons.

***Module II.8 Note:*** *These lectures are intended as optional enrichment modules. Each can be taught as a standalone 90-minute lecture. Full expansion to the standard format (90-min pacing, dual-track content, five-tier pedagogy, Problem Sets A & B, mini-projects) can be carried out on request. Prerequisites listed in the overview table are recommended but not strictly required for strong students.*

## 2.9 Module II.9 — Quantum Information and Quantum Logic

**Series II / Module II.9** *Quantum Information and Quantum Logic*

**Duration:** 12 lectures × 90 minutes. Prerequisites: Modules I.1–I.3 (bra-ket, density matrices, measurement); Module II.4 (Fock space and many-body formalism) recommended. This module develops the mathematical language and operational framework of quantum information processing from first principles.

## Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Qubits, Qutrits, and Qudits | Two-level quantum systems; Bloch sphere; generalisation to $d$-level; Hilbert space $\mathbb{C}^d$ |
| L02 | Quantum Gates and Circuits | Unitary gates; Pauli, Hadamard, CNOT, Toffoli; universality; circuit model |
| L03 | Entanglement: Measures and Classification | Bell states; Schmidt rank; concurrence; entanglement of formation; PPT criterion |
| L04 | Quantum Teleportation and Superdense Coding | Bell measurement; classical communication; channel capacity; no-cloning |
| L05 | Quantum Error Correction I | Bit-flip and phase-flip codes; stabiliser formalism; syndrome measurement |
| L06 | Quantum Error Correction II | Shor code; CSS codes; fault-tolerant thresholds; surface code overview |
| L07 | Quantum Channels and CPTP Maps | Kraus operators; Stinespring dilation; depolarising, dephasing, amplitude-damping channels |
| L08 | Quantum Entropy and Information | von Neumann entropy; quantum mutual information; conditional entropy; strong subadditivity |
| L09 | Quantum Algorithms I | Deutsch–Jozsa; Bernstein–Vazirani; Simon's algorithm; query complexity |
| L10 | Quantum Algorithms II | Quantum Fourier transform; phase estimation; Shor's factoring algorithm |
| L11 | Quantum Algorithms III | Grover's search; amplitude amplification; quantum walks; HHL algorithm |
| L12 | Quantum Complexity and Limits | BQP; QMA; quantum advantage; limits of quantum speedup; NISQ outlook |

***Module II.9 Reading:*** *Nielsen & Chuang, Quantum Computation and Quantum Information (Cambridge) — primary reference; Preskill, Lecture Notes on Quantum Computation (available online); Wilde, Quantum Information Theory, 2nd ed. (Cambridge).*

## 2.10 Module II.10 — Signature Systems of Quantum Information and Quantum Logic

**Series II / Module II.10** *Signature Systems of Quantum Information and Quantum Logic*

**Duration:** 12 lectures × 90 minutes. Each lecture focuses on one paradigmatic physical or logical system that exemplifies the principles of Module II.9. Lectures are largely self-contained and may be selected as case studies. Prerequisite: Module II.9.

### Module Overview

| Lec. | Title | Key System / Focus |
|---|---|---|
| L01 | The Spin-$\frac{1}{2}$ Qubit | Physical qubit; Larmor precession; NMR qubit; Bloch sphere dynamics |
| L02 | The Superconducting Transmon Qubit | Josephson junction; charge qubit; transmon regime; dispersive readout |
| L03 | The Trapped-Ion Qubit | Laser–ion interaction; Mølmer–Sørensen gate; NIST and Oxford experiments |
| L04 | The Photonic Qubit | Dual-rail encoding; KLM scheme; boson sampling; photonic cluster states |
| L05 | Quantum Key Distribution (BB84 and E91) | Prepare-and-measure; entanglement-based; security proof; device-independence |
| L06 | The Quantum Repeater | Entanglement swapping; quantum memory; DLCZ protocol; Bell-state analyser |
| L07 | Quantum Error Correction: The Surface Code | Planar code; stabilisers; logical qubit; threshold theorem; Google/IBM experiments |
| L08 | Measurement-Based Quantum Computation | Cluster states; one-way quantum computer; graph states; adaptive measurements |
| L09 | Quantum Simulation: Bose–Hubbard Realisation | Optical lattice; superfluid–Mott transition; analogue quantum simulation |
| L10 | The Variational Quantum Eigensolver (VQE) | Parameterised ansatz; UCCSD; classical optimiser; near-term quantum chemistry |
| L11 | Quantum Sensing and Metrology | Ramsey interferometry; quantum Fisher information; Heisenberg limit; GHZ state |
| L12 | Topological Quantum Computation Preview | Anyons; Majorana modes; non-Abelian braiding; fault-tolerance without error correction |

***Module II.10 Reading:*** *Aaronson, Quantum Computing Since Democritus; Ladd et al., Nature* ***464****, 45 (2010) — physical implementations; Fowler et al., Phys. Rev. A* ***86****, 032324 (2012) — surface code.*

## 2.11 Module II.11 — The Tenfold Way and Symmetry Classification of Topological Phases

**Series II / Module II.11** *The Tenfold Way and Symmetry Classification of Topological Phases*

**Duration:** 12 lectures × 90 minutes. Prerequisites: Modules II.3 (angular momentum, time-reversal), II.4 (second quantisation), II.7 (statistical mechanics); Module II.9 recommended. This module develops the complete classification of non-interacting fermionic topological phases by symmetry class and spatial dimension.

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Symmetry Classes: Time-Reversal, Particle–Hole, Chiral | Antiunitary symmetries; $\hat{T}^2 = \pm 1$, $\hat{C}^2 = \pm 1$; chiral $\hat{S} = \hat{T}\hat{C}$; Wigner's theorem revisited |
| L02 | Cartan's Classification and the Tenfold Way | Ten Altland–Zirnbauer symmetry classes; Cartan labels (A, AI, AII, …, CI); table of classes |
| L03 | Topological Invariants in 0D and 1D | $\mathbb{Z}$ and $\mathbb{Z}_2$ invariants; SSH model; winding number; Zak phase; Majorana end modes |
| L04 | Topological Invariants in 2D | Chern number; TKNN formula; $\mathbb{Z}_2$ Kane–Mele invariant; time-reversal polarisation |
| L05 | The Periodic Table of Topological Insulators | Classifying space; Bott periodicity; $(d, \delta)$-dimensional table; eight-fold periodicity in $d$ |
| L06 | Bulk–Boundary Correspondence | Spectral flow; index theorem; edge states as topological zero modes; K-theory sketch |
| L07 | Interacting Topological Phases and SPT Orders | Beyond free fermions; group cohomology; SPT phases; symmetry-protected vs. intrinsic topological order |
| L08 | Topological Semimetals and Weyl Physics | Weyl nodes as monopoles; surface Fermi arcs; chiral anomaly; Nielsen–Ninomiya theorem |
| L09 | Floquet Topological Phases | Periodically driven systems; anomalous Floquet topological insulator; $\pi$-mode edge states |
| L10 | Topological Superconductors | Class D and DIII; $p$-wave pairing; Majorana zero modes; Kitaev chain classification |
| L11 | Disorder and Topological Robustness | Anderson localisation; topological protection; symmetry-class-dependent DOS at zero energy |
| L12 | Experimental Signatures and Outlook | ARPES of topological surface states; transport anomalies; non-Abelian braiding experiments |

***Module II.11 Reading:*** *Chiu et al., Rev. Mod. Phys.* ***88****, 035005 (2016) — tenfold way review; Hasan & Kane, Rev. Mod. Phys.* ***82****, 3045 (2010) — topological insulators; Altland & Zirnbauer, Phys. Rev. B* ***55****, 1142 (1997) — original classification.*

## 2.12 Module II.12 — Signature Physical Systems of the Tenfold Way

**Series II / Module II.12** *Signature Physical Systems of the Tenfold Way*

**Duration:** 12 lectures × 90 minutes. Each lecture presents one canonical physical system that realises a distinct symmetry class or spatial dimension of the tenfold way classification. Lectures are self-contained case studies and may be selected individually. Prerequisite: Module II.11.

### Module Overview

| Lec. | Title | System / Class / Key Physics |
|---|---|---|
| L01 | The SSH Model (Class BDI, 1D) | Polyacetylene; dimerised chain; winding number $\in \mathbb{Z}$; edge modes; solitons |
| L02 | The Kitaev Chain (Class D, 1D) | $p$-wave superconductor; Majorana fermions; topological $\mathbb{Z}_2$ invariant; non-Abelian statistics |
| L03 | Integer Quantum Hall Effect (Class A, 2D) | Landau levels; Chern number = Hall conductance; TKNN; edge states; plateau robustness |
| L04 | The Quantum Spin Hall Insulator (Class AII, 2D) | HgTe/CdTe quantum well; Kane–Mele model; $\mathbb{Z}_2$ invariant; helical edge states; experiment |
| L05 | The 3D Topological Insulator (Class AII, 3D) | $Bi_2Se_3$; surface Dirac cone; spin-momentum locking; ARPES observation; gapping by magnetism |
| L06 | Topological Crystalline Insulators (Class A with $C_{4v}$) | SnTe; mirror Chern number; crystalline symmetry protection; surface states on high-symmetry faces |
| L07 | The Kitaev Honeycomb Model (Class D, 2D) | Exactly solvable spin liquid; Majorana bands; abelian and non-Abelian phases; $\mathbb{Z}_2$ flux |
| L08 | Helical Topological Superconductors (Class DIII, 3D) | Time-reversal invariant $p$-wave; surface Majorana cone; $\mathbb{Z}$ classification; $^3$He-B analogy |
| L09 | Weyl Semimetals (Class A, 3D) | TaAs; Weyl nodes; Fermi arcs; chiral anomaly; negative magnetoresistance; ARPES |
| L10 | Floquet Topological Insulator (Class A, Driven 2D) | Circularly polarised light on graphene; anomalous edge states; Floquet Chern insulator |
| L11 | Higher-Order Topological Insulators (HOTI) | Corner and hinge states; quadrupole insulator; breathing kagome; $C_4$-protected |
| L12 | Interacting Topological Phases: Fractional QHE | Laughlin state; composite fermions; $\nu = 5/2$ non-Abelian state; experimental signatures |

***Module II.12 Reading:*** *Chiu et al., Rev. Mod. Phys.* ***88****, 035005 (2016); Ando & Fu, Annu. Rev. Condens. Matter Phys.* ***6****, 361 (2015) — topological crystalline; Armitage, Mele & Vishwanath, Rev. Mod. Phys.* ***90****, 015001 (2018) — Weyl semimetals.*

SERIES III

# Molecular Electronic Structure & Quantum Chemistry

*Three modules — 31 lectures × 90 minutes*

*Audience: Graduate students in chemistry and chemical physics*

*Prerequisites: Series I required; Modules II.1 and II.4 strongly recommended*

| Module | Title | Lectures |
|---|---|---|
| **III.1** | Basic Premises of Quantum Chemistry | 3 |
| | Molecular Hamiltonian ($\hat{T}_n + \hat{T}_e + \hat{V}_{ne} + \hat{V}_{ee} + \hat{V}_{nn}$); Born–Oppenheimer approximation; adiabatic potential energy surfaces; non-adiabatic couplings; Feynman–Hellmann theorem; force calculations. | |
| **III.2** | Principles of Molecular and Atomic Electronic Structure | 16 |
| | Hydrogenic systems; polyelectronic wavefunctions; spin orbitals; Slater determinants; Hartree–Fock; Roothaan–Hall; Koopman's theorem; configuration interaction; orbital theories; natural and localised orbitals; basis sets (STO, GTO, Pople, Dunning, Karlsruhe); molecular integrals. | |
| **III.3** | Density Functional Theory and Beyond | 12 |
| | Hohenberg–Kohn theorems; Thomas–Fermi–Weizsäcker; Kohn–Sham DFT; XC functionals (LDA, GGA, hybrid, double-hybrid); Fukui functions; tight-binding; Hückel/Möbius; crystal/ligand field theory; full CI QMC; diffusion QMC; method comparison and outlook. | |
| | **Total** | **31** |

*Focus: first-principles molecular physics · From exact wavefunction methods to DFT · Computational & theoretical depth*

# 3 Series III — Molecular Electronic Structure and Quantum Chemistry

**Series III Overview**

**Prerequisite:** Series I required; Series II Modules II.1, II.4 strongly recommended.
**Structure:** Three modules totalling approximately 31 lectures.
**Core aim:** Rigorous, graduate-level understanding of molecular electronic structure from first principles through to modern computational methods.

| Module | Title | Lectures |
|---|---|---|
| III.1 | Basic Premises of Quantum Chemistry | 3 |
| III.2 | Principles of Molecular and Atomic Electronic Structure | 16 |
| III.3 | Density Functional Theory and Beyond | 12 |
| | **Total** | **31** |

## 3.1 Module III.1 — Basic Premises of Quantum Chemistry

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | The Molecular Hamiltonian | $\hat{T}_n + \hat{T}_e + \hat{V}_{ne} + \hat{V}_{ee} + \hat{V}_{nn}$; atomic units; degrees of freedom; symmetries |
| L02 | Born–Oppenheimer Approximation | Adiabatic PES; non-adiabatic couplings; derivative couplings; conical intersections |
| L03 | Feynman–Hellmann Theorem | $\partial_\lambda E = \langle \partial_\lambda \hat{H} \rangle$; forces; geometry optimisation; analytic gradients |

### Lecture-by-Lecture Syllabi

**Module III.1 / L01: The Molecular Hamiltonian**

**90-min pacing:** (0–20 min) The complete molecular Hamiltonian in SI and atomic units: $\hat{H} = \hat{T}_n + \hat{T}_e + \hat{V}_{ne} + \hat{V}_{ee} + \hat{V}_{nn}$; nuclear KE $\hat{T}_n = -\sum_A \frac{\hbar^2}{2M_A}\nabla_A^2$; electronic KE $\hat{T}_e = -\sum_i \frac{\hbar^2}{2m_e}\nabla_i^2$; electron–nuclear attraction $\hat{V}_{ne} = -\sum_{iA} \frac{Z_A e^2}{r_{iA}}$; electron–electron repulsion $\hat{V}_{ee} = \sum_{i<j} \frac{e^2}{r_{ij}}$; nuclear–nuclear repulsion $\hat{V}_{nn} = \sum_{A<B} \frac{Z_A Z_B e^2}{R_{AB}}$; (20–45 min) Atomic units: Hartree $E_h = 27.211$ eV; Bohr $a_0 = 0.529$ Å; time $\hbar/E_h = 24.2$ as; mass ratio $m_e/M_p \approx 1/1836$; physical intuition for each scale; (45–70 min) Degrees of freedom: $3N_e$ electronic + $3N_n$ nuclear; total $3(N_e + N_n)$ minus COM; translational/rotational invariance; remaining internal coordinates; electronic Schrödinger equation at fixed $\{\mathbf{R}_A\}$; (70–90 min) Symmetries of the molecular Hamiltonian: permutation symmetry of electrons (Pauli); permutation of identical nuclei; point-group symmetry; time-reversal; connection between molecular symmetry and observable selection rules.

**Core learning outcomes:**

- Write the full molecular Hamiltonian and identify each physical contribution.
- Convert between SI and atomic units fluently.
- Count the independent degrees of freedom for a given molecule.
- Identify the symmetries of $\hat{H}$ and their consequences.

**Dual-track content:**

- *Physical/chemical track:* Physical meaning of each term; energy scales ($\hat{V}_{ee}$ vs. $\hat{T}_e$ vs. $\hat{V}_{ne}$ for $H_2$); why $\hat{V}_{ee}$ is the source of difficulty.
- *Mathematical track:* Schrödinger equation as a PDE; separation of COM motion; permutation group $S_{N_e}$; point groups as symmetry of the Hamiltonian operator.

**Tier differentiation:** HS: the molecular Hamiltonian as "the rule book for quantum chemistry." Beg. UG: write $\hat{H}$ in atomic units for $H_2$; count degrees of freedom. Adv. UG: symmetry operators that commute with $\hat{H}$; block-diagonal structure. MSc: permutation symmetry of $\hat{H}$; parity; time-reversal; vibrational normal modes. PhD: Born–Huang expansion; relativistic corrections (Breit–Pauli); quantum electrodynamic corrections.

**Assessment:** Set A — write $\hat{H}$ for $H_2$, $H_2O$, $CH_4$ in atomic units; count internal DOF. Set B — symmetry analysis: which symmetry operators commute with $\hat{H}$ for a given point group? Projects: (simple) plot potential energy terms vs. $r$ for $H_2^+$; (complex) relativistic corrections: mass–velocity, Darwin, spin–orbit for He using perturbation theory.

## Module III.1 / L02: Born–Oppenheimer Approximation and Adiabaticity

**90-min pacing:** (0–20 min) Physical motivation: nuclei are $10^3$–$10^5$ times heavier than electrons; electronic motion is "infinitely fast" on nuclear timescale; decoupling ansatz $\Psi(\mathbf{r}, \mathbf{R}) = \psi_e(\mathbf{r}; \mathbf{R})\chi(\mathbf{R})$; (20–45 min) Formal BO derivation: expand full wavefunction in adiabatic electronic states; project onto single state; leading BO approximation neglects non-adiabatic coupling $\mathbf{d}_{IJ} = \langle\psi_I|\nabla_{\mathbf{R}}|\psi_J\rangle/(E_J - E_I)$; adiabatic potential energy surface (PES) $V_I(\mathbf{R}) = E_I(\mathbf{R}) + V_{nn}(\mathbf{R})$; (45–65 min) Validity of BO: small $m_e/M_n$; large energy gap $E_J - E_I$; breakdown near conical intersections and avoided crossings; derivative couplings and non-adiabatic dynamics; (65–90 min) Beyond BO: Born–Huang expansion; diabatic states; non-adiabatic matrix elements; *ab initio* molecular dynamics; Tully surface hopping; photochemistry and conical intersections (Jahn–Teller, Renner–Teller).

**Core learning outcomes:**

- Derive the BO approximation and state its validity conditions.
- Define the adiabatic PES and the derivative coupling $\mathbf{d}_{IJ}$.
- Explain conical intersections and their role in photochemistry.
- Describe the Born–Huang expansion and diabatic representations.

**Dual-track content:**

- *Chemical track:* PES as the foundation of all molecular structure and dynamics; conical intersections in vision, photosynthesis, and photostability of DNA; Jahn–Teller distortions in transition metal complexes.
- *Mathematical track:* Adiabatic theorem; Berry phase accumulated around a conical intersection; $\mathbf{d}_{IJ}$ as a gauge connection; diabatic vs. adiabatic gauge.

**Tier differentiation:** HS: "nuclei are slow; electrons adjust instantly — that's the Born–Oppenheimer idea." Beg. UG: BO ansatz; PES definition; validity conditions. Adv. UG: formal BO derivation; derivative couplings; conical intersection definition. MSc: Born–Huang expansion; diabatisation; Tully surface hopping. PhD: geometric phase around CI; Longuet-Higgins sign change; exact factorisation of electron–nuclear wavefunction.

**Assessment:** Set A — write BO decoupling ansatz; identify conditions for BO validity; sketch a conical intersection seam. Set B — derive the BO Hamiltonian; show derivative coupling diverges at CI; diabatisation matrix. Projects: (simple) PES scan for $H_2$: BO vs. exact; (complex) conical intersection in $H_3$: geometric phase and Longuet-Higgins theorem.

**Module III.1 / L03: Parametric Dependence and the Feynman–Hellmann Theorem**

**90-min pacing:** (0–20 min) Parametric dependence of the electronic energy: $E(\lambda) = \langle\psi(\lambda)|\hat{H}(\lambda)|\psi(\lambda)\rangle$; first-order change in energy; (20–45 min) Feynman–Hellmann theorem: $\partial E/\partial\lambda = \langle\psi|\partial\hat{H}/\partial\lambda|\psi\rangle$; derivation from stationarity of energy; (45–65 min) Applications: nuclear forces from $\partial E/\partial R_{A\alpha} = -\langle\hat{F}_{A\alpha}\rangle$ (Hellmann–Feynman force); geometry optimisation algorithms (steepest descent, BFGS, L-BFGS); vibrational frequencies from second derivatives (Hessian); (65–90 min) Beyond first order: second derivatives and the coupled-perturbed HF/KS equations; analytic vs. numerical gradients; computational cost; NMR chemical shifts, IR intensities, polarisability as FH applications; connection to molecular response theory.

**Core learning outcomes:**

- Derive the Feynman–Hellmann theorem from variational stationarity.
- Apply FH to compute nuclear forces and relate to geometry optimisation.
- Distinguish analytic gradients from numerical finite differences in cost.
- Describe coupled-perturbed equations for second-order properties.

**Dual-track content:**

- *Computational chemistry track:* Geometry optimisation workflow; transition-state search; IRC; force-field parameterisation from ab initio gradients; NMR shielding from FH.
- *Mathematical track:* FH as the quantum mechanical analogue of Darboux's theorem; Wigner's $2n+1$ rule; Dalgarno–Lewis perturbation theory.

**Tier differentiation:** HS: "the force on a nucleus equals the electrostatic force from the electron density and other nuclei." Beg. UG: FH theorem derivation; HF force on H nucleus; steepest-descent geometry optimisation. Adv. UG: analytic gradient for HF; BFGS; Hessian for frequencies. MSc: coupled-perturbed HF/KS; analytic second derivatives; NMR shielding. PhD: Wigner's $2n+1$ rule; Dalgarno–Lewis; response theory for arbitrary-order properties.

**Assessment:** Set A — FH theorem derivation; nuclear force on H in $H_2$; identify minimum from gradient. Set B — analytic gradient derivation for HF; FH for NMR shielding; cost comparison analytic vs. numerical. Projects: (simple) geometry optimisation of $H_2O$: energy and gradient at each BFGS step; (complex) FH theorem for polarisability: analytic vs. finite-difference comparison for HF/cc-pVDZ.

***Module III.1 Reading:*** *Helgaker, Jørgensen & Olsen, Ch. 1–2; Born & Oppenheimer (1927) original paper; Yarkony, Rev. Mod. Phys.* ***68****, 985 (1996) — conical intersections.*

## 3.2 Module III.2 — Principles of Molecular and Atomic Electronic Structure

## Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | The Quantum Many-Body Problem | Complexity; $N$-representability; curse of dimensionality |
| L02 | Hydrogenic Systems | Exact solution; $Z_{\text{eff}}$; quantum defect; radial nodes |
| L03 | Polyelectronic Systems | He prototype; electron correlation; Coulomb/Fermi hole |
| L04 | Spin Orbitals & Slater Screening | $\chi(\mathbf{x}) = \phi(\mathbf{r})\sigma(s)$; Slater rules; shielding |
| L05 | Antisymmetry & Slater Determinants | Pauli exclusion; Slater determinant; Slater–Condon rules |
| L06 | Restricted and Unrestricted Methods | RHF vs. UHF; spin contamination; ROHF |
| L07 | Variational Methods | Ritz; Hylleraas–Undheim; cusp condition; R12/F12 |
| L08 | Configurations, Aufbau & Virtuals | Aufbau; orbital energies; excited configurations |
| L09 | Configuration Interaction | FCI; CISD; size consistency; Davidson correction |
| L10 | Hartree–Fock & Correlation Energy | SCF; Roothaan–Hall; Koopman's; $E_{\text{corr}}$ |
| L11 | Orbital Theories & Bonding | MO theory; VB theory; Coulson–Fischer; resonance |
| L12 | Natural and Localised Orbitals | NBO; NAO; Foster–Boys; DLPNO connection |
| L13 | Basis Sets I — Function Types | STO; GTO; contraction; minimal; split-valence |
| L14 | Basis Sets II — Standard Families | Pople; Dunning cc-pVXZ; Karlsruhe def2; ECP |
| L15 | Molecular Integral Schemes | Two-electron integrals; Obara–Saika; RI approximation |
| L16 | Glimpse into Solid State | PAW; plane-wave basis; Bloch electrons; band vs. MO |

## Lecture-by-Lecture Syllabi

### Module III.2 / L01: The Quantum Many-Body Problem in Chemistry

**90-min pacing:** (0–20 min) The electronic Schrödinger equation $\hat{H}_e\psi_e = E_e\psi_e$; why it cannot be solved exactly for $N_e > 2$; exponential wall: Hilbert space dimension grows as $\binom{K}{N_e}$ for $K$ basis functions; (20–45 min) $N$-representability: which density matrices can arise from a physical $N$-electron wavefunction; necessary conditions on 1-RDM and 2-RDM; the $v$-representability problem in DFT; (45–65 min) Curse of dimensionality: wavefunction depends on $3N_e$ (+ spin) coordinates; grid methods infeasible; why we need orbital-based methods; (65–90 min) Systematic approximation hierarchy: HF $\rightarrow$ MP2 $\rightarrow$ CCSD(T) $\rightarrow$ FCI; cost scaling; accuracy benchmarks for small molecules; accuracy vs. cost trade-off as the central theme of quantum chemistry.

**Core learning outcomes:**

- Explain the exponential complexity of the $N$-electron problem.
- State the $N$-representability conditions for the 1-RDM.
- Identify the systematic approximation hierarchy in quantum chemistry.
- Quantify the accuracy–cost trade-off for standard methods.

**Tier differentiation:** HS: "quantum chemistry is hard because electrons interact and live in high-dimensional space." Beg. UG: Hilbert space dimension; method hierarchy; scaling

table. Adv. UG: $N$-representability conditions; $v$-representability; curse of dimensionality. MSc: 2-RDM $N$-representability (P,Q,G conditions); Coleman's theorem; SDP. PhD: quantum complexity of FCI; MA-hardness; quantum computing for quantum chemistry. **Assessment:** Set A — Hilbert space dimension for $H_2O$/STO-3G; accuracy table for methods on $H_2$. Set B — $N$-representability conditions for 1-RDM; $v$-representability counterexample. Projects: (simple) dimension and energy hierarchy for $H_2$ from STO-3G to FCI; (complex) SDP lower bound on FCI energy via 2-RDM.

## Module III.2 / L02: Hydrogenic Systems

**90-min pacing:** (0–20 min) Hydrogen-like Hamiltonian $\hat{H} = -\frac{1}{2}\nabla^2 - Z/r$ (atomic units); separation in spherical coordinates; radial and angular parts; (20–45 min) Exact solution: $\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)Y_l^m(\theta,\phi)$; associated Laguerre polynomials for $R_{nl}$; energy $E_n = -Z^2/2n^2$; radial nodes $n-l-1$; angular nodes $l$; (45–65 min) One-electron atomic orbitals: $1s, 2s, 2p, 3s, 3p, 3d$; shapes; nodal surfaces; radial probability density $4\pi r^2|R_{nl}|^2$; expectation values $\langle r\rangle, \langle r^2\rangle, \langle r^{-1}\rangle$; (65–90 min) Effective nuclear charge $Z_{\text{eff}}$ and orbital exponents in multi-electron atoms; quantum defect for alkali atoms; penetration and shielding; connection to Slater orbitals and GTO basis functions.

**Core learning outcomes:**

- Write and interpret the hydrogenic wavefunctions $\psi_{nlm}$.
- Compute radial probability densities and expectation values.
- Explain the concept of effective nuclear charge and quantum defect.
- Connect hydrogenic orbitals to Slater-type and Gaussian-type basis functions.

**Tier differentiation:** HS: shapes and nodal structure of atomic orbitals. Beg. UG: radial wavefunctions; probability density; $\langle r\rangle$; orbital energies. Adv. UG: derivation via Frobenius; quantum defect; $Z_{\text{eff}}$ and shielding. MSc: connection to STO exponents; pseudo-potential core replacement. PhD: relativistic hydrogenic wavefunctions; Dirac equation for $Z_{\text{eff}}\alpha \sim 1$. **Assessment:** Set A — $\psi_{nlm}$ for $1s, 2p_z$; radial density; $\langle r\rangle_{2p}$; energy ordering. Set B — derive $E_n = -Z^2/2n^2$ from termination; $\langle r^{-1}\rangle$ for $1s$. Projects: (simple) radial probability density plotter for all $n \leq 3$ AOs; (complex) quantum defect for Li, Na, K: fit from experimental term values.

## Module III.2 / L03: Polyelectronic Systems and Electron Correlation

**90-min pacing:** (0–20 min) Helium as prototype: $\hat{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - \frac{2}{r_1} - \frac{2}{r_2} + \frac{1}{r_{12}}$; exact solution impossible; perturbation theory ($1/r_{12}$ as perturbation); zeroth-order: product of hydrogenic $1s$; first-order correction $\langle 1/r_{12}\rangle = 5Z/8$; (20–45 min) Variational treatment: $Z_{\text{eff}}$ as variational parameter; optimal $Z_{\text{eff}} = Z - 5/16$; comparison to exact; (45–65 min) Electron correlation: definition $E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}$; static (near-degeneracy) vs. dynamic (short-range) correlation; Coulomb hole and Fermi hole; angular correlation (left–right) in $H_2$; (65–90 min) Survey of correlation effects: dissociation energies; ionisation potentials; electron affinities; geometries; dipole moments; van der Waals interactions (London dispersion = pure correlation).

**Core learning outcomes:**

- Apply PT and variational method to the helium atom.
- Define and distinguish static and dynamic correlation.
- Explain the Coulomb hole and Fermi hole physically.

- Identify properties that are particularly sensitive to electron correlation.

**Tier differentiation:** HS: electrons avoid each other — that's correlation; why chemistry needs it. Beg. UG: He PT and variational; $E_{\text{corr}}$ definition; static vs. dynamic. Adv. UG: Coulomb and Fermi holes; left–right correlation in $H_2$; dispersion. MSc: London dispersion from PT; correlation effects on PES; adiabatic connection. PhD: transcorrelated methods; R12/F12 explicitly correlated wavefunctions; cusp condition. **Assessment:** Set A — He energy from PT and variational; optimal $Z_{\text{eff}}$; $E_{\text{corr}}$ for He. Set B — Coulomb hole derivation; left–right correlation in $H_2$ from CI(2,2). Projects: (simple) He energy: PT vs. variational vs. HF vs. FCI; (complex) $H_2$ correlation energy vs. bond length: dynamic to static crossover.

## Module III.2 / L04: Spin Orbitals, Slater Screening, and Effective Nuclear Charge

**90-min pacing:** (0–20 min) Spin orbitals: $\chi(\mathbf{x}) = \phi(\mathbf{r})\sigma(s)$ where $\sigma \in \{\alpha, \beta\}$; $\mathbf{x} = (\mathbf{r}, s)$; orthonormality; (20–45 min) Slater's rules for $Z_{\text{eff}}$: group electrons by $(1s), (2s, 2p), (3s, 3p), (3d), \ldots$; shielding contributions from inner and same-shell electrons; worked examples for B, Cl, Fe; limitation of Slater rules vs. computed $Z_{\text{eff}}$; (45–65 min) Physical effects captured by $Z_{\text{eff}}$: penetration of $s$ orbitals; orbital contraction; Clementi–Raimondi refined values; connection to orbital exponents in STO/GTO; (65–90 min) Periodic trends from $Z_{\text{eff}}$: atomic radius, ionisation energy, electron affinity, electronegativity along period and down group; comparative analysis with experimental data.

**Core learning outcomes:**

- Define spin orbitals and state their orthonormality relations.
- Apply Slater's rules to compute $Z_{\text{eff}}$ for any element.
- Explain penetration, shielding, and orbital contraction.
- Predict periodic trends from $Z_{\text{eff}}$.

**Tier differentiation:** HS: periodic trends from effective nuclear charge. Beg. UG: spin orbitals; Slater rules for B, Cl, Fe; periodic trends. Adv. UG: Clementi–Raimondi; connection to STO exponents; limitations of Slater. MSc: relativistic contraction of $s, p$ orbitals in heavy atoms; pseudo-potentials from $Z_{\text{eff}}$. PhD: exact $Z_{\text{eff}}$ from computed densities; Kohn–Sham $Z_{\text{eff}}$; relativistic effective core potentials. **Assessment:** Set A — $Z_{\text{eff}}$ for B, Si, Fe by Slater rules; periodic trend in $r_{\text{cov}}$. Set B — Clementi–Raimondi vs. Slater; orbital contraction in 4th period. Projects: (simple) periodic table of $Z_{\text{eff}}$ heatmap using Slater rules; (complex) $Z_{\text{eff}}$ from HF electron density vs. Slater rules for 3d transition metals.

## Module III.2 / L05: Antisymmetry, Hartree Products, and Slater Determinants

**90-min pacing:** (0–20 min) Indistinguishable electrons; Pauli exclusion; antisymmetry postulate $\hat{P}_{ij}\Psi = -\Psi$; (20–45 min) Hartree product: $\Psi_{\text{HP}} = \chi_1(\mathbf{x}_1)\chi_2(\mathbf{x}_2)\cdots\chi_N(\mathbf{x}_N)$ — not antisymmetric; antisymmetriser $\hat{\mathcal{A}} = \frac{1}{\sqrt{N!}}\sum_{\hat{P}}(-1)^p\hat{P}$; Slater determinant $\Phi = \hat{\mathcal{A}}[\chi_1\chi_2\cdots\chi_N]$; (45–65 min) Properties of Slater determinants: Pauli exclusion from determinant vanishing; exchange symmetry; overlap $\langle\Phi|\Phi\rangle = 1$ for orthonormal spin orbitals; orbital rotation and the non-uniqueness of SDs; (65–90 min) Matrix elements of one- and two-body operators between SDs: Slater–Condon rules; $\langle\hat{O}_1\rangle$, $\langle\hat{O}_2\rangle$; application to the HF energy expression.

**Core learning outcomes:**

- Construct a Slater determinant from a set of spin orbitals.

- Prove Pauli exclusion from the determinant structure.
- Apply Slater–Condon rules to evaluate matrix elements.
- Derive the Hartree–Fock energy from Slater–Condon rules.

**Tier differentiation:** HS: "no two electrons can be in the same quantum state" — Pauli exclusion from antisymmetry. Beg. UG: Slater determinant construction; Pauli exclusion; Slater–Condon rules. Adv. UG: antisymmetriser; $N!$ terms; HF energy from Slater–Condon. MSc: generalised Slater–Condon for multi-determinant wavefunctions. PhD: second-quantisation derivation of Slater–Condon; matrix elements for CC theory. **Assessment:** Set A — construct $\Phi$ for He, Li; verify $\langle\Phi|\Phi\rangle = 1$; Pauli exclusion. Set B — Slater–Condon rules derivation; HF energy $E_{\mathrm{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij})$. Projects: (simple) Slater determinant for $H_2$: evaluate all integrals; (complex) Slater–Condon rules for doubly-substituted configuration: matrix element for CISD.

## Module III.2 / L06: Restricted and Unrestricted Methods

**90-min pacing:** (0–20 min) Spin structure of HF: closed-shell (RHF) vs. open-shell; restricted open-shell (ROHF) and unrestricted (UHF); (20–45 min) RHF: $\phi_i^\alpha = \phi_i^\beta$; single set of spatial orbitals; correct spin eigenfunction; Roothaan–Hall equations $\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}$ (details in L10); (45–65 min) UHF: separate $\alpha$ and $\beta$ orbital sets; Pople–Nesbet equations; lower variational energy than RHF; spin contamination $\langle\hat{S}^2\rangle_{\mathrm{UHF}} = S(S+1) + N_\beta - \sum_{ij}|\langle\phi_i^\alpha|\phi_j^\beta\rangle|^2$; (65–90 min) ROHF: constrained optimisation; pure spin eigenfunction; applications to radicals; half-open-shell systems; comparison of RHF, UHF, ROHF for dissociation of $H_2$; open-shell singlet states; broken-symmetry DFT.

**Core learning outcomes:**

- Distinguish RHF, UHF, and ROHF and state their applicability.
- Compute the spin contamination of a UHF wavefunction.
- Explain why UHF gives lower energy than RHF for open-shell systems.
- Describe broken-symmetry DFT for biradicals.

**Tier differentiation:** HS: "restricted" = paired electrons; "unrestricted" = $\alpha$ and $\beta$ electrons optimised independently. Beg. UG: RHF vs. UHF; spin contamination formula; $H_2$ dissociation. Adv. UG: Pople–Nesbet equations; ROHF constraint; spin projection. MSc: broken-symmetry DFT; spin contamination correction; ROHF for excited states. PhD: GHF (generalised HF); spin–orbit coupling; non-collinear DFT. **Assessment:** Set A — $\langle\hat{S}^2\rangle_{\mathrm{UHF}}$ for O atom; energy ordering RHF vs. UHF for $H_2$. Set B — Pople–Nesbet equations derivation; broken-symmetry DFT for $H_2$ at $r = 3$ Å. Projects: (simple) RHF vs. UHF dissociation curve for $H_2$; (complex) spin contamination and spin projection for biradical singlet.

## Module III.2 / L07: Variational Methods

**90-min pacing:** (0–20 min) Variational principle for ground state: $E[\Psi] = \langle\Psi|\hat{H}|\Psi\rangle/\langle\Psi|\Psi\rangle \geq E_0$; (20–45 min) Ritz method: linear ansatz $|\Phi\rangle = \sum_k c_k|\chi_k\rangle$; secular equation $\mathbf{Hc} = E\mathbf{Sc}$; Hylleraas–Undheim theorem: $N$ variational eigenvalues bound the exact ones from above; (45–65 min) Hylleraas function for He: $\psi = e^{-\alpha(r_1+r_2)}(1+\beta r_{12})$; dependence on inter-electron coordinate; cusp condition $(\partial\psi/\partial r_{12})|_{r_{12}=0} = \frac{1}{2}\psi(0)$; (65–90 min) Explicitly correlated methods (R12/F12): Slater geminal $e^{-\gamma r_{12}}$; resolution-of-identity; CABS (complementary auxiliary basis sets); accelerated basis-set convergence.

**Core learning outcomes:**

- Apply the Ritz method to set up and solve the secular equation.
- State the Hylleraas–Undheim theorem and apply it.
- Derive the electron–electron cusp condition.
- Describe explicitly correlated R12/F12 methods.

**Tier differentiation:** HS: variational principle as "upper bound guarantee." Beg. UG: secular equation; He with $Z_{\text{eff}}$; Hylleraas–Undheim. Adv. UG: cusp condition; Hylleraas wavefunction; convergence. MSc: R12/F12 methods; CABS; explicitly correlated coupled cluster. PhD: transcorrelated Hamiltonians; FCIQMC-R12; CCSD(T)-F12 benchmark accuracy. **Assessment:** Set A — secular equation for 2-basis He; Hylleraas–Undheim check; cusp condition. Set B — cusp condition derivation; Hylleraas function for He: compute energy. Projects: (simple) He energy from Hylleraas wavefunctions: convergence with terms; (complex) electron–electron cusp: R12 correction to HF energy for He.

## Module III.2 / L08: Electronic Configurations, Aufbau, and Virtual Orbitals

**90-min pacing:** (0–20 min) Aufbau principle: fill lowest-energy orbitals first; $1s < 2s < 2p < 3s < 3p < 4s < 3d < \ldots$; anomalies (Cr, Cu); Madelung/$(n + l)$ rule; (20–45 min) Orbital energy ordering in polyelectronic atoms: $\varepsilon_i$ from HF; Koopmans' theorem (preview); orbital degeneracy and symmetry; occupations and electron configurations; term symbols (recap from II.3); (45–65 min) Virtual (unoccupied) orbitals: definition; basis-set dependence; canonical virtual orbitals; natural virtual orbitals (NVOs); their role in CI and CC; (65–90 min) Excited configurations: single, double, triple substitutions $|\Phi_i^a\rangle, |\Phi_{ij}^{ab}\rangle, \ldots$; excitation operators $\hat{a}_a^\dagger \hat{a}_i$; Brillouin's theorem; orthogonality to HF ground state; matrix elements for CI.

**Core learning outcomes:**

- Apply the Aufbau principle and explain anomalies for Cr, Cu.
- Construct excited Slater determinants using excitation operators.
- Explain the role of virtual orbitals in post-HF methods.
- State Brillouin's theorem and its implication for CI.

**Tier differentiation:** HS: filling orbitals in order: the Aufbau principle and periodic table. Beg. UG: electron configurations; Aufbau anomalies; excited configurations. Adv. UG: canonical vs. natural virtual orbitals; Brillouin's theorem. MSc: NVOs for local correlation; pair natural orbitals; orbital localisation. PhD: optimal virtual space for DLPNO; projected AOs; pair correlation theories. **Assessment:** Set A — electron configurations for B, Fe, Cr; virtual orbital count for $H_2O$/6-31G*. Set B — Brillouin's theorem derivation; canonical vs. NVO comparison. Projects: (simple) aufbau anomalies: energy rationale for Cr vs. expected; (complex) NVO localisation for $H_2O$: compare canonical vs. Foster–Boys virtual orbitals.

## Module III.2 / L09: Configuration Interaction and the Exact Non-Relativistic Limit

**90-min pacing:** (0–20 min) CI expansion: $|\Psi\rangle = c_0 |\Phi_0\rangle + \sum_{ia} c_i^a |\Phi_i^a\rangle + \sum_{ijab} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \cdots$; FCI = exact non-relativistic ground state within a given basis; (20–45 min) FCI properties: variational; size-consistent; basis-set limit = exact; FCI dimension $\binom{K}{N_\alpha}\binom{K}{N_\beta}$; exponential scaling; (45–65 min) Truncated CI: CIS (singles only; no correlation for ground state by Bril-

louin); CISD; CISDT; size inconsistency of truncated CI; $\langle\Psi_N|\hat{H}|\Psi_N\rangle \neq 2\langle\Psi_{N/2}|\hat{H}|\Psi_{N/2}\rangle$ for non-interacting monomers; Davidson correction $\Delta E_Q \approx (1-c_0^2)E_{\mathrm{CISD}}$; (65–90 min) Selected CI (SCI): CIPSI, ASCI, HCI algorithms; stochastic FCI; FCIQMC; convergence toward FCI; comparison with CCSD(T).

**Core learning outcomes:**

- Write the FCI expansion and state its properties.
- Explain size inconsistency of truncated CI and apply the Davidson correction.
- Describe selected CI algorithms (CIPSI, HCI).
- Compare FCI, CISD, and CCSD(T) in accuracy and scaling.

**Tier differentiation:** HS: FCI as the "exact answer" and why it's too expensive. Beg. UG: CI expansion; FCI dimension; size inconsistency. Adv. UG: Davidson correction; CISD scaling; selected CI idea. MSc: CIPSI; FCIQMC; initiator approximation. PhD: SCI extrapolation to FCI; stochastic-CASSCF; heat-bath CI. **Assessment:** Set A — FCI dimension for $H_2$/STO-3G; size inconsistency for $He_2$; Davidson correction. Set B — prove size inconsistency of CISD; CIPSI selection criterion. Projects: (simple) FCI vs. CISD for $H_2$: energy and $c_0^2$ vs. bond length; (complex) selected CI for $N_2$: CIPSI convergence toward FCI.

## Module III.2 / L10: Hartree–Fock Theory, Koopman's Theorem, and Correlation Energy

**90-min pacing:** (0–20 min) HF as variational optimisation of a Slater determinant; energy $E_{\mathrm{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij})$; Coulomb $J_{ij} = \langle ij|ij\rangle$ and exchange $K_{ij} = \langle ij|ji\rangle$; (20–45 min) Fock operator $\hat{F} = \hat{h} + \sum_j(\hat{J}_j - \hat{K}_j)$; HF equations $\hat{F}\phi_i = \varepsilon_i\phi_i$; Roothaan–Hall in AO basis: $\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}$; density matrix $\mathbf{P} = 2\mathbf{C}_{\mathrm{occ}}\mathbf{C}_{\mathrm{occ}}^{\dagger}$; DIIS convergence acceleration; (45–65 min) Koopman's theorem: $\mathrm{IP}_i \approx -\varepsilon_i$ (frozen-orbital approximation); $\mathrm{EA}_a \approx -\varepsilon_a$; orbital relaxation and correlation corrections; (65–90 min) Correlation energy $E_{\mathrm{corr}} = E_{\mathrm{exact}} - E_{\mathrm{HF}}$; typical magnitudes ($\sim$1 eV per electron pair); dynamic and static contributions; the $T_1$ and $D_1$ diagnostic for multi-reference character; post-HF methods as corrections to HF.

**Core learning outcomes:**

- Derive the HF equations from the variational principle.
- Set up and solve the Roothaan–Hall equations in an AO basis.
- Apply Koopman's theorem to estimate IPs and EAs.
- Define correlation energy and identify multi-reference character from diagnostics.

**Tier differentiation:** HS: "HF gives 99% of the energy but the remaining 1% determines chemistry." Beg. UG: Fock operator; DIIS; Koopman's theorem; $E_{\mathrm{corr}}$ definition. Adv. UG: Roothaan–Hall derivation; density matrix; Brillouin's theorem. MSc: $T_1/D_1$ diagnostics; orbital relaxation vs. correlation; OMP2. PhD: Brueckner orbitals; orbital-optimised CC; HF saddle point. **Assessment:** Set A — HF energy for He; Koopman's IPs; $E_{\mathrm{corr}}$ for Ne. Set B — Roothaan–Hall derivation; density matrix idempotency; DIIS error vector. Projects: (simple) HF SCF for $H_2O$/STO-3G: convergence and Koopman's IPs; (complex) correlation energy per electron pair across noble gas series.

## Module III.2 / L11: Orbital Theories and Bonding Models

**90-min pacing:** (0–20 min) Two paradigms: MO theory (delocalised) and Valence Bond theory (localised, resonance); complementary descriptions; (20–45 min) MO theory: LCAO–

MO; bonding, antibonding, non-bonding MOs; correlation diagrams; Walsh diagrams; orbital symmetry and frontier MO theory; HOMO–LUMO gap; (45–65 min) VB theory: Heitler–London wavefunction for $H_2$; ionic and covalent resonance structures; Coulson–Fischer wavefunction: intermediate between MO and VB; resonance in benzene; (65–90 min) Modern VB methods: spin-coupled VB; VBSCF; BOVB; GVB; comparison to CASSCF; VB interpretation of chemical bonding.

**Core learning outcomes:**

- Construct and interpret MO correlation diagrams.
- Write the Heitler–London and Coulson–Fischer wavefunctions for $H_2$.
- Explain resonance structures in VB theory.
- Compare MO and VB descriptions and identify when each is more appropriate.

**Tier differentiation:** HS: bonding and antibonding orbitals; resonance structures. Beg. UG: MO correlation diagrams; Heitler–London; resonance in benzene. Adv. UG: Coulson–Fischer; GVB; MO–VB connection. MSc: VBSCF; spin-coupled VB; BOVB. PhD: modern VB theories; VB interpretation of aromaticity; VB-CI. **Assessment:** Set A — MO diagram for $N_2$, $O_2$, $F_2$; VB resonance for benzene; Heitler–London for $H_2$. Set B — Coulson–Fischer wavefunction: ionic/covalent mixing; GVB pair functions. Projects: (simple) MO vs. VB energy for $H_2$: ionic content vs. $R$; (complex) VBSCF for allyl radical: spin-coupled vs. delocalised picture.

## Module III.2 / L12: Natural and Localised Orbitals

**90-min pacing:** (0–20 min) Why localised orbitals? Chemical intuition; transferability; local correlation methods; (20–45 min) Natural orbitals (NOs) from 1-RDM: eigenfunctions of $\gamma(\mathbf{r};\mathbf{r}')$; occupation numbers $n_i \in [0,2]$; multi-reference diagnostics from NOONs; (45–65 min) Natural Bond Orbitals (NBOs): Lewis-structure-based orthogonal localisation; core, lone-pair, and bond NBOs; NBO charge analysis; Natural Population Analysis (NPA); Natural Atomic Orbitals (NAOs); (65–90 min) Localisation methods for MOs: Foster–Boys ($\sum_i \langle r^2\rangle_i - \langle r\rangle_i^2$ minimised); Pipek–Mezey (maximise atomic populations); Edmiston–Ruedenberg (maximise self-repulsion); comparison; use in DLPNO-CCSD(T) and pair natural orbital methods.

**Core learning outcomes:**

- Compute NOONs and interpret them as a multi-reference diagnostic.
- Describe the NBO procedure and NPA charge analysis.
- Apply Foster–Boys and Pipek–Mezey localisation.
- Connect localised orbitals to local correlation methods.

**Tier differentiation:** HS: localised orbitals as "lone pairs and bonds" — connecting quantum chemistry to Lewis structures. Beg. UG: NOs and NOONs; NBO for water; NPA charges. Adv. UG: Foster–Boys localisation; NOON diagnostic; Pipek–Mezey. MSc: DLPNO connection; PNOs from 2-RDM; BOYS localisation gradient. PhD: intrinsic bond orbitals (IBO); AVAS; automatic active space selection from NOONs. **Assessment:** Set A — NBO for $H_2O$: identify lone pairs; NPA charges; NOON for $H_2$ at $2R_e$. Set B — Foster–Boys localisation criterion; derivation of gradient; comparison with Pipek–Mezey. Projects: (simple) NBO/NPA analysis for a series of molecules; (complex) NOON diagnostic: HF vs. FCI NOONs for $N_2$ dissociation.

## Module III.2 / L13: Basis Sets I — Types of Functions

**90-min pacing:** (0–20 min) Why basis sets? LCAO expansion; finite basis approximation; basis set error = incompleteness; (20–45 min) Slater-type orbitals (STO): $\chi_{nlm}^{\mathrm{STO}} = Nr^{n-1}e^{-\zeta r}Y_l^m$; correct cusp and asymptotic; expensive two-electron integrals (Boys 1950 Gaussian workaround); Gaussian-type orbitals (GTO): $\chi_{lmn}^{\mathrm{GTO}} = Nx^l y^m z^n e^{-\alpha r^2}$; product theorem; (45–65 min) Contraction: $\phi_\mu = \sum_k d_{\mu k}\chi_k^{\mathrm{GTO}}$; segmented vs. general contraction; minimal basis (1 function/occupied AO); split-valence (double/triple zeta); polarisation ($d$ on heavy, $p$ on H); diffuse functions (for anions and Rydbergs); (65–90 min) Quality indicators: basis set superposition error (BSSE); counterpoise correction (Boys–Bernardi); basis set incompleteness error (BSIE); CBS extrapolation; approximate completeness via RI/density fitting.

**Core learning outcomes:**

- Define STO and GTO and explain why GTOs are used in practice.
- Describe contraction schemes and the meaning of split-valence, polarised, and diffuse.
- Define BSSE and apply the counterpoise correction.
- Explain BSIE and CBS extrapolation.

**Tier differentiation:** HS: basis functions as the "building blocks" for molecular orbitals. Beg. UG: STO vs. GTO; contraction; minimal and split-valence; BSSE. Adv. UG: BSSE counterpoise; BSIE vs. correlation error; RI approximation. MSc: general contraction; CBS extrapolation formulae; resolution-of-identity. PhD: basis set design principles; even-tempered series; optimisation of exponents for FCI. **Assessment:** Set A — GTO product theorem; contraction for 6-31G*; BSSE for $He_2$ at 3 Å. Set B — STO-nG contraction: fit STO with $n$ Gaussians; counterpoise correction derivation. Projects: (simple) BSSE for $H_2O$ dimer vs. basis set; (complex) CBS extrapolation for He correlation energy: Dunning cc-pVDZ through cc-pV5Z.

## Module III.2 / L14: Basis Sets II — Standard Basis Families

**90-min pacing:** (0–20 min) Overview of the major families: Pople, Dunning correlation-consistent, Karlsruhe def2, ANO, Sapporo; design philosophy of each; (20–45 min) Pople basis sets: 3-21G, 6-31G*, 6-311++G**; notation; polarisation and diffuse; historical prevalence; (45–65 min) Dunning cc-pVXZ ($X = D, T, Q, 5, 6$): hierarchical; designed for CBS extrapolation; aug-cc-pVXZ (with diffuse); cc-pCVXZ (core–valence); cc-pwCVXZ (weighted); (65–90 min) Karlsruhe def2-SVP, def2-TZVP, def2-QZVP: efficient segmented contraction; ECP for $Z \geq 37$; Ahlrichs ECPs; Dunning ECPs (cc-pVXZ-PP); comparison on a benchmark set; practical choice guide.

**Core learning outcomes:**

- Decode basis set notation (Pople, Dunning, def2).
- Select an appropriate basis set for a given property and accuracy target.
- Explain effective core potentials and when to use them.
- Apply CBS extrapolation using the Dunning cc-pVXZ hierarchy.

**Tier differentiation:** HS: practical guide to basis set selection. Beg. UG: decode 6-31G*, cc-pVDZ, def2-TZVP; select for geometry vs. energy. Adv. UG: CBS extrapolation; ECP choice; core–valence cc-pCVXZ. MSc: ANO basis sets; CABS for F12; basis set optimisation. PhD: basis set extrapolation theory; explicitly correlated basis-set convergence; periodic basis sets. **Assessment:** Set A — decode basis set notation; select basis for $H_2O$ geometry

vs. interaction energy. Set B — CBS extrapolation for Ne: Helgaker formula; core–valence effect on Cu ionisation. Projects: (simple) basis set convergence for $H_2O$: geometry and energy from STO-3G to aug-cc-pV5Z; (complex) ECP vs. all-electron for Au: relativistic effects on bond length.

## Module III.2 / L15: Molecular Integral Schemes

**90-min pacing:** (0–20 min) The two-electron integral bottleneck: $(\mu\nu|\lambda\sigma) = \int\int \phi_\mu(\mathbf{r}_1)\phi_\nu(\mathbf{r}_1)\frac{1}{r_{12}}\phi_\lambda(\mathbf{r}_2)\phi_\sigma(\mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2$; formally $\mathcal{O}(K^4)$; (20–45 min) Obara–Saika recurrence relations: generate higher angular momentum integrals from $s$-type; vertical and horizontal recurrence; computational savings; (45–65 min) Resolution-of-identity (RI/density fitting) approximation: $(\mu\nu|\lambda\sigma) \approx \sum_{PQ}(\mu\nu|P)(P|Q)^{-1}(Q|\lambda\sigma)$; auxiliary basis $\{P\}$; reduces to $\mathcal{O}(K^3)$; Coulomb and exchange RI; (65–90 min) Modern integral algorithms: linear-scaling methods (J-engine, LinK); screening (Schwarz); GPU implementations; integral-direct algorithms; resolution of identity for MP2 and CC; RIJCOSX for large molecules.

**Core learning outcomes:**

- Write the two-electron repulsion integral and identify its computational complexity.
- Describe the Obara–Saika recurrence and its efficiency gain.
- Explain the RI approximation and its error and cost.
- Describe Schwarz screening and linear-scaling integral algorithms.

**Tier differentiation:** HS: molecular integrals as the computational cost of quantum chemistry. Beg. UG: two-electron integral definition; $\mathcal{O}(K^4)$ scaling; RI approximation. Adv. UG: Obara–Saika recurrence; Schwarz screening; RI error analysis. MSc: RIJCOSX; GPU implementation; linear-scaling density fitting. PhD: Boys function; McMurchie–Davidson; Gill's PRISM; adaptive integral algorithms. **Assessment:** Set A — count two-electron integrals for $H_2O$/STO-3G; RI approximation error estimate. Set B — Obara–Saika for $s$-$p$ integrals; Schwarz screening criterion derivation. Projects: (simple) RI error vs. auxiliary basis size for HF/def2-TZVP; (complex) timing benchmark: conventional vs. RI-MP2 for linear alkane chain.

## Module III.2 / L16: Glimpse into the Solid State

**90-min pacing:** (0–20 min) Periodic systems: Bloch's theorem $\psi_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}$; crystal momentum; band structure vs. MO picture; (20–45 min) Plane-wave basis: $\phi_{\mathbf{G}}(\mathbf{r}) = e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}/\sqrt{\Omega}$; kinetic energy cutoff $E_{\text{cut}}$; completeness and convergence; (45–65 min) Projector augmented-wave (PAW) method: all-electron quality with plane-wave efficiency; PAW transformation; frozen-core PAW; comparison with pseudopotentials (norm-conserving, ultrasoft); (65–90 min) GTO basis sets for solids (Crystal, CP2K): comparison with plane waves; advantages and disadvantages; k-point sampling; Gamma-point approximation; periodic MP2 and CC with GTOs; Γ-point CCSD(T) for solids; outlook.

**Core learning outcomes:**

- State Bloch's theorem and explain crystal momentum and band structure.
- Describe plane-wave basis sets and the role of the energy cutoff.
- Explain the PAW method and compare with pseudopotentials.
- Compare GTO and plane-wave approaches for periodic quantum chemistry.

**Tier differentiation:** HS: from molecules to crystals — how symmetry simplifies the problem. Beg. UG: Bloch's theorem; plane-wave basis; band structure concept. Adv. UG: PAW

method; GTO vs. PW; k-point sampling. MSc: periodic MP2; CCSD(T) for solids; GTO-based periodic codes. PhD: diagrammatic MBPT for periodic systems; stochastic CCSD(T) for solids; quantum Monte Carlo for bulk. **Assessment:** Set A — Bloch states for 1D chain; plane-wave cutoff for Si; band gap vs. k-point density. Set B — PAW transformation derivation; GTO vs. PW cost comparison for periodic HF. Projects: (simple) tight-binding band structure for graphene; (complex) periodic HF/GTO vs. DFT/PW for LiH crystal: lattice constant and band gap.

***Module III.2 Reading:*** *Helgaker, Jørgensen & Olsen, Molecular Electronic-Structure Theory (Wiley) — primary reference; Szabo & Ostlund, Modern Quantum Chemistry (Dover); Jensen, Introduction to Computational Chemistry; Cramer, Essentials of Computational Chemistry.*

## 3.3 Module III.3 — Density Functional Theory and Beyond

### Module Overview

| Lec. | Title | Key Concepts |
|---|---|---|
| L01 | Hohenberg–Kohn Theorems | HK1/HK2; $v$-representability; universal functional |
| L02 | Thomas–Fermi–Weizsäcker | Orbital-free DFT; TF kinetic energy; gradient correction |
| L03 | Kohn–Sham DFT | KS equations; non-interacting reference; XC functional; SCF |
| L04 | XC Functionals — Jacob's Ladder | LDA; GGA; meta-GGA; hybrid; range-separated; double-hybrid |
| L05 | Useful DFT Quantities | Fukui functions; chemical hardness; dual descriptor; CDFT |
| L06 | Tight-Binding & Effective Hamiltonians | LCAO; Slater–Koster; SSH model; topological TBM |
| L07 | Hückel and Möbius Aromaticity | Hückel MO; $4n+2$ rule; Möbius; Woodward–Hoffmann |
| L08 | Crystal Field Theory | $d$-orbital splitting; $\Delta_o$; high-spin/low-spin; Jahn–Teller |
| L09 | Ligand Field Theory | Nephelauxetic; AOM; Tanabe–Sugano; Racah parameters |
| L10 | Full CI QMC (FCIQMC) | Booth–Alavi algorithm; walker dynamics; sign problem; initiator |
| L11 | Diffusion QMC (DMC) | Fixed-node; branching walk; trial wavefunction; pseudopotentials |
| L12 | Synthesis and Outlook | Method comparison; strong correlation; excited states; periodic; ML |

### Lecture-by-Lecture Syllabi

**Module III.3 / L01: Hohenberg–Kohn Theorems**

**90-min pacing:** (0–20 min) Motivation: the electron density $\rho(\mathbf{r})$ is a 3D function vs. the $3N_e$-dimensional wavefunction; if $\rho$ determines everything, enormous simplification; (20–45 min) HK Theorem 1: for any system of $N$ electrons under an external potential $v_{\text{ext}}$, the ground-state density $\rho_0(\mathbf{r})$ uniquely determines $v_{\text{ext}}(\mathbf{r})$ (up to a constant); proof by reductio ad absurdum; (45–65 min) HK Theorem 2: the exact ground-state energy is the minimum of $E_v[\rho] = F_{\text{HK}}[\rho] + \int v_{\text{ext}}\rho\, d\mathbf{r}$ over all $v$-representable densities; universal functional $F_{\text{HK}}[\rho] = T[\rho] + V_{ee}[\rho]$; (65–90 min) Limitations: $v$-representability (not all $\rho$ arise from a ground state of some $v_{\text{ext}}$); $N$-representability of densities; the Levy–Lieb constrained-

search reformulation avoids $v$-representability; excited states (Gross–Oliveira–Kohn); ensemble DFT.

**Core learning outcomes:**

- Prove the first HK theorem by contradiction.
- State the second HK theorem and identify the universal functional.
- Explain the $v$-representability problem.
- Describe the Levy–Lieb constrained-search and its advantage.

**Tier differentiation:** HS: "the electron density contains all information about a molecule." Beg. UG: HK1 proof sketch; HK2 variational principle; $F_{\mathrm{HK}}[\rho]$. Adv. UG: $v$-representability; Levy–Lieb; ensemble DFT. MSc: Lieb's convex conjugate; subdifferential; functional derivatives in DFT. PhD: Moreau–Yosida regularisation; N-representability of $\rho$; adiabatic connection. **Assessment:** Set A — prove HK1 by contradiction; identify $F_{\mathrm{HK}}$ for He; variational principle check. Set B — Levy–Lieb constrained search derivation; $v$-representability counterexample. Projects: (simple) HK1 visualisation: two different $v_{\mathrm{ext}}$ cannot give same $\rho_0$; (complex) Levy–Lieb: evaluate $F_{\mathrm{HK}}[\rho]$ for He by constrained optimisation.

## Module III.3 / L02: Thomas–Fermi–Weizsäcker Theory

**90-min pacing:** (0–20 min) Orbital-free DFT: express total energy entirely as a functional of $\rho(\mathbf{r})$; Thomas–Fermi kinetic energy $T_{\mathrm{TF}}[\rho] = C_F \int \rho^{5/3} d\mathbf{r}$ from homogeneous electron gas; (20–45 min) Thomas–Fermi total energy $E_{\mathrm{TF}}[\rho] = T_{\mathrm{TF}} + \int v_{\mathrm{ext}}\rho\, d\mathbf{r} + J[\rho] + E_{\mathrm{xc}}^{\mathrm{LDA}}[\rho]$; Euler–Lagrange equation; Thomas–Fermi equation $\mu = C_F\rho^{2/3} + v_{\mathrm{ext}} + v_H + v_{\mathrm{xc}}$; (45–65 min) Weizsäcker correction $T_W[\rho] = \frac{1}{8}\int |\nabla\rho|^2/\rho\, d\mathbf{r}$; gradient expansion (GEA); performance: wrong shell structure, no binding for atoms, no molecular binding; (65–90 min) Modern orbital-free DFT: Wang–Teter; Carter kinetic energy functionals; application to large-scale simulations of metals; keDF machine learning; limitations for covalent systems.

**Core learning outcomes:**

- Derive the TF kinetic energy from the homogeneous electron gas.
- Set up and solve the TF Euler–Lagrange equation.
- State the Weizsäcker correction and explain its physical origin.
- Describe the limitations of TF theory and modern orbital-free DFT.

**Tier differentiation:** HS: "Thomas–Fermi is the simplest density functional theory." Beg. UG: TF kinetic energy; TF equation; Weizsäcker; known failures. Adv. UG: gradient expansion; $T_W$ derivation; comparison with KS. MSc: Wang–Teter functional; orbital-free MD for metals. PhD: conjoint gradient expansion; non-local kinetic energy functionals; ML-keDF. **Assessment:** Set A — TF equation for H atom; Weizsäcker correction; identify TF failures.

Set B — derive $T_{\mathrm{TF}}$ from HEG; gradient expansion to second order. Projects: (simple) TF density for He: compare to exact; (complex) orbital-free DFT for Al: Wang–Teter vs. KS.

## Module III.3 / L03: Kohn–Sham DFT

**90-min pacing:** (0–20 min) Kohn–Sham idea: introduce a non-interacting reference system with the same density as the interacting system; KS kinetic energy $T_s[\rho] = \sum_i \langle\phi_i|-\frac{1}{2}\nabla^2|\phi_i\rangle$; (20–45 min) KS equations: $[-\frac{1}{2}\nabla^2 + v_{\mathrm{eff}}(\mathbf{r})]\phi_i(\mathbf{r}) = \varepsilon_i\phi_i(\mathbf{r})$ with $v_{\mathrm{eff}} = v_{\mathrm{ext}} + v_H + v_{\mathrm{xc}}$; XC potential $v_{\mathrm{xc}} = \delta E_{\mathrm{xc}}/\delta\rho$; self-consistent field loop; (45–65 min) Adiabatic connection formula: $E_{\mathrm{xc}} = \int_0^1 d\lambda \langle\Psi^\lambda|\hat{V}_{ee} - \hat{J}|\Psi^\lambda\rangle$; XC hole; exact conditions on $E_{\mathrm{xc}}$: negativity of XC

hole, sum rule $\int \rho_{\mathrm{xc}} d\mathbf{r} = -1$; (65–90 min) KS DFT in practice: basis set expansion of KS orbitals; Coulomb and XC integrals; DIIS; convergence; computational scaling $\mathcal{O}(K^3)$ for hybrid; $\mathcal{O}(K)$ linear-scaling DFT; comparison with HF.

**Core learning outcomes:**

- Derive the KS equations from the HK variational principle.
- Define the XC energy and potential and their exact properties.
- Describe the KS SCF algorithm and its scaling.
- Explain the adiabatic connection and the XC hole.

**Tier differentiation:** HS: "KS DFT maps the interacting problem onto non-interacting electrons in an effective potential." Beg. UG: KS equations; effective potential; SCF loop; LDA. Adv. UG: adiabatic connection; XC hole; exact conditions. MSc: XC sum rule; self-interaction error; linear-response TDDFT. PhD: ensemble KS DFT; optimised effective potential (OEP); exact exchange. **Assessment:** Set A — KS equations for He with LDA; identify $v_{\mathrm{xc}}$; SCF convergence check. Set B — derive KS equations from $E[\rho]$; adiabatic connection formula. Projects: (simple) KS SCF for $H_2O$ with LDA: density and KS eigenvalues; (complex) adiabatic connection integrand for He: $\lambda$-dependence of $E_{\mathrm{xc}}^{\lambda}$.

## Module III.3 / L04: Overview of XC Functionals — Jacob's Ladder

**90-min pacing:** (0–20 min) Perdew's Jacob's ladder: LDA, GGA, meta-GGA, hybrid GGA, double hybrid — each rung adds one more ingredient; (20–45 min) LDA: $E_{\mathrm{xc}}^{\mathrm{LDA}} = \int \epsilon_{\mathrm{xc}}^{\mathrm{HEG}}(\rho)\rho\, d\mathbf{r}$; Dirac exchange; Vosko–Wilk–Nusair correlation; PW92; accuracy and systematic overbinding; GGA: $E_{\mathrm{xc}}^{\mathrm{GGA}}[\rho, |\nabla\rho|]$; PBE (parameter-free); B88 exchange; LYP correlation; B3LYP (empirical hybrid); (45–65 min) meta-GGA: kinetic energy density $\tau = \sum_i |\nabla\phi_i|^2/2$; TPSS; SCAN; MN15; range-separated hybrids: $v_{ee} = \frac{1-\mathrm{erf}(\omega r)}{r} + \frac{\mathrm{erf}(\omega r)}{r}$; CAM-B3LYP; HSE06; (65–90 min) Double hybrids: $E_{\mathrm{xc}}^{\mathrm{DH}} = a_x E_x^{\mathrm{HF}} + (1-a_x)E_x^{\mathrm{GGA}} + a_c E_c^{\mathrm{GGA}} + b_c E_c^{\mathrm{MP2}}$; B2-PLYP; $\mathcal{O}(N^5)$ scaling; accuracy compared to CCSD(T); dispersion corrections: D3, D3BJ, MBD, VV10; practical guidance for choosing functionals.

**Core learning outcomes:**

- Describe each rung of Jacob's ladder and its new ingredient.
- State the key properties and failures of LDA, PBE, and B3LYP.
- Explain range-separated hybrids and when to use them.
- Choose an appropriate functional for a given property and accuracy target.

**Tier differentiation:** HS: Jacob's ladder as a guide to increasingly accurate DFT. Beg. UG: LDA, PBE, B3LYP; when to use each; dispersion corrections. Adv. UG: meta-GGA; range-separated hybrids; B2-PLYP double hybrid. MSc: SCAN; MBD dispersion; GMTKN55 benchmark comparison. PhD: self-interaction correction; OEP exact exchange; DFT+U; ACFDT-RPA. **Assessment:** Set A — identify rung for SCAN, CAM-B3LYP, B2-PLYP; choose functional for H-bond energy. Set B — PBE exchange enhancement factor derivation; B2-PLYP energy expression. Projects: (simple) Jacob's ladder benchmark for atomisation energies: LDA/PBE/B3LYP/B2-PLYP vs. CCSD(T); (complex) dispersion corrections for rare-gas dimers: D3 vs. MBD vs. VV10.

## Module III.3 / L05: Useful DFT Quantities — Fukui Functions and Reactivity Indices

**90-min pacing:** (0–20 min) Conceptual DFT: chemical potential $\mu = (\partial E/\partial N)_v = -\chi$ (negative of electronegativity); chemical hardness $\eta = \frac{1}{2}(\partial^2 E/\partial N^2)_v$; (20–45 min) Fukui functions: $f^+(\mathbf{r}) = \rho_{N+1}(\mathbf{r}) - \rho_N(\mathbf{r})$ (nucleophilic attack site); $f^-(\mathbf{r}) = \rho_N(\mathbf{r}) - \rho_{N-1}(\mathbf{r})$ (electrophilic attack); $f^0 = \frac{1}{2}(f^+ + f^-)$ (radical); finite-difference approximation; Fukui from frontier orbitals; (45–65 min) Yang–Parr local softness $s^\pm(\mathbf{r}) = Sf^\pm(\mathbf{r})$; electrophilicity index $\omega = \mu^2/2\eta$; hardness–softness complementarity; HSAB principle; (65–90 min) Dual descriptor $\Delta f(\mathbf{r}) = f^+ - f^-$; identifies nucleophilic ($\Delta f < 0$) and electrophilic ($\Delta f > 0$) sites simultaneously; constrained DFT (CDFT) for charge-transfer states; non-covalent interaction (NCI) plots; reduced density gradient; applications in organic and organometallic reactivity.

**Core learning outcomes:**

- Define chemical potential, hardness, and electronegativity in DFT.
- Compute Fukui functions $f^\pm$ from frontier orbital densities.
- Apply the HSAB principle using hardness and Fukui functions.
- Interpret the dual descriptor and NCI plots for reactivity analysis.

**Tier differentiation:** HS: "Fukui functions point to where a molecule is most reactive." Beg. UG: $f^\pm$ from frontier orbitals; HOMO/LUMO as reactivity sites; HSAB. Adv. UG: Fukui from density difference; dual descriptor; electrophilicity. MSc: CDFT for CT states; NCI analysis; chemical softness and polarisability. PhD: DFT $N$-derivative discontinuity; derivative of chemical potential; ensemble DFT for Fukui. **Assessment:** Set A — compute $f^\pm$ from HOMO/LUMO for formaldehyde; hardness for hard/soft acids. Set B — dual descriptor for acrolein: identify nucleophilic and electrophilic sites. Projects: (simple) Fukui function map for a series of carbonyl compounds; (complex) CDFT for charge transfer in push-pull chromophore: ground vs. CT state densities.

## Module III.3 / L06: Tight-Binding and Effective Hamiltonians

**90-min pacing:** (0–20 min) LCAO tight-binding Hamiltonian: $H_{ij} = \langle\phi_i|\hat{H}|\phi_j\rangle$; hopping $t_{ij}$ and on-site $\varepsilon_i$; connection to DFT Hamiltonian matrix elements; (20–45 min) Slater–Koster two-centre approximation: parametrise $H_{ij}$ by bond type ($ss\sigma$, $sp\sigma$, $pp\sigma$, $pp\pi$, ...) and distance; tables of Slater–Koster integrals; applications: Si band structure, graphene Dirac cone; (45–65 min) Su–Schrieffer–Heeger (SSH) model: 1D chain with alternating hoppings $t, t'$; topological insulator; bulk-boundary correspondence; chiral symmetry; edge states; (65–90 min) DFTB (density functional tight-binding): repulsive pair potential from DFT; self-consistent charge (SCC-DFTB); applications to bio- and nano-systems; GFN-xTB for non-covalent interactions; parametrisation strategies.

**Core learning outcomes:**

- Set up a tight-binding Hamiltonian using Slater–Koster parameters.
- Compute the band structure for a 1D chain and graphene.
- Describe the SSH model and identify its topological edge states.
- Explain DFTB and its relation to DFT.

**Tier differentiation:** HS: tight-binding as a simple model for electrons in solids. Beg. UG: 1D chain; hopping; band structure; graphene Dirac cone. Adv. UG: Slater–Koster; SSH model; topological edge states. MSc: SCC-DFTB; GFN-xTB; applications to bio-systems. PhD: DFTB parametrisation; non-orthogonal TB; Berry phase in TB models. **Assessment:**

Set A — 1D TB chain: band structure and DOS; Dirac cone for graphene. Set B — SSH model: bulk-boundary correspondence; edge state energies. Projects: (simple) TB band structure for graphene and hBN: compare Dirac cones; (complex) SSH model: topological phase transition as $t'/t$ varies.

## Module III.3 / L07: Hückel and Möbius Aromaticity

**90-min pacing:** (0–20 min) Simple Hückel MO theory: $\alpha$ and $\beta$ parameters; secular determinant; MO energies and coefficients for ethylene, allyl, butadiene; (20–45 min) Benzene and aromaticity: $4n + 2$ Hückel rule; delocalisation energy; resonance energy; anti-aromaticity ($4n$); NICS (nucleus-independent chemical shift) as probe; (45–65 min) Möbius topology: $4n$ electrons aromatic on Möbius ring; $4n+2$ anti-aromatic; Heilbronner's prediction (1964); twisted annulene experiments; (65–90 min) Woodward–Hoffmann rules: orbital symmetry conservation; pericyclic reactions; thermal and photochemical; Frontier Molecular Orbital (FMO) theory; HOMO–LUMO interactions; electrocyclic reactions; cycloadditions; sigmatropic rearrangements.

**Core learning outcomes:**

- Apply Hückel theory to compute MO energies for $\pi$ systems.
- State Hückel's $4n + 2$ rule and explain aromaticity/anti-aromaticity.
- Describe Möbius aromaticity and contrast with Hückel.
- Apply Woodward–Hoffmann rules to predict pericyclic reaction outcomes.

**Tier differentiation:** HS: aromaticity from Hückel rule; why benzene is stable. Beg. UG: Hückel for ethylene–benzene; aromatic stabilisation energy; W–H rules. Adv. UG: NICS; Möbius topology; FMO theory for cycloadditions. MSc: beyond Hückel: extended Hückel; aromatic transition states. PhD: magnetic aromaticity indices; ACID plots; spherical aromaticity (fullerenes). **Assessment:** Set A — Hückel MO energies for benzene; resonance energy; W–H rule for [4+2]. Set B — Möbius aromatic [8]annulene: Hückel vs. Möbius electron count. Projects: (simple) Hückel MO solver for arbitrary acyclic and cyclic $\pi$ systems; (complex) Woodward–Hoffmann correlation diagram for thermal vs. photochemical ring closure.

## Module III.3 / L08: Crystal Field Theory

**90-min pacing:** (0–20 min) Transition metal complexes: $d$ orbitals in the field of ligands; point-charge model; electrostatic perturbation of $d$-orbital energies; (20–45 min) Octahedral field: $O_h$ point group; $d$ orbitals split into $t_{2g}$ ($d_{xy}, d_{xz}, d_{yz}$) and $e_g$ ($d_{z^2}, d_{x^2-y^2}$); crystal field splitting $\Delta_o = 10Dq$; energy: $t_{2g} = -4Dq$, $e_g = +6Dq$; (45–65 min) High-spin vs. low-spin: pairing energy $P$ vs. $\Delta_o$; configuration $d^4$–$d^7$ have two possibilities; spectrochemical series $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CO < CN^-$; CFSE (crystal field stabilisation energy); (65–90 min) Tetrahedral field: $\Delta_t = \frac{4}{9}\Delta_o$; square-planar field: strong field; Jahn–Teller effect: distortion for $e_g^1$, $e_g^3$ ($d^9$, $Cu^{2+}$); elongation vs. compression; applications: colour of complexes, magnetic moments.

**Core learning outcomes:**

- Derive the $d$-orbital splitting in octahedral and tetrahedral fields.
- Determine high-spin vs. low-spin configurations from $\Delta_o$ and $P$.
- Apply the spectrochemical series.
- Explain the Jahn–Teller effect and identify susceptible configurations.

**Tier differentiation:** HS: why transition metal complexes are coloured; high-spin vs. low-spin. Beg. UG: octahedral splitting; $t_{2g}/e_g$; CFSE; spin states. Adv. UG: tetrahedral; square planar; Jahn–Teller; spectrochemical series. MSc: $10Dq$ from group theory; selection rules for $d$–$d$ transitions; Orgel diagrams. PhD: CFT from first principles (DFT+U); multiplet theory; dynamical Jahn–Teller. **Assessment:** Set A — $d$ orbital splitting for $Fe^{3+}$ in octahedral field; CFSE for high-spin $d^6$; $\Delta_t$ from $\Delta_o$. Set B — Jahn–Teller distortion for $Cu^{2+}$; derive $t_{2g}$ and $e_g$ energies from point-charge model. Projects: (simple) colour of octahedral $Ti^{3+}$ complex from $\Delta_o$; (complex) Jahn–Teller distortion in $CuF_2$: Q-mode potential energy surface.

## Module III.3 / L09: Ligand Field Theory

**90-min pacing:** (0–20 min) LFT vs. CFT: LFT treats covalency; MO description of metal–ligand bonding; $\sigma$ and $\pi$ donor/acceptor ligands; (20–45 min) Angular overlap model (AOM): $e_\sigma, e_\pi$ parameters; energy from angular overlap geometry factors $F_{lm}(\theta, \phi)$; comparison with Slater–Koster; applications for arbitrary geometries; (45–65 min) Tanabe–Sugano diagrams: energy levels for $d^n$ configurations in octahedral field; Racah parameters $A, B, C$ (electron repulsion); spin-orbit coupling; term symbols for free ion and complex; (65–90 min) Nephelauxetic effect: reduction of Racah $B$ parameter in complex vs. free ion; $\beta = B_{\text{complex}}/B_{\text{free}}$; covalency measure; connection to charge transfer; magnetic moments and spin-orbit coupling in heavy metal complexes; $g$-tensor from EPR.

**Core learning outcomes:**

- Apply the angular overlap model to compute $d$-orbital energies.
- Read and interpret Tanabe–Sugano diagrams.
- Define the nephelauxetic effect and its relation to covalency.
- Compute magnetic moments and $g$-tensors from LFT.

**Tier differentiation:** HS: LFT as the bridge between crystal field and full MO theory. Beg. UG: AOM; Tanabe–Sugano for $Cr^{3+}$; nephelauxetic ratio. Adv. UG: $e_\sigma, e_\pi$ for common ligands; Racah parameters; spin-orbit. MSc: $g$-tensor from EPR; zero-field splitting; spin Hamiltonian. PhD: ab initio LFT (AILFT); multireference CASSCF for spin–orbit; exchange coupling in dinuclear complexes. **Assessment:** Set A — Tanabe–Sugano for $d^3$ in octahedral: identify ground state and lowest transitions. Set B — AOM energies for a trigonal complex; nephelauxetic $\beta$ from UV-vis data. Projects: (simple) Tanabe–Sugano diagram for $Cr^{3+}$: spin-allowed transitions vs. $\Delta/B$; (complex) AILFT from CASSCF: extract $e_\sigma, e_\pi$ for $[Fe(CN)_6]^{3-}$.

## Module III.3 / L10: Full CI Quantum Monte Carlo (FCIQMC)

**90-min pacing:** (0–20 min) Motivation: FCI is exact but exponentially expensive; FCIQMC (Booth & Alavi, 2009) stochastically samples the FCI space without storing all amplitudes; (20–45 min) FCIQMC algorithm: represent $|\Psi\rangle = \sum_i c_i |D_i\rangle$ by signed walkers on Slater determinants; imaginary-time propagation $\partial_\tau c_i = -\hat{H}_{ij} c_j$; three steps per iteration: (1) spawn: walker at $D_i$ creates offspring at $D_j$ with probability $\propto |H_{ij}|$; (2) death: remove walker with probability $\propto E_S - H_{ii}$; (3) annihilation: opposite-sign walkers on same $D_i$ cancel; (45–65 min) Sign problem: positive and negative walkers must be sampled; without intervention the walk decays to a superposition; initiator approximation ($i$-FCIQMC): only "initiator" determinants (with $|c_i| > n_a$) can spawn to unvisited determi-

nants; controlled sign error; (65–90 min) Semistochastic FCIQMC: deterministic subspace for the most important determinants; convergence improvement; FCIQMC benchmarks: carbon dimer (Booth 2013), chromium dimer, iron–sulfur clusters; accuracy vs. CCSD(T) and DMC; GPU acceleration.

**Core learning outcomes:**

- Describe the FCIQMC algorithm: spawn, death, and annihilation steps.
- Explain the sign problem and the initiator approximation.
- Describe semistochastic FCIQMC and its convergence improvement.
- Place FCIQMC in the accuracy hierarchy vs. CCSD(T) and DMC.

**Tier differentiation:** HS: "FCIQMC is a random walk that converges to the exact ground state." Beg. UG: imaginary-time propagation; walker concept; annihilation. Adv. UG: sign problem; initiator approximation; error analysis. MSc: semistochastic FCIQMC; benchmark applications; GPU acceleration. PhD: FCIQMC for excited states; tailored-CC from FCIQMC; FCIQMC-R12. **Assessment:** Set A — FCIQMC steps for $H_2$: spawn/death probabilities; initiator threshold. Set B — initiator approximation error scaling; semistochastic subspace selection criterion. Projects: (simple) FCIQMC for $H_4$ square: energy vs. walker number; (complex) i-FCIQMC for $N_2$ at stretched geometry: convergence with initiator threshold.

## Module III.3 / L11: Diffusion Quantum Monte Carlo for Molecules

**90-min pacing:** (0–20 min) Recap (from II.1 L06): DMC as imaginary-time projection; $-\partial_\tau \Psi = (\hat{H} - E_T)\Psi$; walker diffusion and branching; (20–45 min) Trial wavefunctions for molecular DMC: Slater–Jastrow $\Psi_T = \mathcal{A}[\Phi]\exp(J)$; Jastrow factor: electron–electron, electron–nuclear, three-body; cusp conditions; optimisation via VMC; fixed-node approximation: constrain walkers to nodal surface of $\Psi_T$; (45–65 min) Quality of nodes: HF nodes vs. CASSCF nodes for multi-reference systems; nodal bias; sign problem and fixed-node error as the main source of DMC error; (65–90 min) Pseudopotentials in DMC: non-locality problem; T-moves; Casula's locality approximation; backflow transformation; DMC for molecular energetics: benchmarks for atomisation energies, reaction barriers; comparison with CCSD(T) and FCIQMC; GPU implementation (CASINO, QMCPack, TurboRVB).

**Core learning outcomes:**

- Construct a Slater–Jastrow trial wavefunction and optimise its parameters.
- Explain the fixed-node approximation and identify its main sources of error.
- Describe T-moves for non-local pseudopotentials.
- Assess DMC accuracy for molecular energetics vs. other methods.

**Tier differentiation:** HS: DMC as "the most accurate method for large molecules." Beg. UG: Slater–Jastrow; cusp conditions; fixed-node approximation. Adv. UG: node quality from HF vs. CASSCF; T-moves derivation; pseudopotentials. MSc: backflow transformation; DMC benchmarks; GPU workflow. PhD: exact nodes from FCI; multi-determinant Jastrow; neural-network trial functions. **Assessment:** Set A — Jastrow factor construction; cusp condition for $H_2$; fixed-node energy bound. Set B — T-moves derivation; fixed-node error analysis for $N_2$ at $r = 2R_e$. Projects: (simple) DMC for $H_2$: fixed-node energy vs. HF and CASSCF nodes; (complex) node quality study: HF vs. CASSCF(6,6) nodes for $O_3$ dissociation.

## Module III.3 / L12: Synthesis and Outlook

**90-min pacing:** (0–20 min) Complete method hierarchy for molecular electronic structure: HF → DFT → MP2 → CCSD → CCSD(T) → CASSCF → NEVPT2 → MRCI → FCIQMC → DMC; accuracy and cost at each level; (20–40 min) Open challenges: (1) strong correlation in transition metal and $f$-element chemistry; (2) excited states: TDDFT, EOM-CC, MR methods, quantum computing; (3) relativistic effects: scalar and spin–orbit; four-component DFT; X2C; (20–40 min continued) (4) periodic systems: DFT+U, hybrid DFT, periodic CCSD(T), stochastic CCSD(T), FCIQMC for solids; (5) large molecules and biological systems: DLPNO-CCSD(T), DFTB, ML potentials; (65–80 min) Future directions: machine learning for electronic structure (SchNet, PaiNN, DeepMind AlphaFold-like potentials); neural-network wavefunctions (DeepWF, FermiNet, PauliNet); quantum computing (VQE, QITE, QPE for fault-tolerant); (80–90 min) Synthesis: how all the methods taught in Series III fit together; choosing the right method; the promise of exact quantum simulation.

**Core learning outcomes:**

- Place all quantum chemistry methods in a unified accuracy–cost hierarchy.
- Identify the open challenges and describe current frontiers.
- Describe neural-network wavefunctions and their advantages.
- Outline the quantum computing approach to quantum chemistry.

**Tier differentiation:** HS: from $H_2$ to drug design — quantum chemistry shapes the modern world. Beg. UG: method hierarchy; practical choice guide; exciting frontiers. Adv. UG: open challenges in detail; ML potentials; VQE concept. MSc: neural-network wavefunctions; periodic CCSD(T); relativistic DFT. PhD: quantum complexity; resource estimation for QPE; FermiNet for correlated systems. **Assessment:** Set A — method selection for: (i) organocatalysis barrier, (ii) $Mn_4O_4$ cluster, (iii) singlet oxygen, (iv) protein binding energy. Set B — VQE ansatz for $H_2$: UCCSD circuit; FermiNet architecture for helium. Projects: (simple) decision-tree: method selection for 10 representative chemical problems; (complex) FermiNet energy for He and Li: variational quality vs. classical wavefunction ansatz.

***Module III.3 Reading:*** *Koch & Holthausen, A Chemist's Guide to DFT (Wiley); Parr & Yang, Density-Functional Theory of Atoms and Molecules (Oxford); Booth, Thom & Alavi, J. Chem. Phys.* ***131****, 054106 (2009) — FCIQMC; Foulkes et al., Rev. Mod. Phys.* ***73****, 33 (2001) — QMC review.*

# Appendix A — Programme Roadmap

| Module | Title (abbreviated) | Core Prerequisites | Lectures |
|---|---|---|---|
| **I.1** | State Space, Operators, Bra-Ket | Linear algebra, classical mechanics | 6 |
| **I.2** | Bra-Ket Notation (Dirac + Sakurai) | I.1 (concurrent or prior) | 12 |
| **I.3** | Dynamics, Symmetry, Perturbation Theory | I.1; I.2 recommended | 10 |
| **I.4** | Paradigm Physical Systems | I.1–I.3 | 12 |
| **I.5** | Introductory Physical Systems (Menu) | I.1–I.3 | 12 |
| **II.1** | Ground-State Methods | Series I | 12 |
| **II.2** | Path Integrals | Series I | 12 |
| **II.3** | Angular Momentum | Series I | 12 |
| **II.4** | Second Quantisation | Series I, II.3 | 12 |
| **II.5** | NEGF | II.4, II.7 | 12 |
| **II.6** | Light-Matter/Spectroscopy | Series I, II.4 | 12 |
| **II.7** | Quantum Stat. Mech. | Series I | 12 |
| **II.8** | Advanced Physical Systems | I.1–I.4, Series II | 12 |
| **II.9** | Quantum Information & Logic | II.4; I.1–I.3 | 12 |
| **II.10** | Signature QI Systems | II.9 | 12 |
| **II.11** | The Tenfold Way | II.3, II.4, II.7 | 12 |
| **II.12** | Signature Tenfold Way Systems | II.11 | 12 |
| **III.1** | Basic Premises | Series I | 3 |
| **III.2** | Electronic Structure | Series I, II.4 (rec.) | 16 |
| **III.3** | DFT and Beyond | III.1, III.2 | 12 |
| | **Grand total** | | **216** |

# Appendix B — Suggested Textbooks

## Quantum Mechanics — Foundations

- Dirac, P.A.M. — *The Principles of Quantum Mechanics*, 4th ed. (Oxford)
- Sakurai, J.J. & Napolitano, J. — *Modern Quantum Mechanics*, 3rd ed.
- Weinberg, S. — *Lectures on Quantum Mechanics*, 2nd ed.
- Galindo, A. & Pascual, P. — *Quantum Mechanics* (2 vols., Springer)

## Many-Body and Advanced Topics

- Fetter, A.L. & Walecka, J.D. — *Quantum Theory of Many-Particle Systems*

- Bruus, H. & Flensberg, K. — *Introduction to Many-Body Quantum Theory in Condensed Matter Physics*
- Stefanucci, G. & van Leeuwen, R. — *Nonequilibrium Many-Body Theory of Quantum Systems* (Cambridge)
- Negele, J.W. & Orland, H. — *Quantum Many-Particle Systems*


## Quantum Chemistry and Electronic Structure


- Szabo, A. & Ostlund, N.S. — *Modern Quantum Chemistry*
- Helgaker, T., Jorgensen, P. & Olsen, J. — *Molecular Electronic-Structure Theory* (Wiley)
- Jensen, F. — *Introduction to Computational Chemistry*, 3rd ed.
- Koch, W. & Holthausen, M.C. — *A Chemist's Guide to Density Functional Theory*
- Martin, R.M., Reining, L. & Ceperley, D. — *Interacting Electrons* (Cambridge)